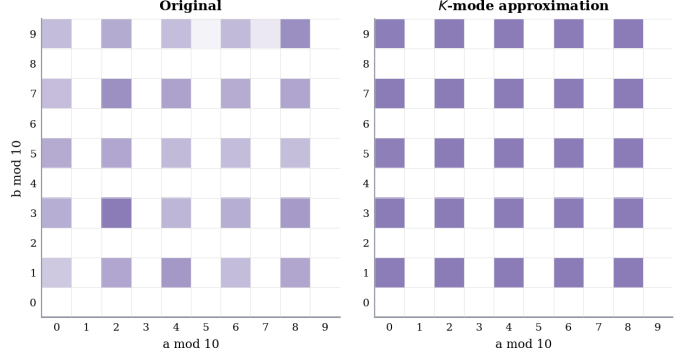


L12_F1468 $K = 8, R^2 = 0.023$ *col_parity: 33%; parity: 31%; row_parity: 31%; mixed: 4%*

$$f(a, b) \approx 0.161 + 0.322 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.311 \cos\left(\frac{2\pi(5a)}{10}\right) \\ + 0.311 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.03 \cos\left(\frac{2\pi(2a)}{10} - \pi\right) + 0.03 \cos\left(\frac{2\pi(2a+5b)}{10}\right) \\ + 0.028 \cos\left(\frac{2\pi(-3a)}{10} + 3.033\right) + 0.028 \cos\left(\frac{2\pi(-3a+5b)}{10} - 0.109\right) + 0.02 \cos\left(\frac{2\pi(5a+3b)}{10} - 0.174\right)$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	33%
parity	$(-1)^{a+b}$	31%
$\text{row}_{\text{parity}}$	—	31%
mixed	—	4%
row	$f(a)$	1%

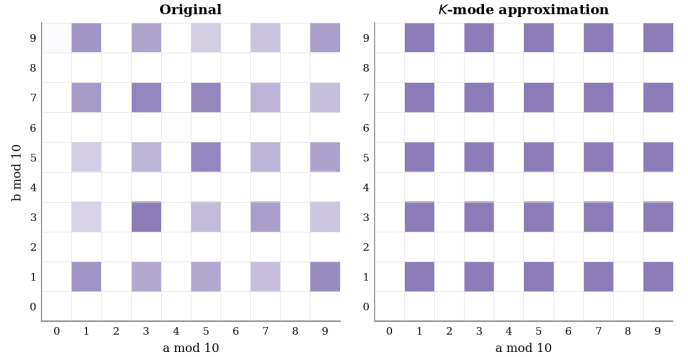


Left: original. Right: K -mode approx.

L21_F30390 $K = 8, R^2 = 0.032$ *col_parity: 30%; parity: 30%; row_parity: 30%; mixed: 6%*

$$f(a, b) \approx 0.164 + 0.327 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.326 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\ + 0.326 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) + 0.036 \cos\left(\frac{2\pi(-a-b)}{10} + 0.083\right) + 0.036 \cos\left(\frac{2\pi(a-4b)}{10} + 3.057\right) \\ + 0.035 \cos\left(\frac{2\pi(4a+4b)}{10} + 0.104\right) + 0.035 \cos\left(\frac{2\pi(4a-b)}{10} - 3.037\right) + 0.028 \cos\left(\frac{2\pi(3a+2b)}{10} + \frac{3\pi}{4}\right)$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	30%
parity	$(-1)^{a+b}$	30%
$\text{row}_{\text{parity}}$	—	30%
mixed	—	6%
sum	$\cos(2\pi k(a+b)/N)$	2%
diff	$\cos(2\pi k(b-a)/N)$	1%

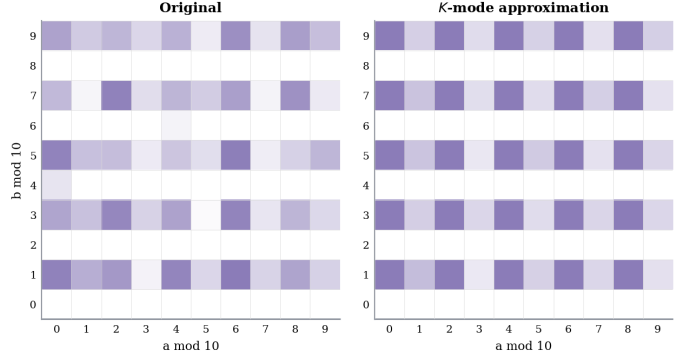


Left: original. Right: K -mode approx.

L22_F158216 $K = 8, R^2 = 0.036$ *col_parity: 61%; row_parity: 13%; parity: 12%; mixed: 9%*

$$f(a, b) \approx 0.258 + 0.505 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.231 \cos\left(\frac{2\pi(5a)}{10}\right) \\ + 0.219 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.042 \cos\left(\frac{2\pi(2a)}{10} - \frac{\pi}{4}\right) + 0.039 \cos\left(\frac{2\pi(2a+5b)}{10} + \frac{2\pi}{3}\right) \\ + 0.038 \cos\left(\frac{2\pi(2a+3b)}{10} - 2.994\right) + 0.038 \cos\left(\frac{2\pi(4a+5b)}{10} + 0.31\right) + 0.037 \cos\left(\frac{2\pi(2a-2b)}{10}\right)$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	61%
$\text{row}_{\text{parity}}$	—	13%
parity	$(-1)^{a+b}$	12%
mixed	—	9%
row	$f(a)$	2%
col	$f(b)$	1%
sum	$\cos(2\pi k(a+b)/N)$	1%
diff	$\cos(2\pi k(b-a)/N)$	1%

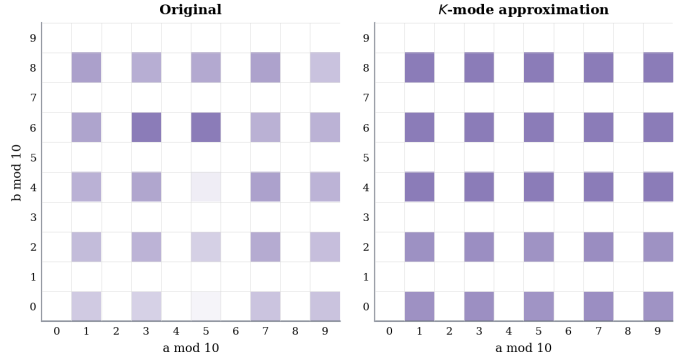


Left: original. Right: K -mode approx.

L12_F128110 $K = 8, R^2 = 0.060$ parity: 28%; row_parity: 28%; col_parity: 28%; mixed: 9%; col: 4%

$$\begin{aligned}
f(a, b) \approx & 0.14 + 0.28 \cos\left(\frac{2\pi(5b)}{10}\right) + 0.28 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) \\
& + 0.28 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.044 \cos\left(\frac{2\pi(4b)}{10} - 2.538\right) + 0.044 \cos\left(\frac{2\pi(5a+4b)}{10} + 0.603\right) \\
& + 0.044 \cos\left(\frac{2\pi(-b)}{10} - 2.538\right) + 0.044 \cos\left(\frac{2\pi(5a-b)}{10} + 0.603\right) + 0.024 \cos\left(\frac{2\pi(2a)}{10} - 2.93\right)
\end{aligned}$$

Dir	Structure	Frac
parity	$(-1)^{a+b}$	28%
$\text{row}_{\text{parity}}$	—	28%
$\text{col}_{\text{parity}}$	—	28%
mixed	—	9%
col	$f(b)$	4%
diff	$\cos(2\pi k(b-a)/N)$	1%

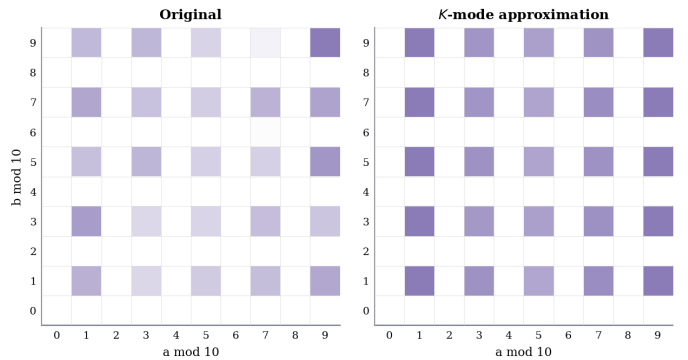


Left: original. Right: K -mode approx.

L12_F37391 $K = 8, R^2 = 0.080$ row_parity: 28%; parity: 28%; col_parity: 28%; mixed: 9%; row: 4%

$$\begin{aligned}
f(a, b) \approx & 0.126 + 0.253 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) + 0.252 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) \\
& + 0.252 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.046 \cos\left(\frac{2\pi(4a+5b)}{10} - 0.09\right) + 0.046 \cos\left(\frac{2\pi(a+5b)}{10} - 3.051\right) \\
& + 0.046 \cos\left(\frac{2\pi(4a)}{10} + 3.037\right) + 0.046 \cos\left(\frac{2\pi(-a)}{10} - 0.104\right) + 0.025 \cos\left(\frac{2\pi(2a+2b)}{10} + 1.875\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{row}_{\text{parity}}$	—	28%
parity	$(-1)^{a+b}$	28%
$\text{col}_{\text{parity}}$	—	28%
mixed	—	9%
row	$f(a)$	4%
sum	$\cos(2\pi k(a+b)/N)$	1%

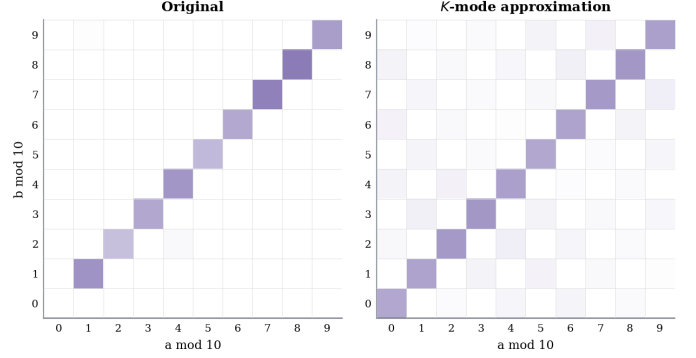


Left: original. Right: K -mode approx.

L13_F57984 $K = 8, R^2 = 0.762$ diff: 75%; mixed: 11%; parity: 10%

$$\begin{aligned}
f(a, b) \approx & 0.067 + 0.134 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.133 \cos\left(\frac{2\pi(4a-4b)}{10}\right) \\
& + 0.133 \cos\left(\frac{2\pi(a-b)}{10}\right) + 0.132 \cos\left(\frac{2\pi(2a-2b)}{10}\right) + 0.132 \cos\left(\frac{2\pi(3a-3b)}{10}\right) \\
& + 0.022 \cos\left(\frac{2\pi(a+b)}{10} + 2.944\right) + 0.022 \cos\left(\frac{2\pi(4a+4b)}{10} - 2.944\right) + 0.022 \cos\left(\frac{2\pi(2b)}{10} + 2.984\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	75%
mixed	—	11%
parity	$(-1)^{a+b}$	10%
sum	$\cos(2\pi k(a+b)/N)$	2%
row	$f(a)$	1%
col	$f(b)$	1%

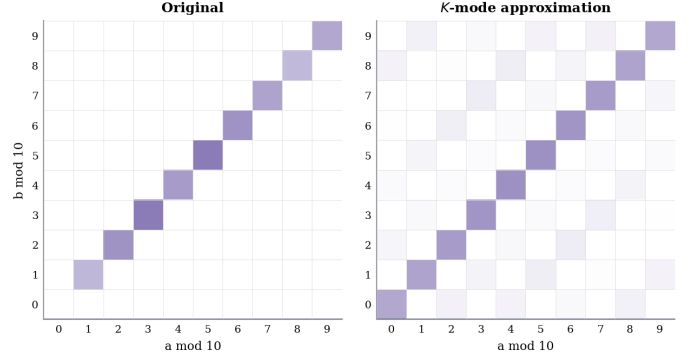


Left: original. Right: K -mode approx.

L13_F33644 $K = 8$, $R^2 = 0.781$ diff: 75%; mixed: 12%; parity: 9%

$$\begin{aligned}
f(a, b) \approx & 0.068 + 0.137 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.137 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\
& + 0.137 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.137 \cos\left(\frac{2\pi(4a-4b)}{10}\right) + 0.137 \cos\left(\frac{2\pi(-a+b)}{10}\right) \\
& + 0.03 \cos\left(\frac{2\pi(-2a+3b)}{10} - 2.838\right) + 0.03 \cos\left(\frac{2\pi(3a-2b)}{10} - 2.838\right) + 0.03 \cos\left(\frac{2\pi(4a+5b)}{10} + 2.837\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	75%
mixed	—	12%
parity	$(-1)^{a+b}$	9%
row	$f(a)$	1%
col	$f(b)$	1%

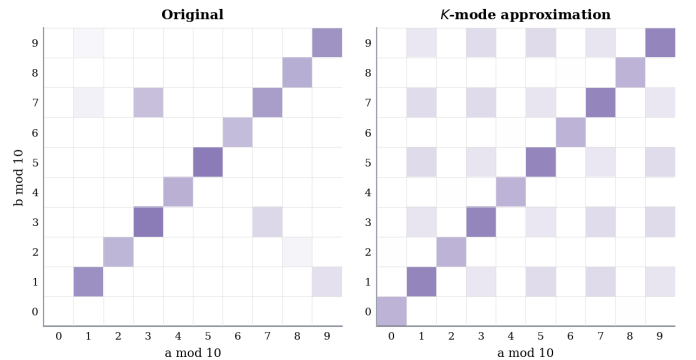


Left: original. Right: K -mode approx.

L20_F2407 $K = 8$, $R^2 = 0.669$ diff: 65%; mixed: 15%; parity: 13%; row-parity: 2%; col-parity: 2%

$$\begin{aligned}
f(a, b) \approx & 0.079 + 0.159 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.135 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\
& + 0.135 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.12 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.041\right) + 0.12 \cos\left(\frac{2\pi(-a+b)}{10} - 0.041\right) \\
& + 0.065 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.065 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) + 0.043 \cos\left(\frac{2\pi(3a+2b)}{10} + 3.096\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	65%
mixed	—	15%
parity	$(-1)^{a+b}$	13%
row _{parity}	—	2%
col _{parity}	—	2%

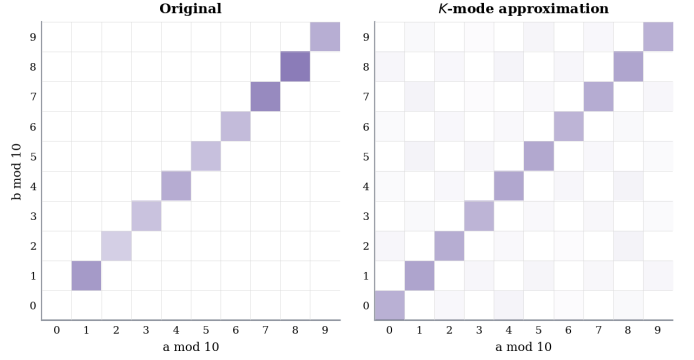


Left: original. Right: K -mode approx.

L14_F153502 $K = 8$, $R^2 = 0.731$ *diff: 71%; mixed: 14%; parity: 9%*

$$f(a, b) \approx 0.057 + 0.114 \cos\left(\frac{2\pi(a-b)}{10}\right) + 0.114 \cos\left(\frac{2\pi(4a-4b)}{10}\right) \\ + 0.114 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.114 \cos\left(\frac{2\pi(-3a+3b)}{10}\right) + 0.114 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\ + 0.021 \cos\left(\frac{2\pi(4a+3b)}{10} + 2.141\right) + 0.021 \cos\left(\frac{2\pi(-2a+5b)}{10} - 2.141\right) + 0.021 \cos\left(\frac{2\pi(3b)}{10} - 2.141\right)$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	71%
mixed	—	14%
parity	$(-1)^{a+b}$	9%
row	$f(a)$	2%
col	$f(b)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%

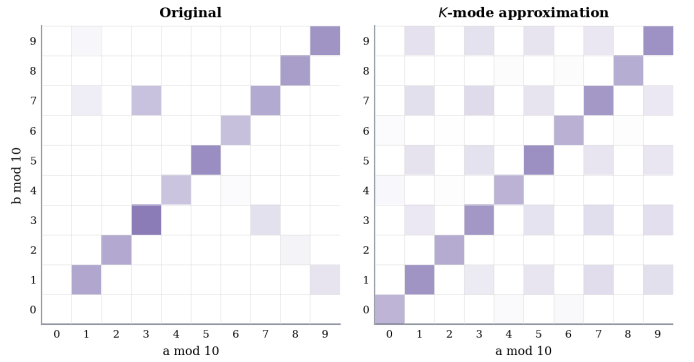


Left: original. Right: K -mode approx.

L21_F1473 $K = 8$, $R^2 = 0.666$ *diff: 65%; mixed: 15%; parity: 13%*

$$f(a, b) \approx 0.075 + 0.151 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.127 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\ + 0.127 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.113 \cos\left(\frac{2\pi(-a+b)}{10}\right) + 0.113 \cos\left(\frac{2\pi(4a-4b)}{10}\right) \\ + 0.053 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.053 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) + 0.035 \cos\left(\frac{2\pi(a-4b)}{10} - 2.885\right)$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	65%
mixed	—	15%
parity	$(-1)^{a+b}$	13%
row _{parity}	—	2%
col _{parity}	—	2%
row	$f(a)$	1%
col	$f(b)$	1%

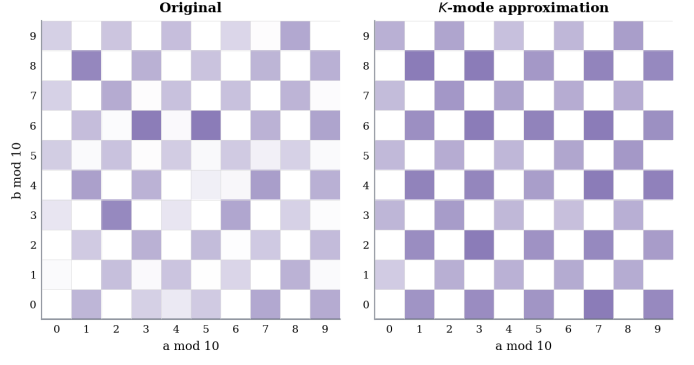


Left: original. Right: K -mode approx.

L13_F75746 $K = 8$, $R^2 = 0.062$ *parity: 76%; mixed: 14%; row: 2%*

$$f(a, b) \approx 0.263 + 0.497 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.077 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) \\ + 0.073 \cos\left(\frac{2\pi(5b)}{10}\right) + 0.047 \cos\left(\frac{2\pi(3a+5b)}{10} - 0.22\right) + 0.046 \cos\left(\frac{2\pi(-4a-3b)}{10} - 1.63\right) \\ + 0.046 \cos\left(\frac{2\pi(2a)}{10} - 3.054\right) + 0.045 \cos\left(\frac{2\pi(a+2b)}{10} + 1.793\right) + 0.04 \cos\left(\frac{2\pi(-b)}{10} - 2.205\right)$$

Dir	Structure	Frac
parity	$(-1)^{a+b}$	76%
mixed	—	14%
row	$f(a)$	2%
row _{parity}	—	2%
col _{parity}	—	2%
col	$f(b)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%
diff	$\cos(2\pi k(b-a)/N)$	1%

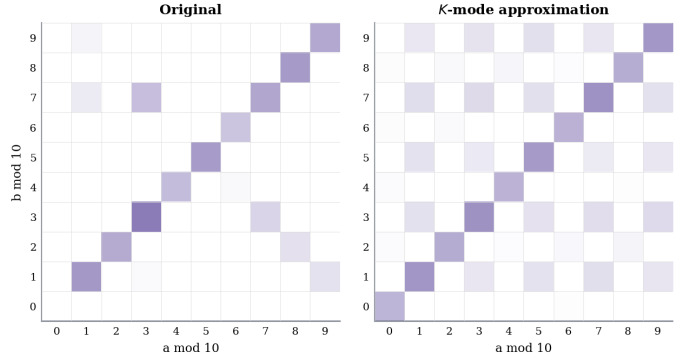


Left: original. Right: K -mode approx.

L19_F26529 $K = 8$, $R^2 = 0.639$ diff: 62%; mixed: 16%; parity: 14%

$$\begin{aligned}
f(a, b) \approx & 0.078 + 0.156 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.126 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\
& + 0.126 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.107 \cos\left(\frac{2\pi(a-b)}{10}\right) + 0.107 \cos\left(\frac{2\pi(4a-4b)}{10}\right) \\
& + 0.051 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.051 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) + 0.037 \cos\left(\frac{2\pi(2b)}{10} - 3.01\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	62%
mixed	—	16%
parity	$(-1)^{a+b}$	14%
row	$f(a)$	2%
col	$f(b)$	2%
row _{parity}	—	1%
col _{parity}	—	1%
sum	$\cos(2\pi k(a+b)/N)$	1%

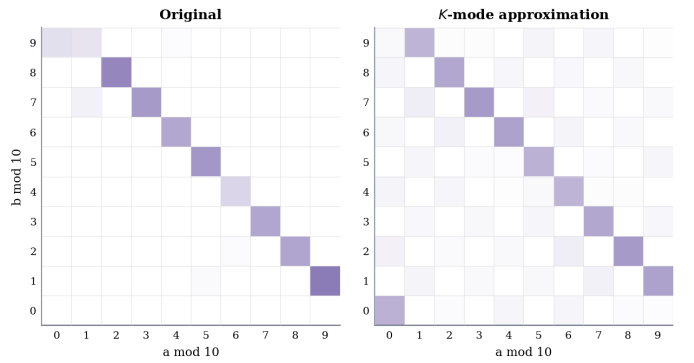


Left: original. Right: K -mode approx.

L14_F114215 $K = 8$, $R^2 = 0.720$ sum: 68%; mixed: 17%; parity: 8%; diff: 3%

$$\begin{aligned}
f(a, b) \approx & 0.064 + 0.124 \cos\left(\frac{2\pi(a+b)}{10} + 0.044\right) + 0.119 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.048\right) \\
& + 0.118 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.117 \cos\left(\frac{2\pi(3a+3b)}{10}\right) + 0.117 \cos\left(\frac{2\pi(4a+4b)}{10}\right) \\
& + 0.036 \cos\left(\frac{2\pi(5a+3b)}{10} + 2.276\right) + 0.033 \cos\left(\frac{2\pi(4a-4b)}{10} - 2.465\right) + 0.032 \cos\left(\frac{2\pi(4a+2b)}{10} + 2.16\right)
\end{aligned}$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	68%
mixed	—	17%
parity	$(-1)^{a+b}$	8%
diff	$\cos(2\pi k(b-a)/N)$	3%
col	$f(b)$	2%
row	$f(a)$	1%

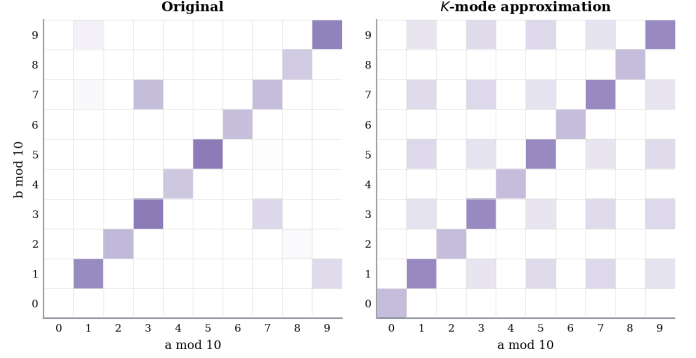


Left: original. Right: K -mode approx.

L22_F27298 $K = 8$, $R^2 = 0.617$ diff: 59%; mixed: 20%; parity: 12%; row_{parity}: 3%; col_{parity}: 3%

$$\begin{aligned}
f(a, b) \approx & 0.074 + 0.149 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.123 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\
& + 0.123 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.112 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.047\right) + 0.112 \cos\left(\frac{2\pi(a-b)}{10} + 0.047\right) \\
& + 0.073 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.073 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) + 0.048 \cos\left(\frac{2\pi(-2a-3b)}{10} - \pi\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	59%
mixed	—	20%
parity	$(-1)^{a+b}$	12%
row _{parity}	—	3%
col _{parity}	—	3%
sum	$\cos(2\pi k(a+b)/N)$	2%

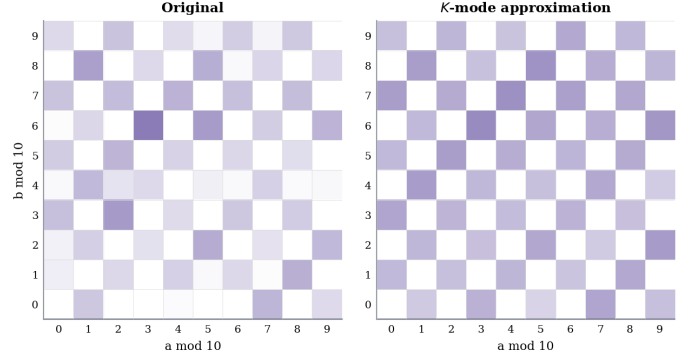


Left: original. Right: K -mode approx.

L19_F97965 $K = 8$, $R^2 = 0.166$ parity: 66%; mixed: 16%; diff: 11%; col: 4%

$$\begin{aligned}
f(a, b) \approx & 0.208 + 0.386 \cos\left(\frac{2\pi(5a+5b)}{10} - \pi\right) + 0.07 \cos\left(\frac{2\pi(2a-2b)}{10} + 3.069\right) \\
& + 0.058 \cos\left(\frac{2\pi(3a-3b)}{10} - 0.069\right) + 0.051 \cos\left(\frac{2\pi(5a+3b)}{10} - \frac{\pi}{4}\right) + 0.05 \cos\left(\frac{2\pi(a-b)}{10} + \frac{\pi}{2}\right) \\
& + 0.044 \cos\left(\frac{2\pi(5a+4b)}{10} + 0.857\right) + 0.042 \cos\left(\frac{2\pi(2a+3b)}{10} + 3.091\right) + 0.04 \cos\left(\frac{2\pi(b)}{10} + 2.434\right)
\end{aligned}$$

Dir	Structure	Frac
parity	$(-1)^{a+b}$	66%
mixed	—	16%
diff	$\cos(2\pi k(b-a)/N)$	11%
col	$f(b)$	4%
sum	$\cos(2\pi k(a+b)/N)$	2%
row	$f(a)$	2%

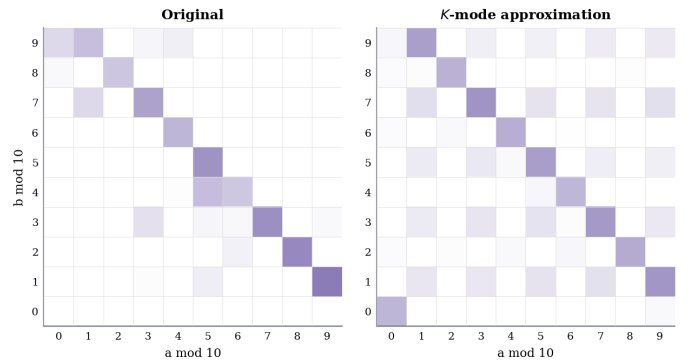


Left: original. Right: K -mode approx.

L14_F45268 $K = 8$, $R^2 = 0.666$ sum: 64%; mixed: 20%; parity: 9%; diff: 2%

$$\begin{aligned}
f(a, b) \approx & 0.082 + 0.135 \cos\left(\frac{2\pi(a+b)}{10} + 0.145\right) + 0.125 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.119 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.131\right) + 0.114 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.172\right) + 0.108 \cos\left(\frac{2\pi(4a+4b)}{10}\right) \\
& + 0.047 \cos\left(\frac{2\pi(5a-3b)}{10} - 2.295\right) + 0.046 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) + 0.045 \cos\left(\frac{2\pi(5b)}{10} - \pi\right)
\end{aligned}$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	64%
mixed	—	20%
parity	$(-1)^{a+b}$	9%
diff	$\cos(2\pi k(b-a)/N)$	2%
col	$f(b)$	2%
row	$f(a)$	1%
row _{parity}	—	1%
col _{parity}	—	1%

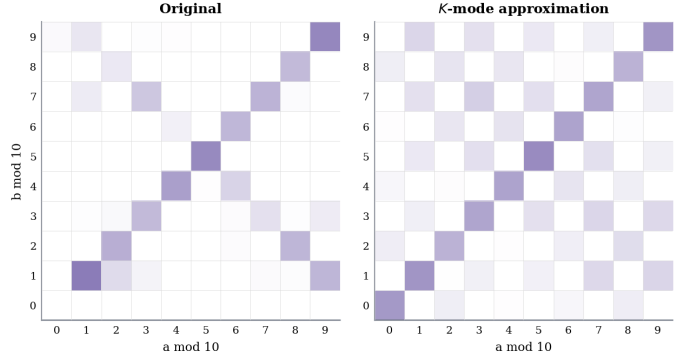


Left: original. Right: K -mode approx.

L16_F131210 $K = 8, R^2 = 0.566$ *diff: 51%; mixed: 18%; parity: 16%; sum: 7%; col: 3%; row: 2%*

$$f(a, b) \approx 0.098 + 0.177 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.119 \cos\left(\frac{2\pi(2a-2b)}{10} - 0.134\right) \\ + 0.114 \cos\left(\frac{2\pi(-3a+3b)}{10}\right) + 0.111 \cos\left(\frac{2\pi(-a+b)}{10}\right) + 0.101 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.118\right) \\ + 0.055 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.076\right) + 0.052 \cos\left(\frac{2\pi(a+b)}{10} - 0.09\right) + 0.052 \cos\left(\frac{2\pi(5b)}{10} + \pi\right)$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	51%
mixed	—	18%
parity	$(-1)^{a+b}$	16%
sum	$\cos(2\pi k(a+b)/N)$	7%
col	$f(b)$	3%
row	$f(a)$	2%
col _{parity}	—	1%

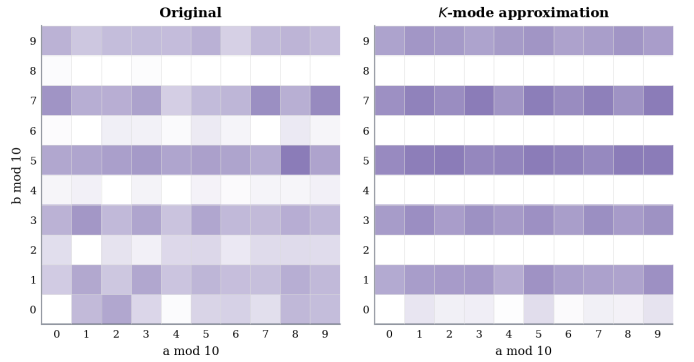


Left: original. Right: K -mode approx.

L00_F83010 $K = 8, R^2 = 0.170$ *col_{parity}: 70%; col: 15%; mixed: 10%; row: 2%*

$$f(a, b) \approx 0.372 + 0.459 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.125 \cos\left(\frac{2\pi(4b)}{10} + 0.396\right) \\ + 0.063 \cos\left(\frac{2\pi(2b)}{10} - 0.631\right) + 0.043 \cos\left(\frac{2\pi(3a-2b)}{10} + \pi\right) + 0.039 \cos\left(\frac{2\pi(3b)}{10} + 0.425\right) \\ + 0.038 \cos\left(\frac{2\pi(5a-3b)}{10} - 2.558\right) + 0.035 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) + 0.034 \cos\left(\frac{2\pi(3a)}{10} + 2.828\right)$$

Dir	Structure	Frac
col _{parity}	—	70%
col	$f(b)$	15%
mixed	—	10%
row	$f(a)$	2%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%

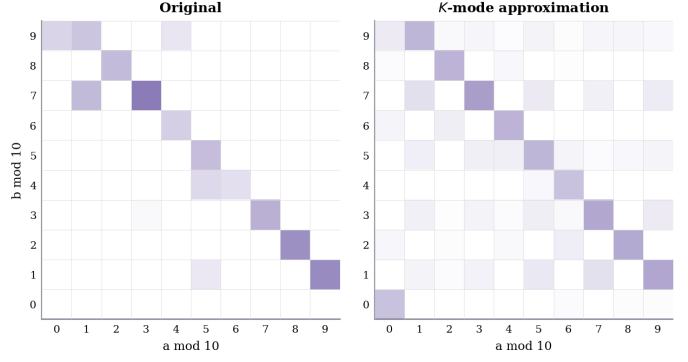


Left: original. Right: K -mode approx.

L15_F85561 $K = 8, R^2 = 0.654$ *sum: 58%; mixed: 24%; parity: 7%; diff: 5%; col: 3%*

$$f(a, b) \approx 0.069 + 0.117 \cos\left(\frac{2\pi(a+b)}{10} + 0.137\right) + 0.106 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\ + 0.102 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.157\right) + 0.1 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.116\right) + 0.099 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.091\right) \\ + 0.058 \cos\left(\frac{2\pi(5a+3b)}{10} + 2.305\right) + 0.048 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.046 \cos\left(\frac{2\pi(4a-4b)}{10} - 2.734\right)$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	58%
mixed	—	24%
parity	$(-1)^{a+b}$	7%
diff	$\cos(2\pi k(b-a)/N)$	5%
col	$f(b)$	3%
col _{parity}	—	2%
row _{parity}	—	1%

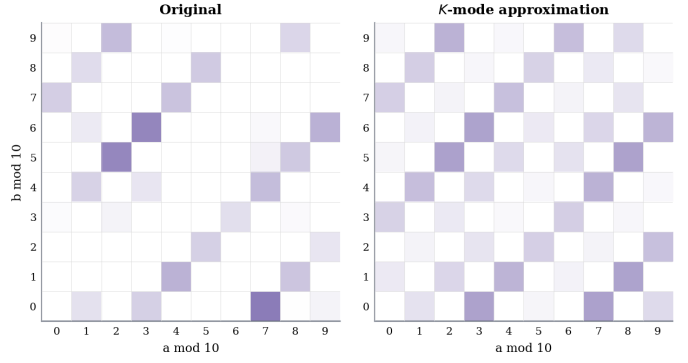


Left: original. Right: K -mode approx.

L13_F107956 $K = 8, R^2 = 0.540$ diff: 44%; parity: 22%; mixed: 20%; sum: 5%; row: 4%; col: 4%

$$\begin{aligned}
f(a, b) \approx & 0.102 + 0.199 \cos\left(\frac{2\pi(5a+5b)}{10} - \pi\right) + 0.135 \cos\left(\frac{2\pi(2a-2b)}{10} + \pi\right) \\
& + 0.132 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.055 \cos\left(\frac{2\pi(2b)}{10} - 0.369\right) + 0.052 \cos\left(\frac{2\pi(2a)}{10} + 2.999\right) \\
& + 0.05 \cos\left(\frac{2\pi(5a+3b)}{10} - 2.742\right) + 0.049 \cos\left(\frac{2\pi(3a+5b)}{10} + 0.199\right) + 0.047 \cos\left(\frac{2\pi(4a-4b)}{10} + 0.353\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	44%
parity	$(-1)^{a+b}$	22%
mixed	—	20%
sum	$\cos(2\pi k(a+b)/N)$	5%
row	$f(a)$	4%
col	$f(b)$	4%

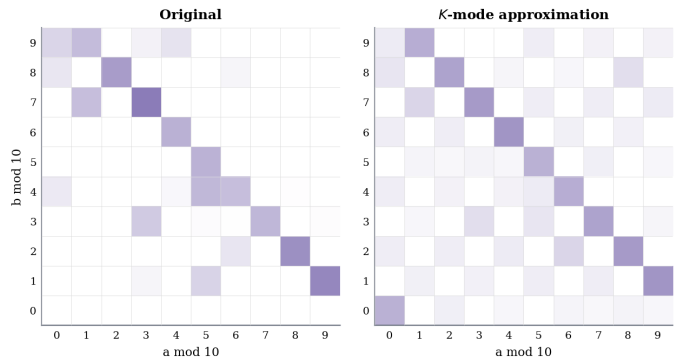


Left: original. Right: K -mode approx.

L14_F107848 $K = 8, R^2 = 0.651$ sum: 56%; mixed: 23%; parity: 11%; diff: 4%; col: 3%

$$\begin{aligned}
f(a, b) \approx & 0.094 + 0.145 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.13 \cos\left(\frac{2\pi(a+b)}{10} + 0.208\right) \\
& + 0.117 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.175\right) + 0.113 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.154\right) + 0.103 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.144\right) \\
& + 0.061 \cos\left(\frac{2\pi(5a+3b)}{10} + 2.479\right) + 0.05 \cos\left(\frac{2\pi(4a+2b)}{10} + 1.85\right) + 0.046 \cos\left(\frac{2\pi(2a-4b)}{10} - 2.726\right)
\end{aligned}$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	56%
mixed	—	23%
parity	$(-1)^{a+b}$	11%
diff	$\cos(2\pi k(b-a)/N)$	4%
col	$f(b)$	3%
row	$f(a)$	2%

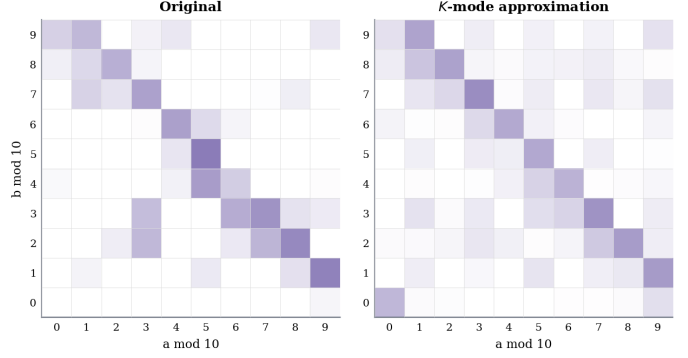


Left: original. Right: K -mode approx.

L13_F53555 $K = 8, R^2 = 0.669$ sum: 62%; mixed: 22%; col: 6%; diff: 4%; parity: 3%

$$\begin{aligned}
f(a, b) \approx & 0.136 + 0.181 \cos\left(\frac{2\pi(a+b)}{10} + 0.242\right) + 0.159 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.328\right) \\
& + 0.1 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.612\right) + 0.082 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.08 \cos\left(\frac{2\pi(4a+4b)}{10} + 0.169\right) \\
& + 0.058 \cos\left(\frac{2\pi(2b)}{10} - \pi\right) + 0.056 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) + 0.054 \cos\left(\frac{2\pi(2a+4b)}{10} - 2.497\right)
\end{aligned}$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	62%
mixed	—	22%
col	$f(b)$	6%
diff	$\cos(2\pi k(b-a)/N)$	4%
parity	$(-1)^{a+b}$	3%
row	$f(a)$	2%
row _{parity}	—	1%

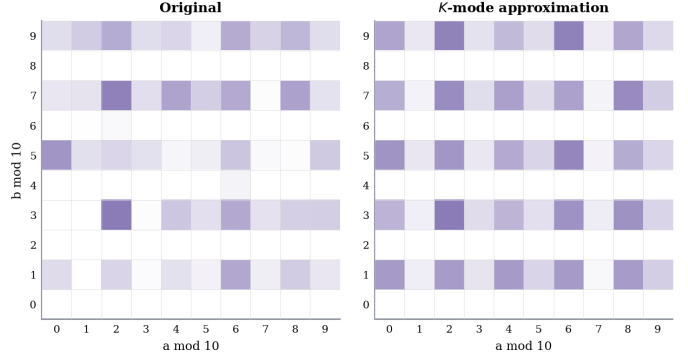


Left: original. Right: K -mode approx.

L14_F129227 $K = 8$, $R^2 = 0.113$ *col-parity: 46%; mixed: 22%; row-parity: 8%; parity: 8%; row: 5%; sum: 4%; col: 4%; diff: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.165 + 0.325 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.137 \cos\left(\frac{2\pi(5a)}{10}\right) \\
& + 0.132 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.062 \cos\left(\frac{2\pi(3a)}{10} + 2.14\right) + 0.058 \cos\left(\frac{2\pi(3a+5b)}{10} - 0.971\right) \\
& + 0.044 \cos\left(\frac{2\pi(3a-2b)}{10} + 0.099\right) + 0.044 \cos\left(\frac{2\pi(2a+2b)}{10} - 0.402\right) + 0.043 \cos\left(\frac{2\pi(2a-3b)}{10} + 2.702\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	46%
mixed	—	22%
row _{parity}	—	8%
parity	$(-1)^{a+b}$	8%
row	$f(a)$	5%
sum	$\cos(2\pi k(a+b)/N)$	4%
col	$f(b)$	4%
diff	$\cos(2\pi k(b-a)/N)$	3%

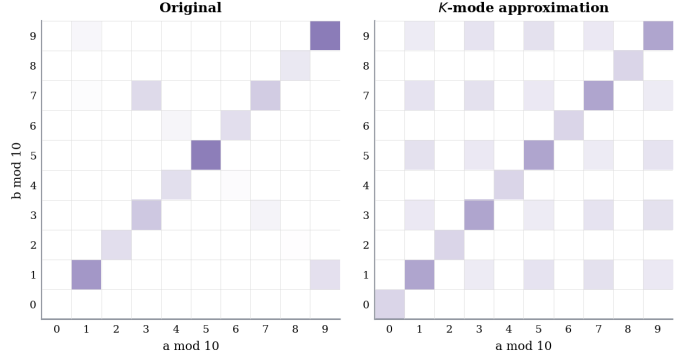


Left: original. Right: K -mode approx.

L17_F1274 $K = 8$, $R^2 = 0.539$ *diff: 49%; mixed: 29%; parity: 9%; row-parity: 4%; col-parity: 4%; sum: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.053 + 0.106 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.086 \cos\left(\frac{2\pi(a-b)}{10} + 0.073\right) \\
& + 0.086 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.073\right) + 0.086 \cos\left(\frac{2\pi(2a-2b)}{10}\right) + 0.086 \cos\left(\frac{2\pi(3a-3b)}{10}\right) \\
& + 0.065 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) + 0.065 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.052 \cos\left(\frac{2\pi(2a+3b)}{10} + \pi\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	49%
mixed	—	29%
parity	$(-1)^{a+b}$	9%
row _{parity}	—	4%
col _{parity}	—	4%
sum	$\cos(2\pi k(a+b)/N)$	3%
row	$f(a)$	1%

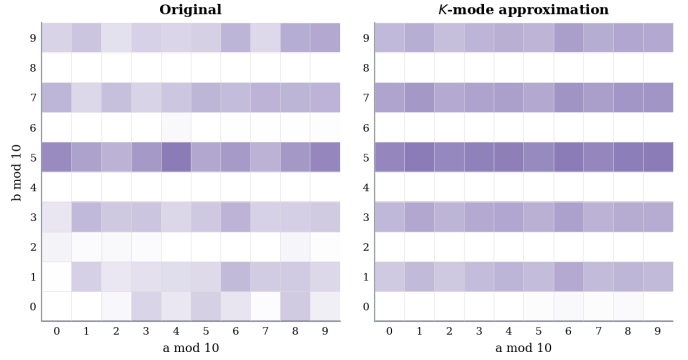


Left: original. Right: K -mode approx.

L00_F86918 $K = 8, R^2 = 0.237$ *col_parity: 66%; col: 21%; mixed: 8%; row: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.256 + 0.431 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.132 \cos\left(\frac{2\pi(4b)}{10} + 0.275\right) \\
& + 0.083 \cos\left(\frac{2\pi(2b)}{10} + 0.089\right) + 0.066 \cos\left(\frac{2\pi(b)}{10} + 2.723\right) + 0.037 \cos\left(\frac{2\pi(4a)}{10} - 2.274\right) \\
& + 0.035 \cos\left(\frac{2\pi(a)}{10} + 1.976\right) + 0.035 \cos\left(\frac{2\pi(a+b)}{10} + \frac{3\pi}{4}\right) + 0.034 \cos\left(\frac{2\pi(4a+5b)}{10} + 0.157\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	66%
col	$f(b)$	21%
mixed	—	8%
row	$f(a)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%
diff	$\cos(2\pi k(b-a)/N)$	1%

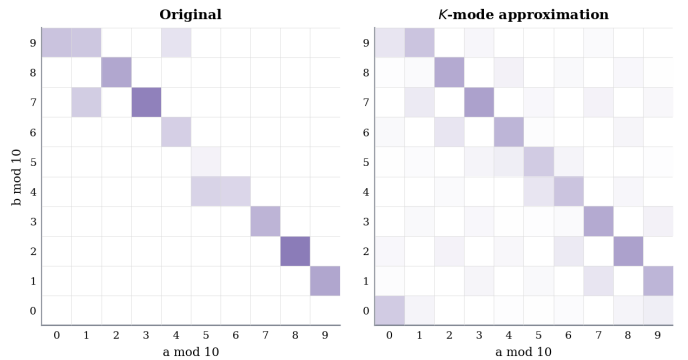


Left: original. Right: K -mode approx.

L17_F109985 $K = 8, R^2 = 0.695$ *sum: 56%; mixed: 26%; diff: 7%; parity: 5%; col: 4%*

$$\begin{aligned}
f(a, b) \approx & 0.064 + 0.116 \cos\left(\frac{2\pi(a+b)}{10} + 0.109\right) + 0.099 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.222\right) \\
& + 0.094 \cos\left(\frac{2\pi(-3a-3b)}{10} - 0.138\right) + 0.092 \cos\left(\frac{2\pi(4a+4b)}{10}\right) + 0.089 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.064 \cos\left(\frac{2\pi(5a+3b)}{10} + 2.504\right) + 0.057 \cos\left(\frac{2\pi(4a-4b)}{10} - 2.799\right) + 0.053 \cos\left(\frac{2\pi(3a+5b)}{10} - 3.082\right)
\end{aligned}$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	56%
mixed	—	26%
diff	$\cos(2\pi k(b-a)/N)$	7%
parity	$(-1)^{a+b}$	5%
col	$f(b)$	4%
row	$f(a)$	1%

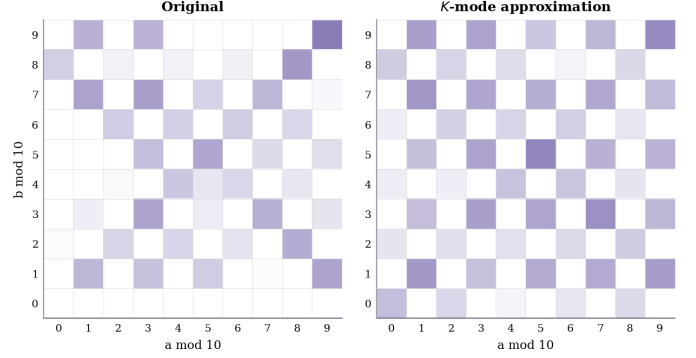


Left: original. Right: K -mode approx.

L16_F34883 $K = 8, R^2 = 0.169$ *parity: 40%; mixed: 24%; diff: 12%; sum: 8%; row: 4%; row_parity: 4%; col_parity: 4%; col: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.159 + 0.31 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.099 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) \\
& + 0.092 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.07 \cos\left(\frac{2\pi(a+b)}{10} + 0.363\right) + 0.065 \cos\left(\frac{2\pi(-a+b)}{10} + 0.402\right) \\
& + 0.063 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.332\right) + 0.06 \cos\left(\frac{2\pi(2a+3b)}{10} - 1.903\right) + 0.059 \cos\left(\frac{2\pi(2a-2b)}{10} + 0.131\right)
\end{aligned}$$

Dir	Structure	Frac
parity	$(-1)^{a+b}$	40%
mixed	—	24%
diff	$\cos(2\pi k(b-a)/N)$	12%
sum	$\cos(2\pi k(a+b)/N)$	8%
row	$f(a)$	4%
row _{parity}	—	4%
col _{parity}	—	4%
col	$f(b)$	3%

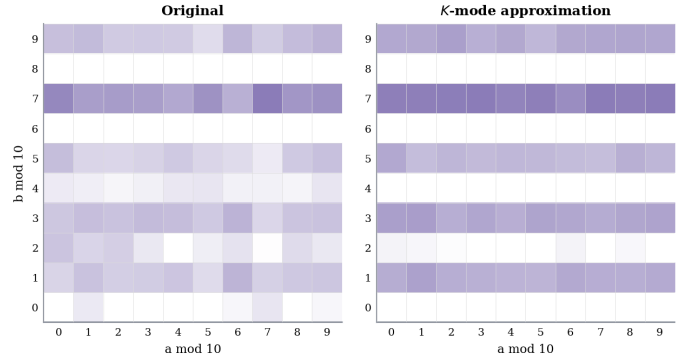


Left: original. Right: K -mode approx.

L00_F23372 $K = 8$, $R^2 = 0.266$ *col_parity: 66%; col: 23%; mixed: 7%*

$$\begin{aligned}
f(a, b) \approx & 0.284 + 0.411 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.124 \cos\left(\frac{2\pi(4b)}{10} + 1.491\right) \\
& + 0.098 \cos\left(\frac{2\pi(2b)}{10} - 2.848\right) + 0.069 \cos\left(\frac{2\pi(3b)}{10} - 0.447\right) + 0.033 \cos\left(\frac{2\pi(a)}{10} + 0.187\right) \\
& + 0.032 \cos\left(\frac{2\pi(4a+b)}{10} + 2.975\right) + 0.032 \cos\left(\frac{2\pi(2a+b)}{10} - \frac{2\pi}{3}\right) + 0.032 \cos\left(\frac{2\pi(5a+3b)}{10} + 2.621\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	66%
col	$f(b)$	23%
mixed	—	7%
row	$f(a)$	1%
diff	$\cos(2\pi k(b-a)/N)$	1%

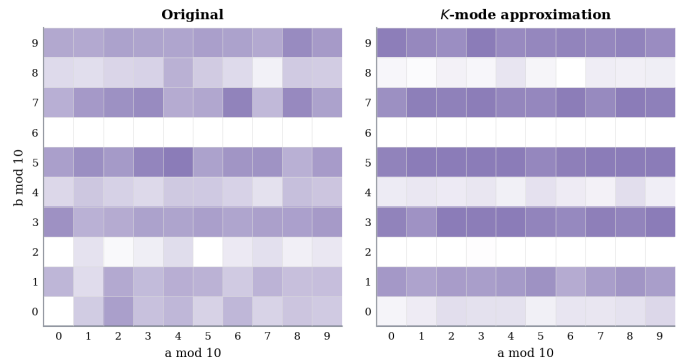


Left: original. Right: K -mode approx.

L00_F109877 $K = 8$, $R^2 = 0.256$ *col_parity: 64%; col: 22%; mixed: 8%; diff: 3%; row: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.473 + 0.444 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.131 \cos\left(\frac{2\pi(2b)}{10} + 1.337\right) \\
& + 0.11 \cos\left(\frac{2\pi(4b)}{10} + 0.336\right) + 0.067 \cos\left(\frac{2\pi(-3b)}{10} + 1.093\right) + 0.05 \cos\left(\frac{2\pi(4a-4b)}{10} + 2.739\right) \\
& + 0.036 \cos\left(\frac{2\pi(-3a-3b)}{10} - 2.019\right) + 0.035 \cos\left(\frac{2\pi(2a)}{10} + 2.17\right) + 0.03 \cos\left(\frac{2\pi(2a-4b)}{10} + 2.873\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	64%
col	$f(b)$	22%
mixed	—	8%
diff	$\cos(2\pi k(b-a)/N)$	3%
row	$f(a)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%

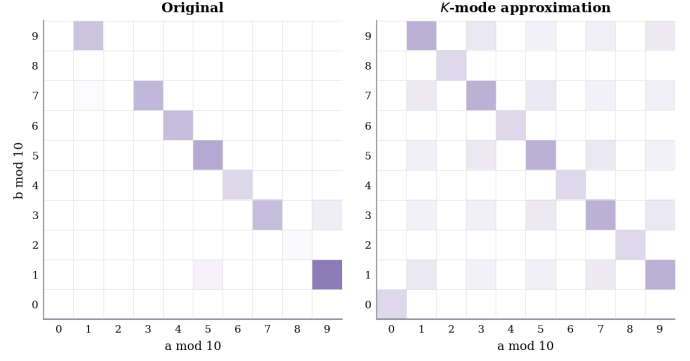


Left: original. Right: K -mode approx.

L15_F25468 $K = 8, R^2 = 0.563$ *sum: 52%; mixed: 28%; parity: 8%; row_parity: 3%; col_parity: 3%; diff: 2%; row: 2%;*

$$f(a, b) \approx 0.042 + 0.085 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.079 \cos\left(\frac{2\pi(4a+4b)}{10}\right) \\ + 0.079 \cos\left(\frac{2\pi(-a-b)}{10}\right) + 0.077 \cos\left(\frac{2\pi(-3a-3b)}{10} - 0.041\right) + 0.077 \cos\left(\frac{2\pi(2a+2b)}{10} - 0.041\right) \\ + 0.051 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.051 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) + 0.045 \cos\left(\frac{2\pi(a-4b)}{10} + \pi\right)$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	52%
mixed	—	28%
parity	$(-1)^{a+b}$	8%
row_parity	—	3%
col_parity	—	3%
diff	$\cos(2\pi k(b-a)/N)$	2%
row	$f(a)$	2%
col	$f(b)$	2%

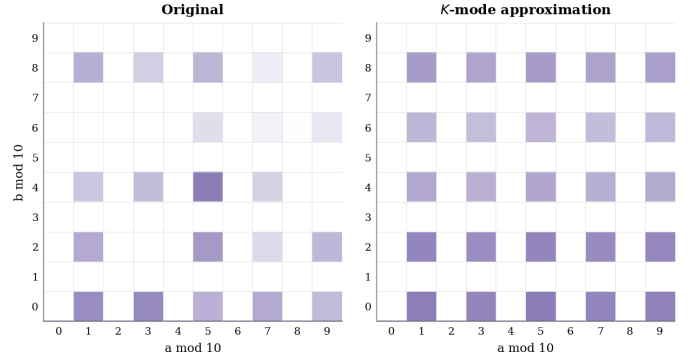


Left: original. Right: K -mode approx.

L21_F40303 $K = 8, R^2 = 0.127$ *mixed: 22%; col_parity: 21%; parity: 21%; row_parity: 21%; col: 8%; row: 4%; diff: 2%*

$$f(a, b) \approx 0.107 + 0.215 \cos\left(\frac{2\pi(5b)}{10}\right) + 0.214 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) \\ + 0.214 \cos\left(\frac{2\pi(5a+5b)}{10} - \pi\right) + 0.056 \cos\left(\frac{2\pi(5a+4b)}{10} - 2.722\right) + 0.056 \cos\left(\frac{2\pi(5a+b)}{10} + 2.722\right) \\ + 0.056 \cos\left(\frac{2\pi(b)}{10} - 0.418\right) + 0.056 \cos\left(\frac{2\pi(4b)}{10} + 0.418\right) + 0.039 \cos\left(\frac{2\pi(-3a)}{10} + 2.79\right)$$

Dir	Structure	Frac
mixed	—	22%
col_parity	—	21%
parity	$(-1)^{a+b}$	21%
row_parity	—	21%
col	$f(b)$	8%
row	$f(a)$	4%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%

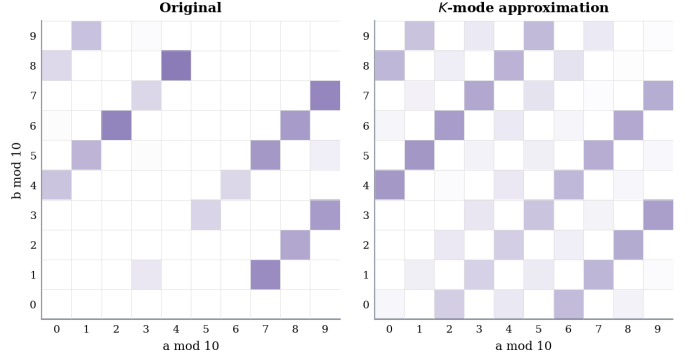


Left: original. Right: K -mode approx.

L13_F56616 $K = 8, R^2 = 0.543$ *diff: 47%; mixed: 26%; parity: 16%; row: 4%; col: 4%*

$$f(a, b) \approx 0.097 + 0.195 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.152 \cos\left(\frac{2\pi(2a-2b)}{10} - 1.767\right) \\ + 0.152 \cos\left(\frac{2\pi(3a-3b)}{10} + 1.767\right) + 0.069 \cos\left(\frac{2\pi(a-b)}{10} - 2.96\right) + 0.069 \cos\left(\frac{2\pi(4a-4b)}{10} + 2.96\right) \\ + 0.052 \cos\left(\frac{2\pi(2a-3b)}{10} + 1.387\right) + 0.052 \cos\left(\frac{2\pi(3a-2b)}{10} - 1.387\right) + 0.05 \cos\left(\frac{2\pi(b)}{10} + 2.753\right)$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	47%
mixed	—	26%
parity	$(-1)^{a+b}$	16%
row	$f(a)$	4%
col	$f(b)$	4%
sum	$\cos(2\pi k(a+b)/N)$	2%

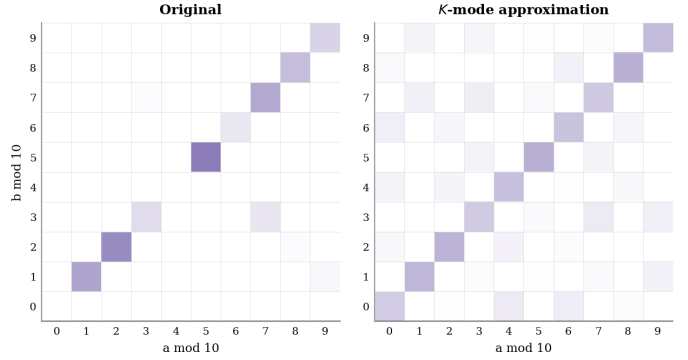


Left: original. Right: K -mode approx.

L18_F15817 $K = 8, R^2 = 0.600$ diff: 53%; mixed: 28%; parity: 8%; row: 3%; sum: 3%; col: 3%

$$\begin{aligned}
f(a, b) \approx & 0.048 + 0.096 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.091 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\
& + 0.091 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.087 \cos\left(\frac{2\pi(a-b)}{10}\right) + 0.087 \cos\left(\frac{2\pi(4a-4b)}{10}\right) \\
& + 0.038 \cos\left(\frac{2\pi(a-4b)}{10} + 3.05\right) + 0.038 \cos\left(\frac{2\pi(4a-b)}{10} - 3.05\right) + 0.034 \cos\left(\frac{2\pi(4a+3b)}{10} + 2.912\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	53%
mixed	—	28%
parity	$(-1)^{a+b}$	8%
row	$f(a)$	3%
sum	$\cos(2\pi k(a+b)/N)$	3%
col	$f(b)$	3%

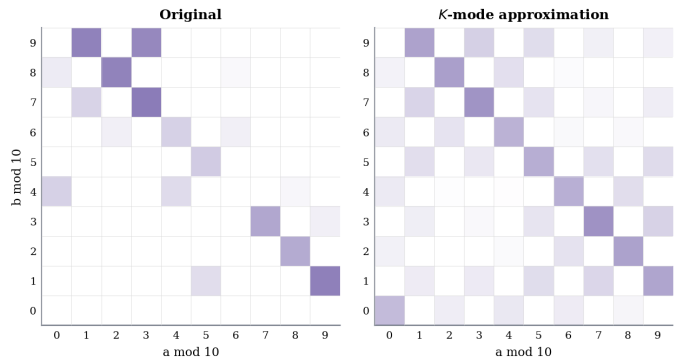


Left: original. Right: K -mode approx.

L15_F153150 $K = 8, R^2 = 0.477$ sum: 42%; mixed: 29%; parity: 14%; diff: 7%; col: 4%; row: 3%

$$\begin{aligned}
f(a, b) \approx & 0.086 + 0.173 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.121 \cos\left(\frac{2\pi(a+b)}{10} - 0.065\right) \\
& + 0.121 \cos\left(\frac{2\pi(4a+4b)}{10} + 0.065\right) + 0.088 \cos\left(\frac{2\pi(2a+2b)}{10}\right) + 0.088 \cos\left(\frac{2\pi(3a+3b)}{10}\right) \\
& + 0.057 \cos\left(\frac{2\pi(a-b)}{10} - 3.03\right) + 0.057 \cos\left(\frac{2\pi(4a-4b)}{10} + 3.03\right) + 0.051 \cos\left(\frac{2\pi(5b)}{10} + \pi\right)
\end{aligned}$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	42%
mixed	—	29%
parity	$(-1)^{a+b}$	14%
diff	$\cos(2\pi k(b-a)/N)$	7%
col	$f(b)$	4%
row	$f(a)$	3%
row _{parity}	—	1%
col _{parity}	—	1%

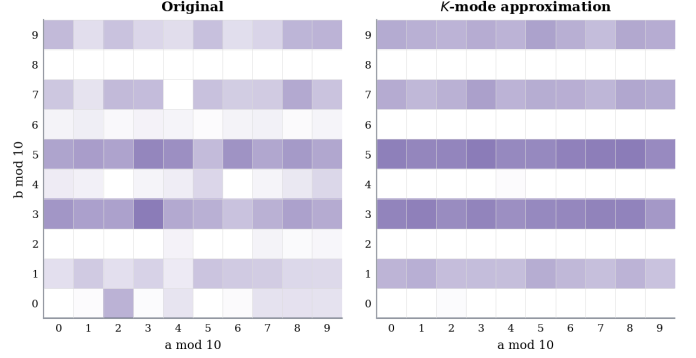


Left: original. Right: K -mode approx.

L00_F144139 $K = 8, R^2 = 0.251$ col_{parity}: 62%; col: 21%; mixed: 12%

$$\begin{aligned}
f(a, b) \approx & 0.294 + 0.429 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.116 \cos\left(\frac{2\pi(b)}{10} - 2.619\right) \\
& + 0.101 \cos\left(\frac{2\pi(4b)}{10} - 0.336\right) + 0.078 \cos\left(\frac{2\pi(2b)}{10} + \frac{\pi}{4}\right) + 0.048 \cos\left(\frac{2\pi(3b)}{10} + \frac{\pi}{3}\right) \\
& + 0.048 \cos\left(\frac{2\pi(4a+5b)}{10} + 2.166\right) + 0.043 \cos\left(\frac{2\pi(2a+4b)}{10} + \pi\right) + 0.037 \cos\left(\frac{2\pi(-3a-b)}{10} + 2.957\right)
\end{aligned}$$

Dir	Structure	Frac
col_{parity}	—	62%
col	$f(b)$	21%
mixed	—	12%
sum	$\cos(2\pi k(a+b)/N)$	2%
row	$f(a)$	2%

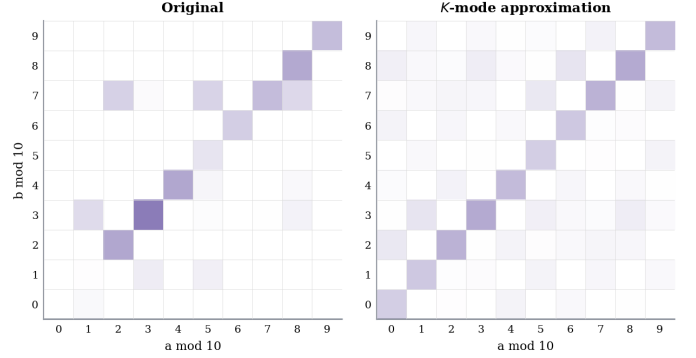


Left: original. Right: K -mode approx.

L11_F43710 $K = 8$, $R^2 = 0.625$ diff: 51%; mixed: 25%; parity: 7%; col: 6%; sum: 6%; row: 4%

$$\begin{aligned}
f(a, b) \approx & 0.064 + 0.096 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.12\right) + 0.094 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.092 \cos\left(\frac{2\pi(2a-2b)}{10}\right) + 0.091 \cos\left(\frac{2\pi(a-b)}{10}\right) + 0.071 \cos\left(\frac{2\pi(3a-3b)}{10} - 0.222\right) \\
& + 0.052 \cos\left(\frac{2\pi(2b)}{10} + 2.932\right) + 0.05 \cos\left(\frac{2\pi(4a-2b)}{10} + 2.449\right) + 0.041 \cos\left(\frac{2\pi(4a+4b)}{10} - 2.97\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	51%
mixed	—	25%
parity	$(-1)^{a+b}$	7%
col	$f(b)$	6%
sum	$\cos(2\pi k(a+b)/N)$	6%
row	$f(a)$	4%

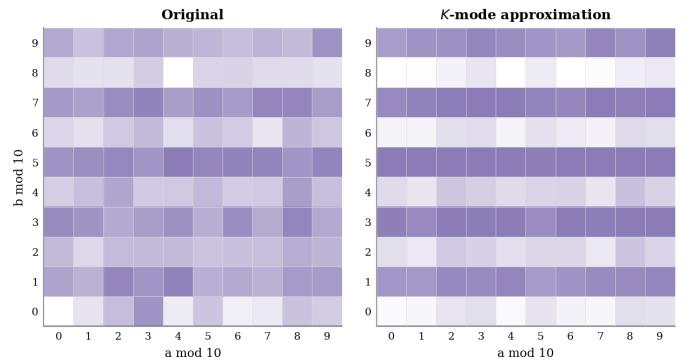


Left: original. Right: K -mode approx.

L00_F76299 $K = 8$, $R^2 = 0.231$ col_parity: 60%; col: 15%; mixed: 15%; row: 6%

$$\begin{aligned}
f(a, b) \approx & 0.574 + 0.375 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.11 \cos\left(\frac{2\pi(b)}{10} - 2.525\right) \\
& + 0.064 \cos\left(\frac{2\pi(4b)}{10} + \frac{\pi}{3}\right) + 0.063 \cos\left(\frac{2\pi(2a)}{10} + \frac{3\pi}{4}\right) + 0.044 \cos\left(\frac{2\pi(3a)}{10} + 2.223\right) \\
& + 0.043 \cos\left(\frac{2\pi(5a+b)}{10} - 2.015\right) + 0.041 \cos\left(\frac{2\pi(3a+5b)}{10} + 2.56\right) + 0.04 \cos\left(\frac{2\pi(4a+5b)}{10} - 0.844\right)
\end{aligned}$$

Dir	Structure	Frac
col_{parity}	—	60%
col	$f(b)$	15%
mixed	—	15%
row	$f(a)$	6%
sum	$\cos(2\pi k(a+b)/N)$	2%
diff	$\cos(2\pi k(b-a)/N)$	2%

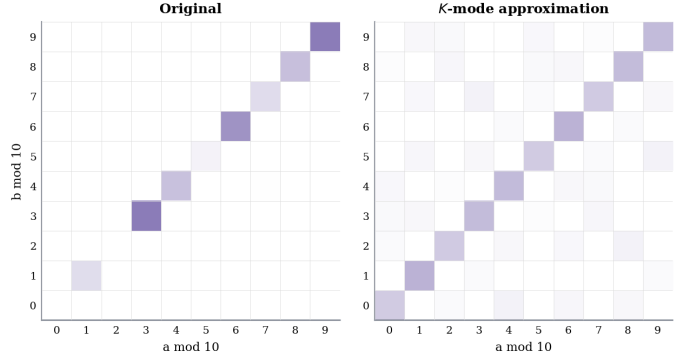


Left: original. Right: K -mode approx.

L18_F3053 $K = 8, R^2 = 0.558$ *diff: 50%; mixed: 29%; parity: 6%; sum: 6%; col: 4%; row: 4%*

$$f(a, b) \approx 0.044 + 0.088 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.088 \cos\left(\frac{2\pi(4a-4b)}{10}\right) \\ + 0.088 \cos\left(\frac{2\pi(a-b)}{10}\right) + 0.088 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.088 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\ + 0.034 \cos\left(\frac{2\pi(-a+5b)}{10} - 2.562\right) + 0.034 \cos\left(\frac{2\pi(-3a-b)}{10} + 2.567\right) + 0.034 \cos\left(\frac{2\pi(4a+2b)}{10} + 2.565\right)$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	50%
mixed	—	29%
parity	$(-1)^{a+b}$	6%
sum	$\cos(2\pi k(a+b)/N)$	6%
col	$f(b)$	4%
row	$f(a)$	4%

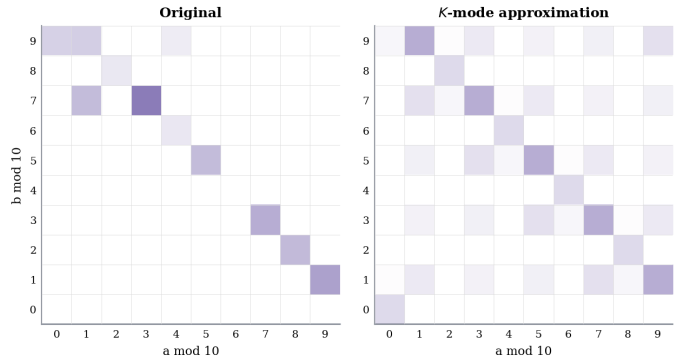


Left: original. Right: K -mode approx.

L16_F48874 $K = 8, R^2 = 0.520$ *sum: 46%; mixed: 34%; parity: 6%; col_parity: 4%; diff: 4%; col: 3%*

$$f(a, b) \approx 0.051 + 0.09 \cos\left(\frac{2\pi(a+b)}{10} + 0.123\right) + 0.082 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\ + 0.081 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.103\right) + 0.075 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.193\right) + 0.074 \cos\left(\frac{2\pi(3a+3b)}{10}\right) \\ + 0.067 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.056 \cos\left(\frac{2\pi(a-4b)}{10} - 2.942\right) + 0.047 \cos\left(\frac{2\pi(5a)}{10} + \pi\right)$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	46%
mixed	—	34%
parity	$(-1)^{a+b}$	6%
col_parity	—	4%
diff	$\cos(2\pi k(b-a)/N)$	4%
col	$f(b)$	3%
row_parity	—	2%
row	$f(a)$	1%

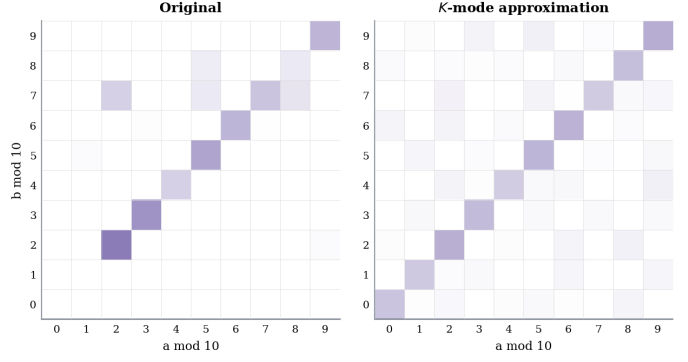


Left: original. Right: K -mode approx.

L06_F24782 $K = 8, R^2 = 0.626$ *diff: 55%; mixed: 28%; parity: 5%; row: 5%; col: 4%*

$$f(a, b) \approx 0.056 + 0.098 \cos\left(\frac{2\pi(4a-4b)}{10}\right) + 0.095 \cos\left(\frac{2\pi(2a-2b)}{10} - 0.051\right) \\ + 0.088 \cos\left(\frac{2\pi(a-b)}{10} + 0.052\right) + 0.085 \cos\left(\frac{2\pi(-3a+3b)}{10} + 0.089\right) + 0.081 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\ + 0.042 \cos\left(\frac{2\pi(3a+4b)}{10} - 1.998\right) + 0.04 \cos\left(\frac{2\pi(3a)}{10} + 2.253\right) + 0.037 \cos\left(\frac{2\pi(5a+2b)}{10} - 1.959\right)$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	55%
mixed	—	28%
parity	$(-1)^{a+b}$	5%
row	$f(a)$	5%
col	$f(b)$	4%
sum	$\cos(2\pi k(a+b)/N)$	1%

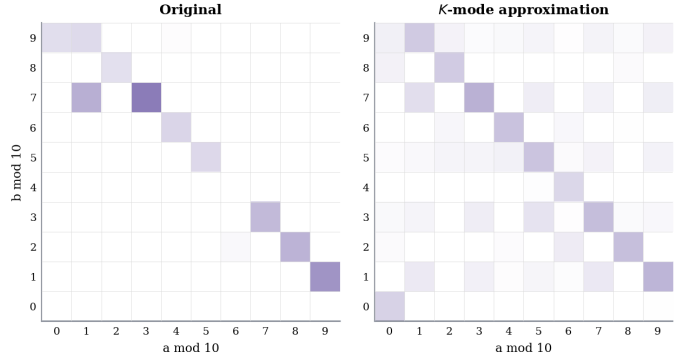


Left: original. Right: K -mode approx.

L17_F33509 $K = 8$, $R^2 = 0.535$ *sum: 45%; mixed: 32%; parity: 7%; diff: 6%; col: 5%; col_parity: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.05 + 0.09 \cos\left(\frac{2\pi(a+b)}{10} + 0.169\right) + 0.089 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.081 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.123\right) + 0.072 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.177\right) + 0.069 \cos\left(\frac{2\pi(3a+3b)}{10} - 0.041\right) \\
& + 0.053 \cos\left(\frac{2\pi(5a+3b)}{10} + 2.315\right) + 0.052 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.047 \cos\left(\frac{2\pi(4a+2b)}{10} + 1.966\right)
\end{aligned}$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	45%
mixed	—	32%
parity	$(-1)^{a+b}$	7%
diff	$\cos(2\pi k(b-a)/N)$	6%
col	$f(b)$	5%
col_parity	—	2%
row	$f(a)$	2%
row_parity	—	2%

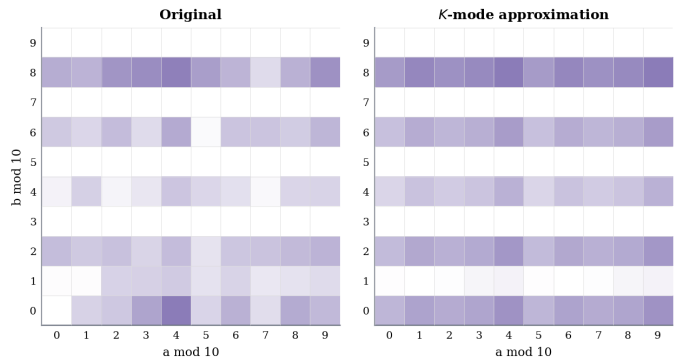


Left: original. Right: K -mode approx.

L00_F149167 $K = 8$, $R^2 = 0.270$ *col_parity: 56%; col: 20%; mixed: 13%; row: 6%; diff: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.248 + 0.402 \cos\left(\frac{2\pi(5b)}{10}\right) + 0.106 \cos\left(\frac{2\pi(-b)}{10} - 0.361\right) \\
& + 0.096 \cos\left(\frac{2\pi(4b)}{10} - 1.326\right) + 0.074 \cos\left(\frac{2\pi(3b)}{10} - 2.533\right) + 0.061 \cos\left(\frac{2\pi(2a)}{10} + 1.811\right) \\
& + 0.06 \cos\left(\frac{2\pi(4a)}{10} + 3.014\right) + 0.057 \cos\left(\frac{2\pi(2b)}{10} - 2.166\right) + 0.049 \cos\left(\frac{2\pi(4a+5b)}{10} + 3.098\right)
\end{aligned}$$

Dir	Structure	Frac
col_parity	—	56%
col	$f(b)$	20%
mixed	—	13%
row	$f(a)$	6%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%

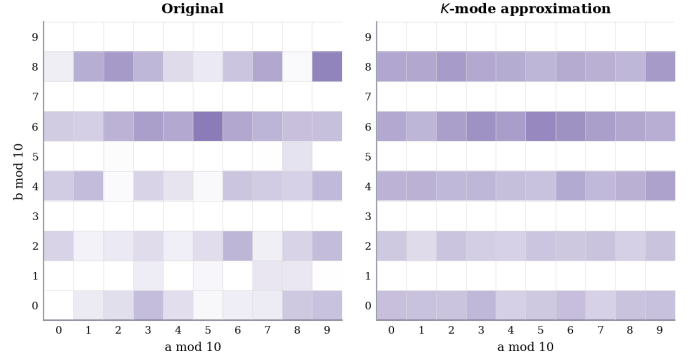


Left: original. Right: K -mode approx.

L00_F17257 $K = 8$, $R^2 = 0.206$ *col_parity: 53%; mixed: 23%; col: 14%; row: 3%; diff: 2%; sum: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.197 + 0.361 \cos\left(\frac{2\pi(5b)}{10}\right) + 0.094 \cos\left(\frac{2\pi(4b)}{10} - 2.26\right) \\
& + 0.077 \cos\left(\frac{2\pi(b)}{10} + 2.279\right) + 0.052 \cos\left(\frac{2\pi(4a+2b)}{10} - 1.516\right) + 0.049 \cos\left(\frac{2\pi(3a+5b)}{10} + 1.481\right) \\
& + 0.042 \cos\left(\frac{2\pi(a+3b)}{10} - 1.646\right) + 0.042 \cos\left(\frac{2\pi(a-4b)}{10} - 0.4\right) + 0.041 \cos\left(\frac{2\pi(a+2b)}{10} + 1.772\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	53%
mixed	—	23%
col	$f(b)$	14%
row	$f(a)$	3%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%

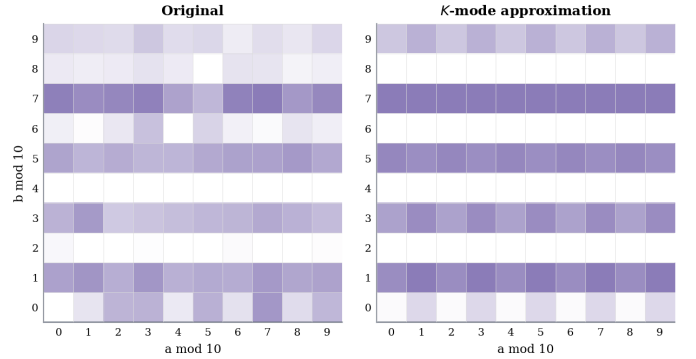


Left: original. Right: K -mode approx.

L00_F158018 $K = 8, R^2 = 0.317$ $\text{col}_{\text{parity}}$: 57%; col : 30%; mixed : 10%

$$\begin{aligned}
f(a, b) \approx & 0.375 + 0.465 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.143 \cos\left(\frac{2\pi(3b)}{10} - \frac{\pi}{4}\right) \\
& + 0.132 \cos\left(\frac{2\pi(2b)}{10} - 1.384\right) + 0.128 \cos\left(\frac{2\pi(4b)}{10} - 0.289\right) + 0.052 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) \\
& + 0.052 \cos\left(\frac{2\pi(-b)}{10} - 1.517\right) + 0.041 \cos\left(\frac{2\pi(5a-b)}{10} - 3.051\right) + 0.039 \cos\left(\frac{2\pi(5a-3b)}{10} + 2.897\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	57%
col	$f(b)$	30%
mixed	—	10%
row	$f(a)$	1%

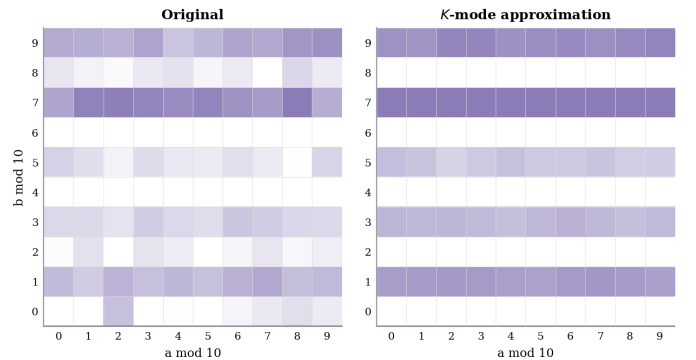


Left: original. Right: K -mode approx.

L00_F69087 $K = 8, R^2 = 0.357$ $\text{col}_{\text{parity}}$: 57%; col : 34%; mixed : 8%

$$\begin{aligned}
f(a, b) \approx & 0.293 + 0.439 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.17 \cos\left(\frac{2\pi(-b)}{10} - \frac{\pi}{4}\right) \\
& + 0.118 \cos\left(\frac{2\pi(4b)}{10} + 1.914\right) + 0.1 \cos\left(\frac{2\pi(2b)}{10} - 2.781\right) + 0.072 \cos\left(\frac{2\pi(-3b)}{10} + 0.644\right) \\
& + 0.032 \cos\left(\frac{2\pi(-2a-b)}{10} - \pi\right) + 0.032 \cos\left(\frac{2\pi(3a+3b)}{10} + 2.992\right) + 0.031 \cos\left(\frac{2\pi(3a-b)}{10} + \frac{2\pi}{3}\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	57%
col	$f(b)$	34%
mixed	—	8%

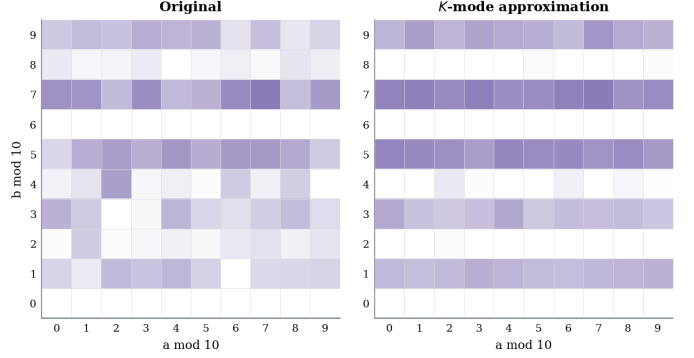


Left: original. Right: K -mode approx.

L00_F95984 $K = 8, R^2 = 0.269$ *col_parity: 53%; col: 22%; mixed: 20%; sum: 3%*

$$f(a, b) \approx 0.292 + 0.4 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.129 \cos\left(\frac{2\pi(4b)}{10} + 1.196\right) \\ + 0.103 \cos\left(\frac{2\pi(b)}{10} + 2.564\right) + 0.067 \cos\left(\frac{2\pi(3b)}{10} - 1.988\right) + 0.051 \cos\left(\frac{2\pi(5a+b)}{10} - \frac{3\pi}{4}\right) \\ + 0.049 \cos\left(\frac{2\pi(2a+4b)}{10} - 0.474\right) + 0.044 \cos\left(\frac{2\pi(3a+5b)}{10} + 2.416\right) + 0.043 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.194\right)$$

Dir	Structure	Frac
col_{parity}	—	53%
col	$f(b)$	22%
mixed	—	20%
sum	$\cos(2\pi k(a+b)/N)$	3%
diff	$\cos(2\pi k(b-a)/N)$	1%

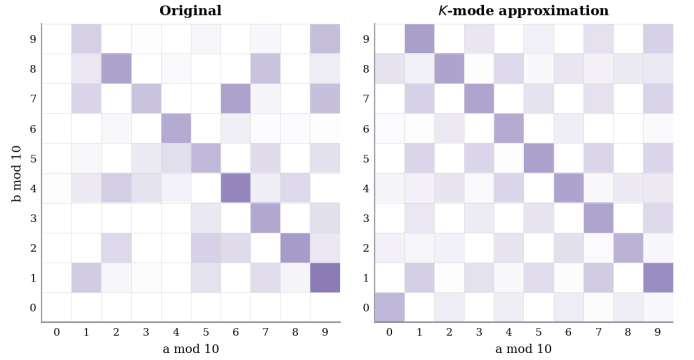


Left: original. Right: K -mode approx.

L16_F119550 $K = 8, R^2 = 0.459$ *sum: 38%; mixed: 26%; parity: 13%; row: 9%; col: 8%; diff: 4%*

$$f(a, b) \approx 0.141 + 0.163 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.118 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.115\right) \\ + 0.102 \cos\left(\frac{2\pi(a+b)}{10} + 0.15\right) + 0.088 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.093\right) + 0.074 \cos\left(\frac{2\pi(2a+2b)}{10}\right) \\ + 0.065 \cos\left(\frac{2\pi(4a)}{10} + 2.653\right) + 0.062 \cos\left(\frac{2\pi(3b)}{10} - 2.141\right) + 0.054 \cos\left(\frac{2\pi(5a)}{10} - \pi\right)$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	38%
mixed	—	26%
parity	$(-1)^{a+b}$	13%
row	$f(a)$	9%
col	$f(b)$	8%
diff	$\cos(2\pi k(b-a)/N)$	4%
row_{parity}	—	1%

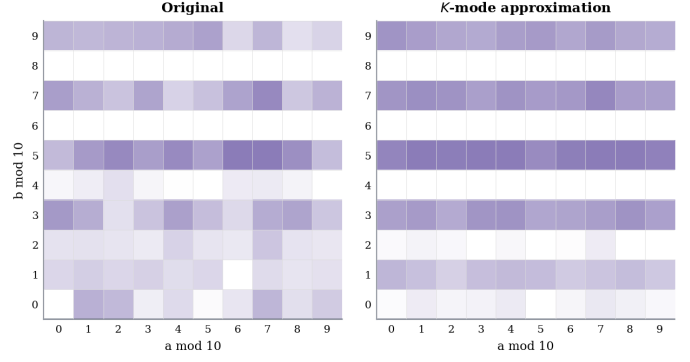


Left: original. Right: K -mode approx.

L00_F66722 $K = 8, R^2 = 0.309$ *col_parity: 52%; col: 27%; mixed: 13%; row: 3%; sum: 3%*

$$f(a, b) \approx 0.33 + 0.41 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.187 \cos\left(\frac{2\pi(4b)}{10} + 0.528\right) \\ + 0.058 \cos\left(\frac{2\pi(2b)}{10} + 0.531\right) + 0.051 \cos\left(\frac{2\pi(3a)}{10} - \frac{\pi}{3}\right) + 0.05 \cos\left(\frac{2\pi(-3b)}{10} - 1.489\right) \\ + 0.049 \cos\left(\frac{2\pi(2a-4b)}{10} - 3.003\right) + 0.047 \cos\left(\frac{2\pi(b)}{10} - 2.777\right) + 0.041 \cos\left(\frac{2\pi(4a+4b)}{10} + 2.963\right)$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	52%
col	$f(b)$	27%
mixed	—	13%
row	$f(a)$	3%
sum	$\cos(2\pi k(a+b)/N)$	3%
diff	$\cos(2\pi k(b-a)/N)$	1%

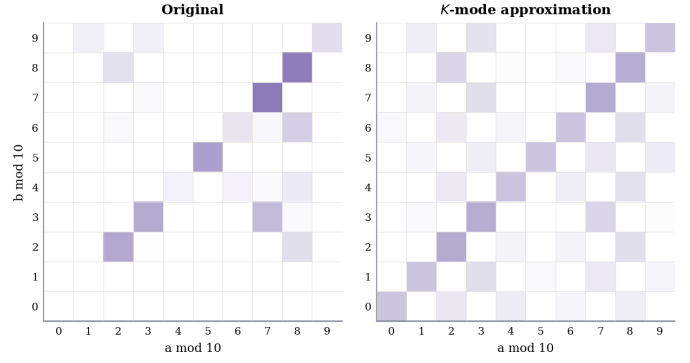


Left: original. Right: K -mode approx.

L15_F154966 $K = 8$, $R^2 = 0.524$ diff: 38%; mixed: 28%; parity: 11%; row: 10%; sum: 7%; col: 5%

$$\begin{aligned}
f(a, b) \approx & 0.066 + 0.127 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.091 \cos\left(\frac{2\pi(2a-2b)}{10} + 0.057\right) \\
& + 0.089 \cos\left(\frac{2\pi(3a-3b)}{10} - 0.078\right) + 0.075 \cos\left(\frac{2\pi(a-b)}{10} - 0.274\right) + 0.074 \cos\left(\frac{2\pi(4a-4b)}{10} + 0.237\right) \\
& + 0.064 \cos\left(\frac{2\pi(2a)}{10} - \pi\right) + 0.059 \cos\left(\frac{2\pi(3a+5b)}{10} - \pi\right) + 0.049 \cos\left(\frac{2\pi(3a+3b)}{10} - 0.156\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	38%
mixed	—	28%
parity	$(-1)^{a+b}$	11%
row	$f(a)$	10%
sum	$\cos(2\pi k(a+b)/N)$	7%
col	$f(b)$	5%

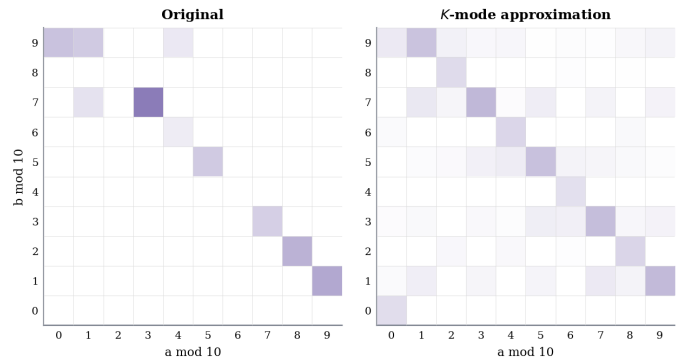


Left: original. Right: K -mode approx.

L18_F19549 $K = 8$, $R^2 = 0.519$ sum: 42%; mixed: 38%; diff: 5%; parity: 4%; col_parity: 4%; col: 4%; row: 2%

$$\begin{aligned}
f(a, b) \approx & 0.044 + 0.079 \cos\left(\frac{2\pi(a+b)}{10} + 0.082\right) + 0.069 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.197\right) \\
& + 0.068 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.122\right) + 0.066 \cos\left(\frac{2\pi(4a+4b)}{10}\right) + 0.063 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.059 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.05 \cos\left(\frac{2\pi(a-4b)}{10} - 3.013\right) + 0.044 \cos\left(\frac{2\pi(5a+3b)}{10} + 2.142\right)
\end{aligned}$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	42%
mixed	—	38%
diff	$\cos(2\pi k(b-a)/N)$	5%
parity	$(-1)^{a+b}$	4%
$\text{col}_{\text{parity}}$	—	4%
col	$f(b)$	4%
row	$f(a)$	2%
$\text{row}_{\text{parity}}$	—	1%

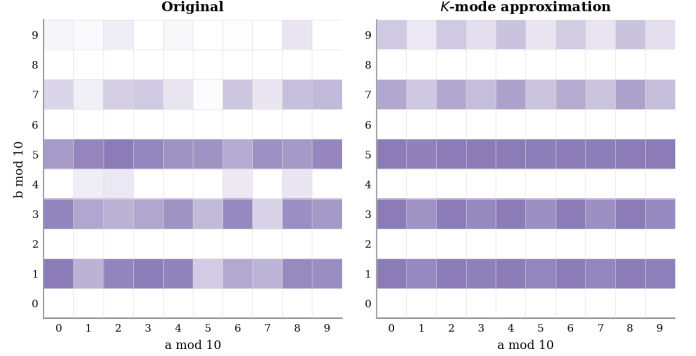


Left: original. Right: K -mode approx.

L00_F32316 $K = 8$, $R^2 = 0.374$ col_parity: 54%; col: 37%; mixed: 6%

$$\begin{aligned}
f(a, b) \approx & 0.275 + 0.524 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.2 \cos\left(\frac{2\pi(b)}{10} - \frac{2\pi}{3}\right) \\
& + 0.176 \cos\left(\frac{2\pi(4b)}{10} - \frac{\pi}{3}\right) + 0.115 \cos\left(\frac{2\pi(-3b)}{10} + 2.03\right) + 0.098 \cos\left(\frac{2\pi(2b)}{10} - 0.924\right) \\
& + 0.061 \cos\left(\frac{2\pi(5a)}{10}\right) + 0.046 \cos\left(\frac{2\pi(5a+5b)}{10} - \pi\right) + 0.04 \cos\left(\frac{2\pi(2a+5b)}{10} - 1.258\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	54%
col	$f(b)$	37%
mixed	—	6%
row	$f(a)$	1%

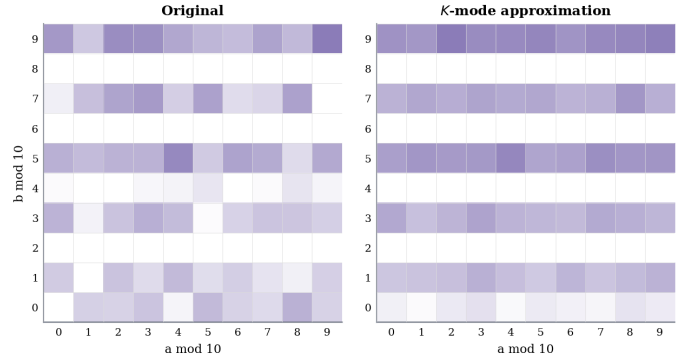


Left: original. Right: K -mode approx.

L00_F123242 $K = 8, R^2 = 0.278$ $\text{col}_{\text{parity}}$: 50%; col : 23%; mixed : 19%; diff : 3%; row : 3%

$$\begin{aligned}
f(a, b) \approx & 0.28 + 0.4 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.14 \cos\left(\frac{2\pi(2b)}{10} + 0.618\right) \\
& + 0.091 \cos\left(\frac{2\pi(4b)}{10} + \frac{\pi}{4}\right) + 0.067 \cos\left(\frac{2\pi(-b)}{10} - \frac{\pi}{4}\right) + 0.06 \cos\left(\frac{2\pi(3b)}{10} + 1.11\right) \\
& + 0.054 \cos\left(\frac{2\pi(3a-b)}{10} + 1.973\right) + 0.052 \cos\left(\frac{2\pi(4a-3b)}{10} - 0.659\right) + 0.048 \cos\left(\frac{2\pi(2a)}{10} + \frac{3\pi}{4}\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	50%
col	$f(b)$	23%
mixed	—	19%
diff	$\cos(2\pi k(b-a)/N)$	3%
row	$f(a)$	3%
sum	$\cos(2\pi k(a+b)/N)$	2%

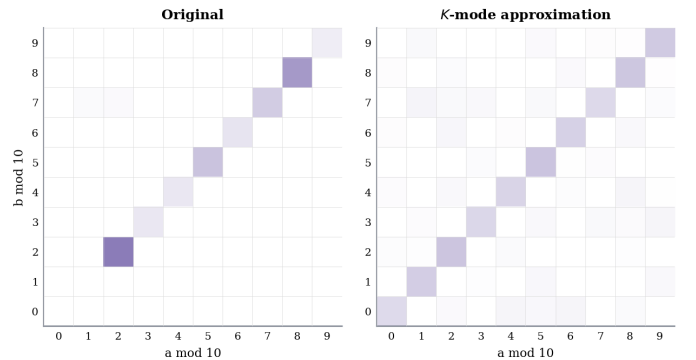


Left: original. Right: K -mode approx.

L11_F60700 $K = 8, R^2 = 0.504$ diff : 44%; mixed : 35%; parity : 6%; sum : 5%; row : 5%; col : 5%

$$\begin{aligned}
f(a, b) \approx & 0.034 + 0.067 \cos\left(\frac{2\pi(2a-2b)}{10}\right) + 0.066 \cos\left(\frac{2\pi(4a-4b)}{10}\right) \\
& + 0.066 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.065 \cos\left(\frac{2\pi(-3a+3b)}{10}\right) + 0.065 \cos\left(\frac{2\pi(a-b)}{10}\right) \\
& + 0.029 \cos\left(\frac{2\pi(-a-2b)}{10} - 3.013\right) + 0.028 \cos\left(\frac{2\pi(-a+4b)}{10} + 2.987\right) + 0.028 \cos\left(\frac{2\pi(3a)}{10} + 3.035\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	44%
mixed	—	35%
parity	$(-1)^{a+b}$	6%
sum	$\cos(2\pi k(a+b)/N)$	5%
row	$f(a)$	5%
col	$f(b)$	5%

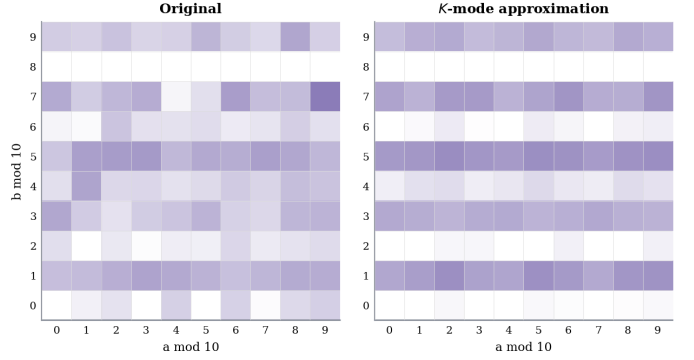


Left: original. Right: K -mode approx.

L00_F16289 $K = 8, R^2 = 0.255$ *col_parity: 50%; mixed: 21%; col: 19%; row: 4%; diff: 4%*

$$f(a, b) \approx 0.355 + 0.336 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.096 \cos\left(\frac{2\pi(b)}{10} - 2.546\right) \\ + 0.084 \cos\left(\frac{2\pi(2b)}{10} - 0.406\right) + 0.053 \cos\left(\frac{2\pi(3a-2b)}{10} - 2.92\right) + 0.052 \cos\left(\frac{2\pi(3b)}{10} - 1.351\right) \\ + 0.052 \cos\left(\frac{2\pi(4b)}{10} + \frac{2\pi}{3}\right) + 0.048 \cos\left(\frac{2\pi(3a)}{10} + \frac{3\pi}{4}\right) + 0.044 \cos\left(\frac{2\pi(-3a+4b)}{10} - 0.284\right)$$

Dir	Structure	Frac
col_parity	—	50%
mixed	—	21%
col	$f(b)$	19%
row	$f(a)$	4%
diff	$\cos(2\pi k(b-a)/N)$	4%

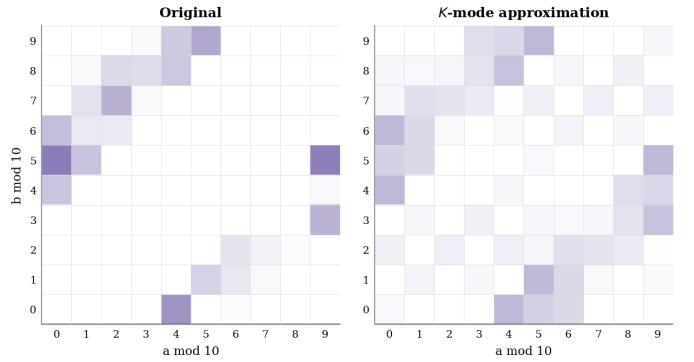


Left: original. Right: K -mode approx.

L13_F132660 $K = 8, R^2 = 0.575$ *diff: 42%; mixed: 30%; sum: 10%; row: 9%; col: 6%; parity: 3%*

$$f(a, b) \approx 0.092 + 0.16 \cos\left(\frac{2\pi(a-b)}{10} - 3.098\right) + 0.095 \cos\left(\frac{2\pi(2a-2b)}{10} + 0.108\right) \\ + 0.07 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.067 \cos\left(\frac{2\pi(2a)}{10} + 0.416\right) + 0.063 \cos\left(\frac{2\pi(-a-b)}{10} + 2.782\right) \\ + 0.056 \cos\left(\frac{2\pi(3a+5b)}{10} - 0.697\right) + 0.052 \cos\left(\frac{2\pi(4a+4b)}{10} + 2.526\right) + 0.049 \cos\left(\frac{2\pi(3a-b)}{10} - 2.684\right)$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	42%
mixed	—	30%
sum	$\cos(2\pi k(a+b)/N)$	10%
row	$f(a)$	9%
col	$f(b)$	6%
parity	$(-1)^{a+b}$	3%

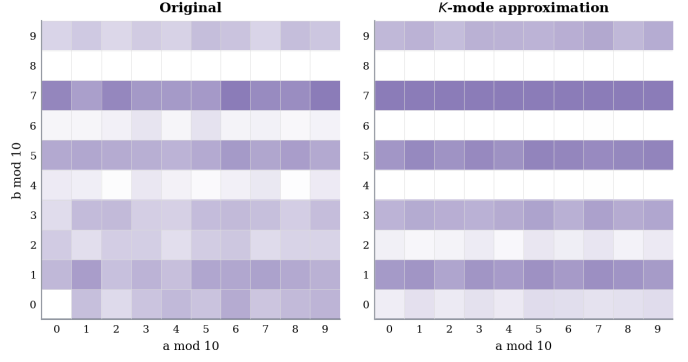


Left: original. Right: K -mode approx.

L00_F146561 $K = 8, R^2 = 0.383$ *col_parity: 53%; col: 37%; mixed: 6%*

$$f(a, b) \approx 0.397 + 0.395 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.171 \cos\left(\frac{2\pi(4b)}{10} + \frac{\pi}{4}\right) \\ + 0.134 \cos\left(\frac{2\pi(2b)}{10} - 1.346\right) + 0.079 \cos\left(\frac{2\pi(3b)}{10} - 0.673\right) + 0.033 \cos\left(\frac{2\pi(a)}{10} + \frac{2\pi}{3}\right) \\ + 0.028 \cos\left(\frac{2\pi(4a+2b)}{10} - 2.183\right) + 0.028 \cos\left(\frac{2\pi(3a+4b)}{10} + 2.555\right) + 0.027 \cos\left(\frac{2\pi(5a)}{10} + \pi\right)$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	53%
col	$f(b)$	37%
mixed	—	6%
row	$f(a)$	2%

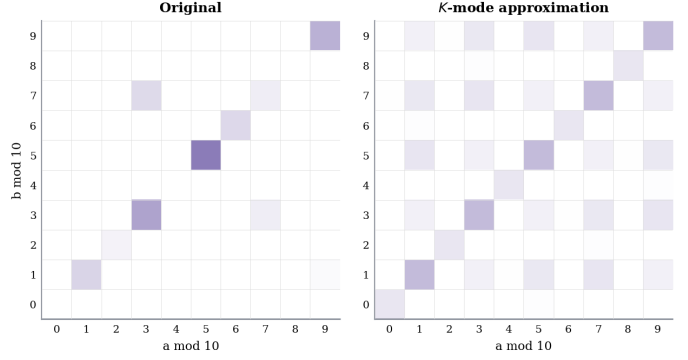


Left: original. Right: K -mode approx.

L23_F126973 $K = 8$, $R^2 = 0.425$ *mixed: 41%; diff: 36%; parity: 6%; sum: 4%; row_parity: 4%; col_parity: 4%; row: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.036 + 0.072 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.065 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\
& + 0.065 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.056 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.056 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) \\
& + 0.056 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.043\right) + 0.056 \cos\left(\frac{2\pi(a-b)}{10} + 0.043\right) + 0.049 \cos\left(\frac{2\pi(3a+2b)}{10} - 3.092\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	41%
diff	$\cos(2\pi k(b-a)/N)$	36%
parity	$(-1)^{a+b}$	6%
sum	$\cos(2\pi k(a+b)/N)$	4%
$\text{row}_{\text{parity}}$	—	4%
$\text{col}_{\text{parity}}$	—	4%
row	$f(a)$	3%
col	$f(b)$	2%

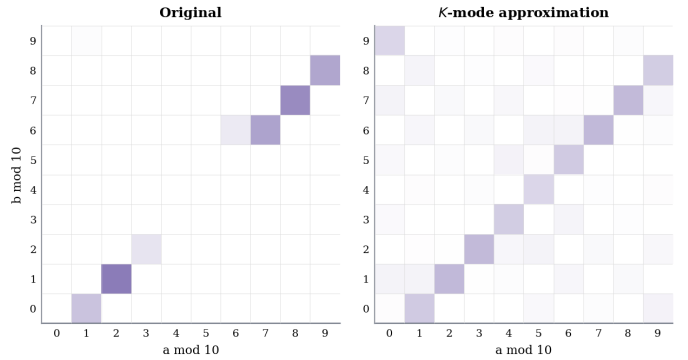


Left: original. Right: K -mode approx.

L13_F129441 $K = 8$, $R^2 = 0.521$ *diff: 43%; mixed: 36%; sum: 6%; col: 5%; parity: 5%; row: 5%*

$$\begin{aligned}
f(a, b) \approx & 0.041 + 0.081 \cos\left(\frac{2\pi(-a+b)}{10} + 0.609\right) + 0.079 \cos\left(\frac{2\pi(2a-2b)}{10} - 1.224\right) \\
& + 0.077 \cos\left(\frac{2\pi(3a-3b)}{10} - 1.852\right) + 0.075 \cos\left(\frac{2\pi(4a-4b)}{10} - 2.492\right) + 0.075 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) \\
& + 0.041 \cos\left(\frac{2\pi(a-3b)}{10} + 1.187\right) + 0.04 \cos\left(\frac{2\pi(2a-4b)}{10} + 0.608\right) + 0.04 \cos\left(\frac{2\pi(2b)}{10} - 1.762\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	43%
mixed	—	36%
sum	$\cos(2\pi k(a+b)/N)$	6%
col	$f(b)$	5%
parity	$(-1)^{a+b}$	5%
row	$f(a)$	5%

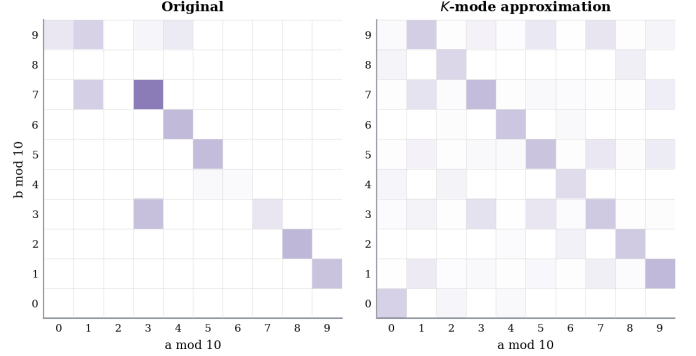


Left: original. Right: K -mode approx.

L17_F89282 $K = 8$, $R^2 = 0.471$ *sum: 39%; mixed: 36%; parity: 7%; row: 6%; col: 4%; diff: 3%; col_parity: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.049 + 0.083 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.071 \cos\left(\frac{2\pi(a+b)}{10}\right) + 0.154 \\
& + 0.07 \cos\left(\frac{2\pi(3a+3b)}{10}\right) + 0.17 + 0.067 \cos\left(\frac{2\pi(2a+2b)}{10}\right) + 0.065 \cos\left(\frac{2\pi(4a+4b)}{10}\right) - 0.177 \\
& + 0.052 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.046 \cos\left(\frac{2\pi(4a+2b)}{10} + 1.657\right) + 0.041 \cos\left(\frac{2\pi(-a-3b)}{10} + 1.611\right)
\end{aligned}$$

Dir	Structure	Frac
sum	$\cos(2\pi k(a+b)/N)$	39%
mixed	—	36%
parity	$(-1)^{a+b}$	7%
row	$f(a)$	6%
col	$f(b)$	4%
diff	$\cos(2\pi k(b-a)/N)$	3%
col _{parity}	—	3%
row _{parity}	—	2%

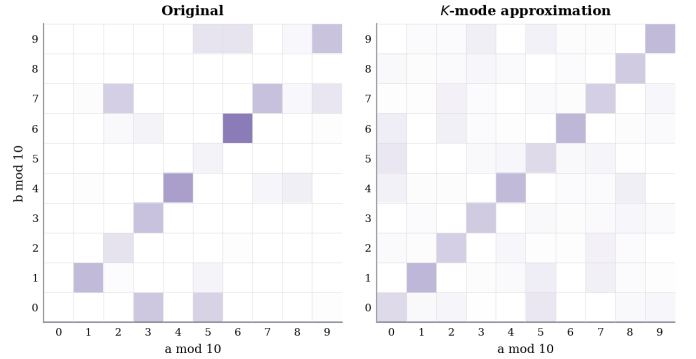


Left: original. Right: K -mode approx.

L18_F96615 $K = 8$, $R^2 = 0.562$ diff: 43%; mixed: 38%; sum: 6%; col: 6%; row: 4%; parity: 3%

$$\begin{aligned}
f(a, b) \approx & 0.063 + 0.089 \cos\left(\frac{2\pi(4a-4b)}{10}\right) + 0.08 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\
& + 0.077 \cos\left(\frac{2\pi(3a-3b)}{10} + 0.059\right) + 0.059 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.056 \cos\left(\frac{2\pi(-a+b)}{10} + 0.133\right) \\
& + 0.052 \cos\left(\frac{2\pi(5a-b)}{10} - 3.065\right) + 0.048 \cos\left(\frac{2\pi(3a+b)}{10} - \pi\right) + 0.045 \cos\left(\frac{2\pi(-3a-3b)}{10} - 2.832\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	43%
mixed	—	38%
sum	$\cos(2\pi k(a+b)/N)$	6%
col	$f(b)$	6%
row	$f(a)$	4%
parity	$(-1)^{a+b}$	3%

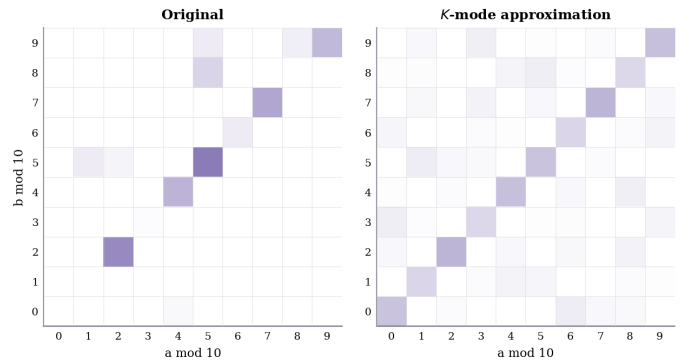


Left: original. Right: K -mode approx.

L07_F77494 $K = 8$, $R^2 = 0.528$ diff: 41%; mixed: 35%; row: 7%; sum: 7%; col: 6%; parity: 5%

$$\begin{aligned}
f(a, b) \approx & 0.047 + 0.085 \cos\left(\frac{2\pi(-3a+3b)}{10}\right) + 0.073 \cos\left(\frac{2\pi(2a-2b)}{10} - 0.094\right) \\
& + 0.073 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.072 \cos\left(\frac{2\pi(4a-4b)}{10} + 0.089\right) + 0.072 \cos\left(\frac{2\pi(a-b)}{10} + 0.168\right) \\
& + 0.051 \cos\left(\frac{2\pi(2a+4b)}{10} - 1.375\right) + 0.048 \cos\left(\frac{2\pi(5a+b)}{10} - 1.368\right) + 0.045 \cos\left(\frac{2\pi(a+3b)}{10} + 1.36\right)
\end{aligned}$$

Dir	Structure	Frac
diff	$\cos(2\pi k(b-a)/N)$	41%
mixed	—	35%
row	$f(a)$	7%
sum	$\cos(2\pi k(a+b)/N)$	7%
col	$f(b)$	6%
parity	$(-1)^{a+b}$	5%

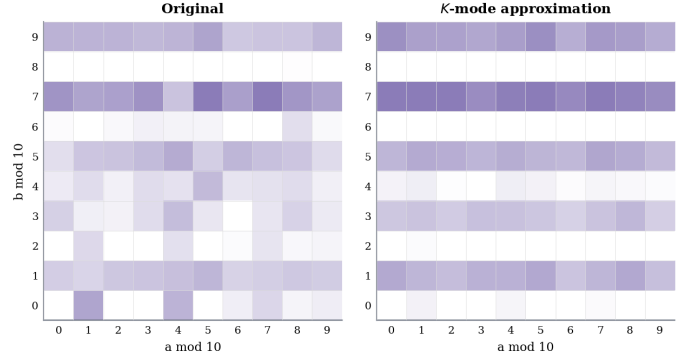


Left: original. Right: K -mode approx.

L00_F67861 $K = 8$, $R^2 = 0.346$ *col_parity: 48%; col: 29%; mixed: 15%; row: 3%; diff: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.297 + 0.361 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.157 \cos\left(\frac{2\pi(4b)}{10} + 1.631\right) \\
& + 0.084 \cos\left(\frac{2\pi(3b)}{10} - 0.84\right) + 0.083 \cos\left(\frac{2\pi(b)}{10} + \frac{\pi}{2}\right) + 0.055 \cos\left(\frac{2\pi(-3a)}{10} + 1.645\right) \\
& + 0.043 \cos\left(\frac{2\pi(4a+5b)}{10} - 3.028\right) + 0.041 \cos\left(\frac{2\pi(4a+4b)}{10} + 2.865\right) + 0.041 \cos\left(\frac{2\pi(2a-4b)}{10} - 2.646\right)
\end{aligned}$$

Dir	Structure	Frac
col_parity	—	48%
col	$f(b)$	29%
mixed	—	15%
row	$f(a)$	3%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%

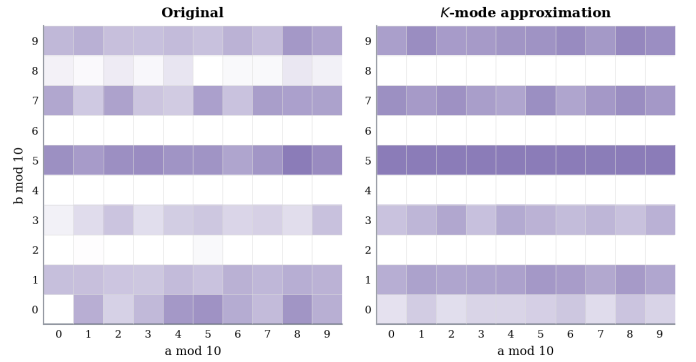


Left: original. Right: K -mode approx.

L00_F154973 $K = 8$, $R^2 = 0.417$ *col_parity: 49%; col: 39%; mixed: 7%; row: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.349 + 0.434 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.181 \cos\left(\frac{2\pi(2b)}{10} - 0.067\right) \\
& + 0.172 \cos\left(\frac{2\pi(4b)}{10} + 0.236\right) + 0.102 \cos\left(\frac{2\pi(-b)}{10} - \frac{\pi}{4}\right) + 0.042 \cos\left(\frac{2\pi(3a)}{10} - 3.02\right) \\
& + 0.04 \cos\left(\frac{2\pi(a+b)}{10} + 2.293\right) + 0.039 \cos\left(\frac{2\pi(4a+2b)}{10} - 1.918\right) + 0.039 \cos\left(\frac{2\pi(4a-b)}{10} - \frac{3\pi}{4}\right)
\end{aligned}$$

Dir	Structure	Frac
col_parity	—	49%
col	$f(b)$	39%
mixed	—	7%
row	$f(a)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%

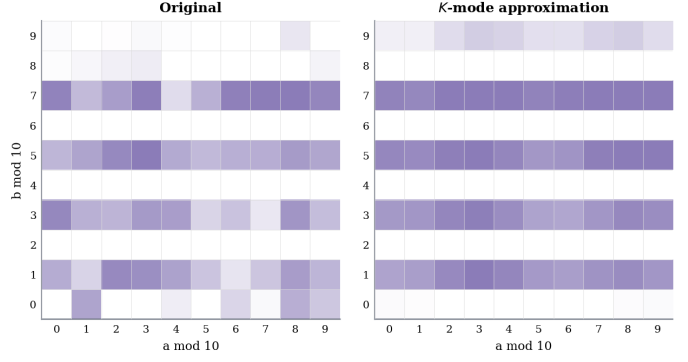


Left: original. Right: K -mode approx.

L00_F75117 $K = 8$, $R^2 = 0.364$ *col_parity: 48%; col: 32%; mixed: 13%; row: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.295 + 0.483 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.177 \cos\left(\frac{2\pi(4b)}{10} - 0.194\right) \\
& + 0.159 \cos\left(\frac{2\pi(-3b)}{10} + \frac{\pi}{3}\right) + 0.125 \cos\left(\frac{2\pi(2b)}{10} - 1.637\right) + 0.08 \cos\left(\frac{2\pi(b)}{10} - 2.926\right) \\
& + 0.063 \cos\left(\frac{2\pi(a-4b)}{10} + 0.316\right) + 0.058 \cos\left(\frac{2\pi(2a)}{10} + 2.545\right) + 0.053 \cos\left(\frac{2\pi(2a+5b)}{10} - 0.708\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	48%
col	$f(b)$	32%
mixed	—	13%
row	$f(a)$	3%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%

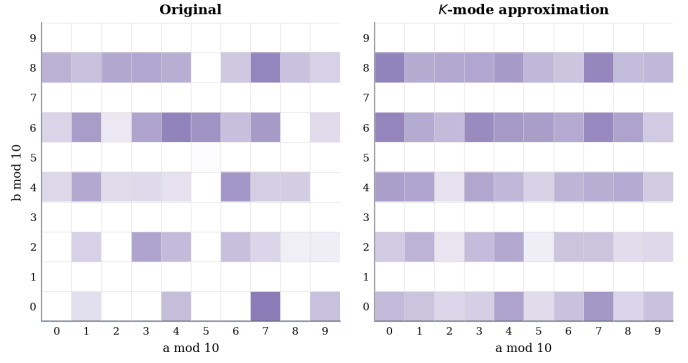


Left: original. Right: K -mode approx.

L09_F107059 $K = 8, R^2 = 0.265$ $\text{col}_{\text{parity}}$: 44%; mixed : 30%; col : 10%; row : 9%; diff : 3%; sum : 3%

$$\begin{aligned}
f(a, b) \approx & 0.185 + 0.368 \cos\left(\frac{2\pi(5b)}{10}\right) + 0.088 \cos\left(\frac{2\pi(3a+5b)}{10} - 0.556\right) \\
& + 0.087 \cos\left(\frac{2\pi(3a)}{10} - 0.563\right) + 0.085 \cos\left(\frac{2\pi(-b)}{10} - 2.139\right) + 0.084 \cos\left(\frac{2\pi(4b)}{10} - \frac{2\pi}{3}\right) \\
& + 0.064 \cos\left(\frac{2\pi(4a+b)}{10} + 2.149\right) + 0.063 \cos\left(\frac{2\pi(4a-4b)}{10} + 2.135\right) + 0.059 \cos\left(\frac{2\pi(a+2b)}{10} + 1.941\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	44%
mixed	—	30%
col	$f(b)$	10%
row	$f(a)$	9%
diff	$\cos(2\pi k(b-a)/N)$	3%
sum	$\cos(2\pi k(a+b)/N)$	3%

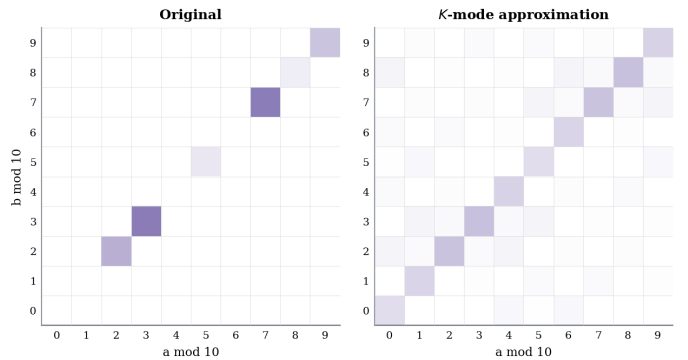


Left: original. Right: K -mode approx.

L14_F161442 $K = 8, R^2 = 0.450$ mixed : 40%; diff : 36%; sum : 7%; parity : 5%; row : 4%; col : 4%

$$\begin{aligned}
f(a, b) \approx & 0.034 + 0.067 \cos\left(\frac{2\pi(a-b)}{10}\right) + 0.067 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\
& + 0.067 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.067 \cos\left(\frac{2\pi(4a-4b)}{10}\right) + 0.067 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.038 \cos\left(\frac{2\pi(2b)}{10} + 3.061\right) + 0.038 \cos\left(\frac{2\pi(a+b)}{10} + 3.061\right) + 0.038 \cos\left(\frac{2\pi(a-3b)}{10} - 3.061\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	40%
diff	$\cos(2\pi k(b-a)/N)$	36%
sum	$\cos(2\pi k(a+b)/N)$	7%
parity	$(-1)^{a+b}$	5%
row	$f(a)$	4%
col	$f(b)$	4%
$\text{row}_{\text{parity}}$	—	1%
$\text{col}_{\text{parity}}$	—	1%

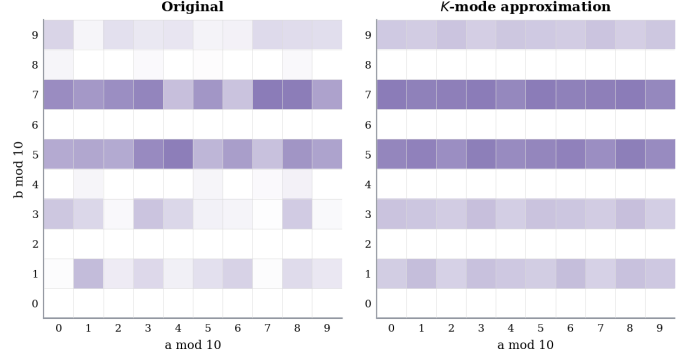


Left: original. Right: K -mode approx.

L00_F116333 $K = 8, R^2 = 0.419$ $\text{col}_{\text{parity}}$: 48%; col : 39%; mixed : 8%

$$\begin{aligned}
f(a, b) \approx & 0.217 + 0.415 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.177 \cos\left(\frac{2\pi(b)}{10} + 2.494\right) \\
& + 0.171 \cos\left(\frac{2\pi(4b)}{10} + 0.673\right) + 0.078 \cos\left(\frac{2\pi(3b)}{10} - 1.654\right) + 0.062 \cos\left(\frac{2\pi(2b)}{10} - 1.505\right) \\
& + 0.04 \cos\left(\frac{2\pi(4a)}{10} - 1.264\right) + 0.039 \cos\left(\frac{2\pi(4a-3b)}{10} - 0.106\right) + 0.038 \cos\left(\frac{2\pi(4a+b)}{10} + 2.309\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	48%
col	$f(b)$	39%
mixed	—	8%
row	$f(a)$	2%
diff	$\cos(2\pi k(b-a)/N)$	1%

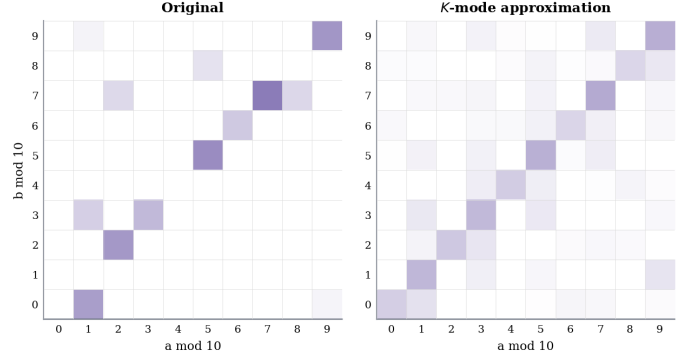


Left: original. Right: K -mode approx.

L07_F78372 $K = 8$, $R^2 = 0.472$ mixed: 42%; diff: 39%; sum: 6%; col: 4%; parity: 3%; row: 3%; row_parity: 2%

$$\begin{aligned}
f(a, b) \approx & 0.065 + 0.102 \cos\left(\frac{2\pi(a-b)}{10}\right) + 0.091 \cos\left(\frac{2\pi(2a-2b)}{10} - 0.192\right) \\
& + 0.081 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.152\right) + 0.075 \cos\left(\frac{2\pi(3a-3b)}{10} - 0.323\right) + 0.064 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.058 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) + 0.057 \cos\left(\frac{2\pi(2a+4b)}{10} - 1.393\right) + 0.056 \cos\left(\frac{2\pi(-4a-b)}{10} - 3.05\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	42%
diff	$\cos(2\pi k(b-a)/N)$	39%
sum	$\cos(2\pi k(a+b)/N)$	6%
col	$f(b)$	4%
parity	$(-1)^{a+b}$	3%
row	$f(a)$	3%
$\text{row}_{\text{parity}}$	—	2%
$\text{col}_{\text{parity}}$	—	1%

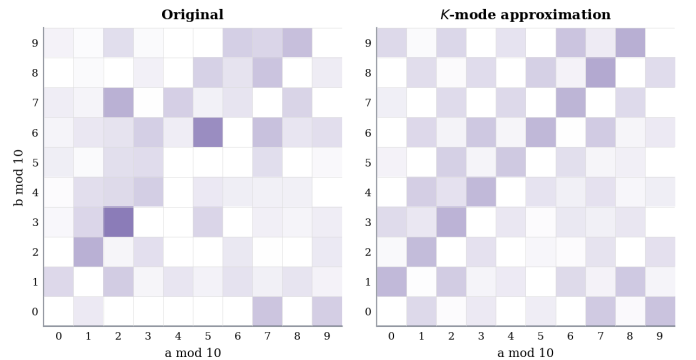


Left: original. Right: K -mode approx.

L14_F46580 $K = 8$, $R^2 = 0.415$ mixed: 30%; diff: 29%; parity: 17%; row: 12%; col: 7%; sum: 4%

$$\begin{aligned}
f(a, b) \approx & 0.158 + 0.152 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.094 \cos\left(\frac{2\pi(a-b)}{10} + 0.951\right) \\
& + 0.081 \cos\left(\frac{2\pi(4a-4b)}{10} + 2.407\right) + 0.069 \cos\left(\frac{2\pi(2a)}{10} - \frac{3\pi}{4}\right) + 0.051 \cos\left(\frac{2\pi(2a-2b)}{10} + 2.169\right) \\
& + 0.047 \cos\left(\frac{2\pi(4b)}{10} - 2.674\right) + 0.044 \cos\left(\frac{2\pi(3a-3b)}{10} + 2.769\right) + 0.043 \cos\left(\frac{2\pi(5a+4b)}{10} + \frac{2\pi}{3}\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	30%
diff	$\cos(2\pi k(b-a)/N)$	29%
parity	$(-1)^{a+b}$	17%
row	$f(a)$	12%
col	$f(b)$	7%
sum	$\cos(2\pi k(a+b)/N)$	4%

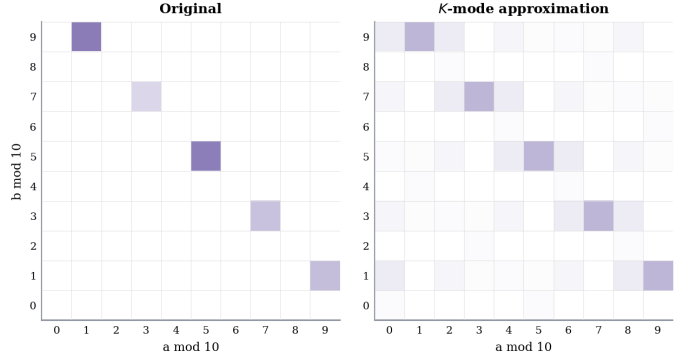


Left: original. Right: K -mode approx.

L20_F120007 $K = 8, R^2 = 0.611$ *mixed: 47%; sum: 35%; parity: 4%; row_parity: 4%; col_parity: 4%*

$$f(a, b) \approx 0.032 + 0.065 \cos\left(\frac{2\pi(2a+2b)}{10}\right) + 0.065 \cos\left(\frac{2\pi(3a-2b)}{10} - \pi\right) \\ + 0.065 \cos\left(\frac{2\pi(-a+4b)}{10} + \pi\right) + 0.065 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.065 \cos\left(\frac{2\pi(a+b)}{10}\right) \\ + 0.065 \cos\left(\frac{2\pi(2a-3b)}{10} - \pi\right) + 0.065 \cos\left(\frac{2\pi(3a+3b)}{10}\right) + 0.065 \cos\left(\frac{2\pi(4a+4b)}{10}\right)$$

Dir	Structure	Frac
mixed	—	47%
sum	$\cos(2\pi k(a+b)/N)$	35%
parity	$(-1)^{a+b}$	4%
row_parity	—	4%
col_parity	—	4%
row	$f(a)$	2%
diff	$\cos(2\pi k(b-a)/N)$	2%
col	$f(b)$	2%

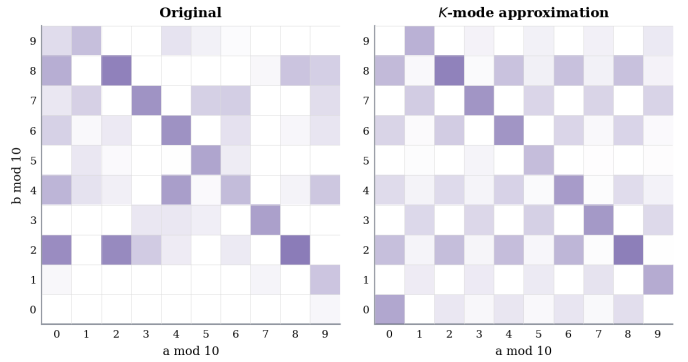


Left: original. Right: K -mode approx.

L17_F99204 $K = 8, R^2 = 0.377$ *mixed: 36%; sum: 27%; parity: 16%; col: 12%; col_parity: 3%; row: 3%*

$$f(a, b) \approx 0.168 + 0.204 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.12 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.128\right) \\ + 0.096 \cos\left(\frac{2\pi(3a+3b)}{10}\right) + 0.092 \cos\left(\frac{2\pi(2b)}{10} + \pi\right) + 0.085 \cos\left(\frac{2\pi(5b)}{10}\right) \\ + 0.081 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.158\right) + 0.08 \cos\left(\frac{2\pi(a+b)}{10} + 0.153\right) + 0.074 \cos\left(\frac{2\pi(5a+3b)}{10} + 2.795\right)$$

Dir	Structure	Frac
mixed	—	36%
sum	$\cos(2\pi k(a+b)/N)$	27%
parity	$(-1)^{a+b}$	16%
col	$f(b)$	12%
col_parity	—	3%
row	$f(a)$	3%
row_parity	—	2%
diff	$\cos(2\pi k(b-a)/N)$	2%

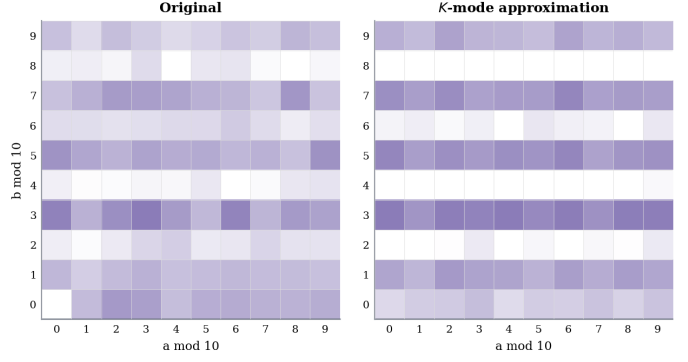


Left: original. Right: K -mode approx.

L00_F130657 $K = 8, R^2 = 0.369$ *col_parity: 46%; col: 34%; mixed: 14%; row: 3%*

$$f(a, b) \approx 0.418 + 0.342 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.146 \cos\left(\frac{2\pi(4b)}{10} - 0.29\right) \\ + 0.126 \cos\left(\frac{2\pi(3b)}{10} + 0.379\right) + 0.063 \cos\left(\frac{2\pi(2b)}{10} - \frac{\pi}{4}\right) + 0.049 \cos\left(\frac{2\pi(5a+5b)}{10} - \pi\right) \\ + 0.049 \cos\left(\frac{2\pi(2a-b)}{10} - 2.971\right) + 0.042 \cos\left(\frac{2\pi(-3a)}{10} - 1.51\right) + 0.041 \cos\left(\frac{2\pi(-b)}{10} + 1.399\right)$$

Dir	Structure	Frac
col_{parity}	—	46%
col	$f(b)$	34%
mixed	—	14%
row	$f(a)$	3%
sum	$\cos(2\pi k(a+b)/N)$	2%

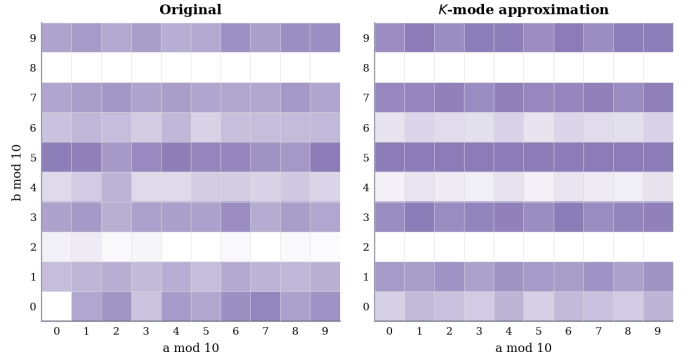


Left: original. Right: K -mode approx.

L00_F154272 $K = 8, R^2 = 0.441$ col_{parity} : 47%; col : 42%; $mixed$: 7%

$$\begin{aligned}
f(a, b) \approx & 0.519 + 0.416 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.207 \cos\left(\frac{2\pi(2b)}{10}\right) \\
& + 0.16 \cos\left(\frac{2\pi(3b)}{10} + 0.396\right) + 0.063 \cos\left(\frac{2\pi(b)}{10} + 2.732\right) + 0.059 \cos\left(\frac{2\pi(4b)}{10} + 0.274\right) \\
& + 0.044 \cos\left(\frac{2\pi(4a)}{10} + 3.101\right) + 0.036 \cos\left(\frac{2\pi(4a-2b)}{10} + 2.732\right) + 0.036 \cos\left(\frac{2\pi(4a-3b)}{10} + 2.56\right)
\end{aligned}$$

Dir	Structure	Frac
col_{parity}	—	47%
col	$f(b)$	42%
mixed	—	7%
row	$f(a)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%

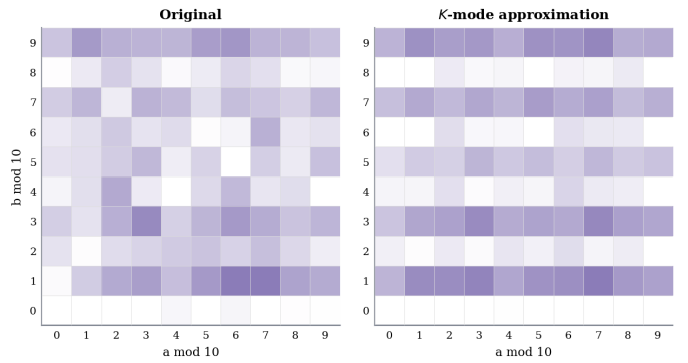


Left: original. Right: K -mode approx.

L23_F93464 $K = 8, R^2 = 0.314$ col_{parity} : 38%; $mixed$: 21%; col : 21%; row : 9%; sum : 5%; $diff$: 3%

$$\begin{aligned}
f(a, b) \approx & 0.346 + 0.311 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.131 \cos\left(\frac{2\pi(4b)}{10} - \pi\right) \\
& + 0.075 \cos\left(\frac{2\pi(2a)}{10} - 2.591\right) + 0.07 \cos\left(\frac{2\pi(2a+4b)}{10} + 1.403\right) + 0.067 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.064 \cos\left(\frac{2\pi(a)}{10} + 3.028\right) + 0.063 \cos\left(\frac{2\pi(2b)}{10} - 3.017\right) + 0.063 \cos\left(\frac{2\pi(b)}{10} - 0.922\right)
\end{aligned}$$

Dir	Structure	Frac
col_{parity}	—	38%
mixed	—	21%
col	$f(b)$	21%
row	$f(a)$	9%
sum	$\cos(2\pi k(a+b)/N)$	5%
diff	$\cos(2\pi k(b-a)/N)$	3%
parity	$(-1)^{a+b}$	2%

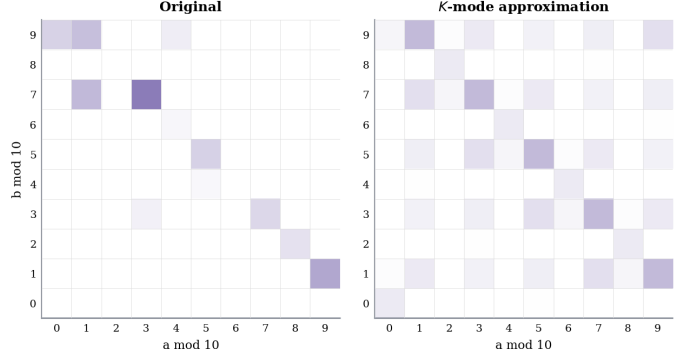


Left: original. Right: K -mode approx.

L16_F113794 $K = 8, R^2 = 0.398$ $mixed$: 44%; sum : 33%; col_{parity} : 6%; $parity$: 5%; col : 4%; row_{parity} : 3%; row : 3%; $diff$: 2%

$$\begin{aligned}
f(a, b) \approx & 0.043 + 0.071 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.07 \cos\left(\frac{2\pi(a+b)}{10} + 0.194\right) \\
& + 0.064 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.058 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.162\right) + 0.057 \cos\left(\frac{2\pi(a-4b)}{10} - 2.925\right) \\
& + 0.056 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.243\right) + 0.054 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) + 0.054 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.096\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	44%
sum	$\cos(2\pi k(a+b)/N)$	33%
col _{parity}	—	6%
parity	$(-1)^{a+b}$	5%
col	$f(b)$	4%
row _{parity}	—	3%
row	$f(a)$	3%
diff	$\cos(2\pi k(b-a)/N)$	2%

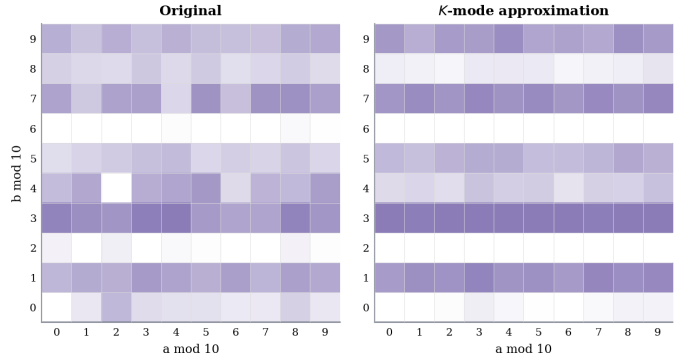


Left: original. Right: K -mode approx.

L00_F85933 $K = 8$, $R^2 = 0.419$ *col_parity: 46%; col: 38%; mixed: 11%; row: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.422 + 0.386 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.174 \cos\left(\frac{2\pi(-3b)}{10} + 0.856\right) \\
& + 0.156 \cos\left(\frac{2\pi(2b)}{10} + \frac{2\pi}{3}\right) + 0.062 \cos\left(\frac{2\pi(4b)}{10} - 2.406\right) + 0.057 \cos\left(\frac{2\pi(b)}{10} - 1.447\right) \\
& + 0.051 \cos\left(\frac{2\pi(2a)}{10} + 1.953\right) + 0.05 \cos\left(\frac{2\pi(5a-3b)}{10} - 1.888\right) + 0.035 \cos\left(\frac{2\pi(2a+2b)}{10} + 2.574\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	46%
col	$f(b)$	38%
mixed	—	11%
row	$f(a)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%
diff	$\cos(2\pi k(b-a)/N)$	1%

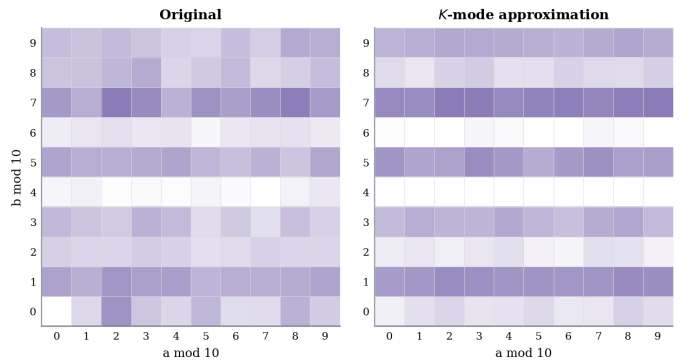


Left: original. Right: K -mode approx.

L00_F45180 $K = 8$, $R^2 = 0.399$ *col_parity: 44%; col: 35%; mixed: 13%; row: 4%; diff: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.436 + 0.311 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.114 \cos\left(\frac{2\pi(b)}{10} + 0.856\right) \\
& + 0.098 \cos\left(\frac{2\pi(4b)}{10} + 0.113\right) + 0.096 \cos\left(\frac{2\pi(2b)}{10} - 1.982\right) + 0.079 \cos\left(\frac{2\pi(3b)}{10} - 1.868\right) \\
& + 0.045 \cos\left(\frac{2\pi(3a+b)}{10} - \pi\right) + 0.043 \cos\left(\frac{2\pi(3a-3b)}{10} - 2.974\right) + 0.041 \cos\left(\frac{2\pi(2a)}{10} + 2.663\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	44%
col	$f(b)$	35%
mixed	—	13%
row	$f(a)$	4%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%

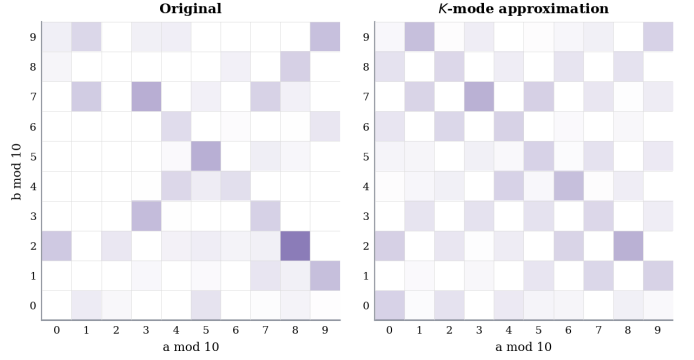


Left: original. Right: K -mode approx.

L17_F134881 $K = 8, R^2 = 0.389$ *mixed: 40%; sum: 24%; parity: 14%; diff: 8%; col: 7%; row: 5%*

$$\begin{aligned}
f(a, b) \approx & 0.096 + 0.129 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.068 \cos\left(\frac{2\pi(-a-b)}{10} - 0.335\right) \\
& + 0.067 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.206\right) + 0.058 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.189\right) + 0.055 \cos\left(\frac{2\pi(a-3b)}{10} - 1.834\right) \\
& + 0.052 \cos\left(\frac{2\pi(5a+3b)}{10} + 2.897\right) + 0.048 \cos\left(\frac{2\pi(4b)}{10} + 1.431\right) + 0.047 \cos\left(\frac{2\pi(4a-2b)}{10} + 1.889\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	40%
sum	$\cos(2\pi k(a+b)/N)$	24%
parity	$(-1)^{a+b}$	14%
diff	$\cos(2\pi k(b-a)/N)$	8%
col	$f(b)$	7%
row	$f(a)$	5%
row _{parity}	—	1%

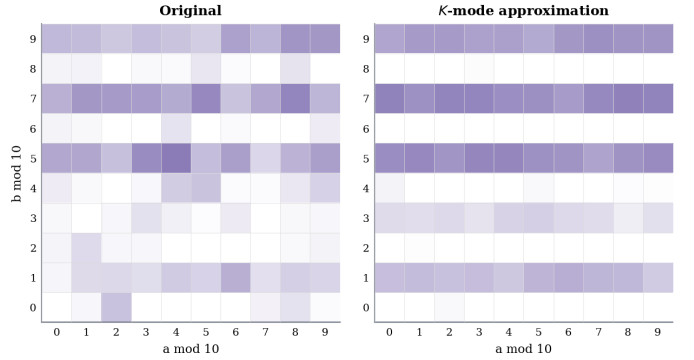


Left: original. Right: K -mode approx.

L00_F94708 $K = 8, R^2 = 0.404$ *col_parity: 44%; col: 36%; mixed: 14%; sum: 2%; row: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.277 + 0.375 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.162 \cos\left(\frac{2\pi(4b)}{10} + 1.412\right) \\
& + 0.144 \cos\left(\frac{2\pi(b)}{10} + 1.893\right) + 0.07 \cos\left(\frac{2\pi(-3b)}{10} + \frac{3\pi}{4}\right) + 0.069 \cos\left(\frac{2\pi(2b)}{10} - 0.048\right) \\
& + 0.054 \cos\left(\frac{2\pi(2a+b)}{10} - 2.227\right) + 0.052 \cos\left(\frac{2\pi(a-4b)}{10} - 1.311\right) + 0.043 \cos\left(\frac{2\pi(-4a+3b)}{10} - 0.927\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	44%
col	$f(b)$	36%
mixed	—	14%
sum	$\cos(2\pi k(a+b)/N)$	2%
row	$f(a)$	2%
diff	$\cos(2\pi k(b-a)/N)$	1%

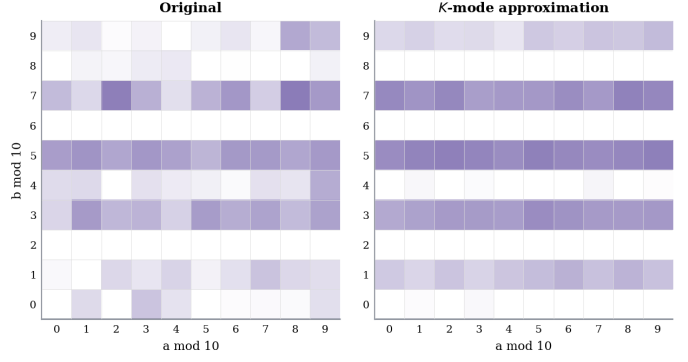


Left: original. Right: K -mode approx.

L00_F114267 $K = 8, R^2 = 0.364$ *col_parity: 45%; col: 32%; mixed: 17%; row: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.276 + 0.39 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.157 \cos\left(\frac{2\pi(b)}{10} - 3.069\right) \\
& + 0.147 \cos\left(\frac{2\pi(4b)}{10} + 0.117\right) + 0.076 \cos\left(\frac{2\pi(3b)}{10} - \frac{\pi}{4}\right) + 0.06 \cos\left(\frac{2\pi(a-4b)}{10} - 1.392\right) \\
& + 0.047 \cos\left(\frac{2\pi(3a+5b)}{10} - 0.595\right) + 0.047 \cos\left(\frac{2\pi(5a-2b)}{10} + 1.904\right) + 0.046 \cos\left(\frac{2\pi(a+5b)}{10} - \frac{\pi}{2}\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	45%
col	$f(b)$	32%
mixed	—	17%
row	$f(a)$	3%
diff	$\cos(2\pi k(b-a)/N)$	2%

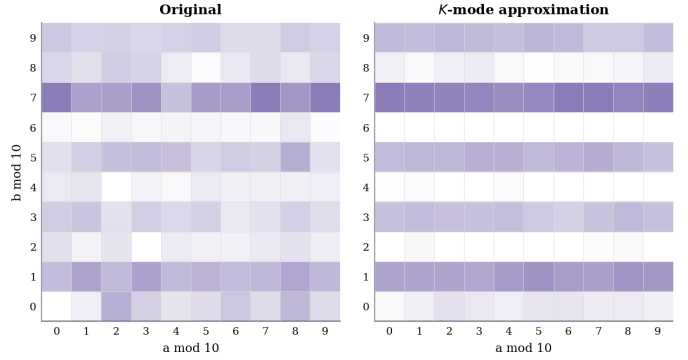


Left: original. Right: K -mode approx.

L00_F150759 $K = 8, R^2 = 0.409$ $\text{col}_{\text{parity}}$: 44%; col : 37%; mixed : 13%; row : 2%

$$\begin{aligned}
f(a, b) \approx & 0.333 + 0.311 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.132 \cos\left(\frac{2\pi(3b)}{10} - 1.159\right) \\
& + 0.096 \cos\left(\frac{2\pi(-b)}{10} - 0.95\right) + 0.088 \cos\left(\frac{2\pi(2b)}{10} - 1.913\right) + 0.075 \cos\left(\frac{2\pi(4b)}{10} + \frac{\pi}{4}\right) \\
& + 0.042 \cos\left(\frac{2\pi(a-2b)}{10} - 2.917\right) + 0.039 \cos\left(\frac{2\pi(-3a-b)}{10} - 2.527\right) + 0.035 \cos\left(\frac{2\pi(2a-4b)}{10} - 2.734\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	44%
col	$f(b)$	37%
mixed	—	13%
row	$f(a)$	2%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%

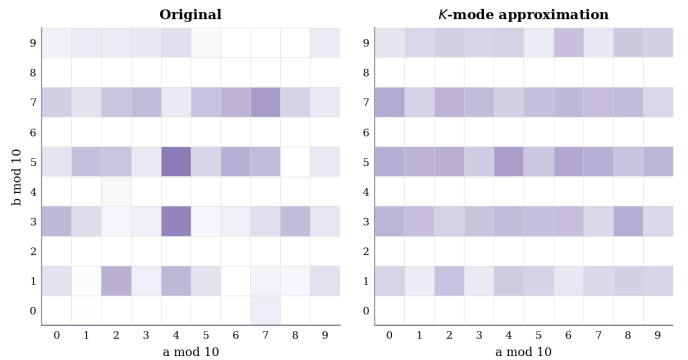


Left: original. Right: K -mode approx.

L00_F27190 $K = 8, R^2 = 0.210$ $\text{col}_{\text{parity}}$: 41%; mixed : 34%; col : 13%; row : 6%; diff : 3%

$$\begin{aligned}
f(a, b) \approx & 0.144 + 0.28 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.073 \cos\left(\frac{2\pi(4b)}{10} + 0.098\right) \\
& + 0.069 \cos\left(\frac{2\pi(b)}{10} + 3.06\right) + 0.052 \cos\left(\frac{2\pi(3a-4b)}{10} - 0.437\right) + 0.051 \cos\left(\frac{2\pi(4a-2b)}{10} + \frac{3\pi}{4}\right) \\
& + 0.051 \cos\left(\frac{2\pi(4a+3b)}{10} - 0.689\right) + 0.049 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.047 \cos\left(\frac{2\pi(2a+4b)}{10} - \frac{\pi}{4}\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	41%
mixed	—	34%
col	$f(b)$	13%
row	$f(a)$	6%
diff	$\cos(2\pi k(b-a)/N)$	3%
sum	$\cos(2\pi k(a+b)/N)$	2%
parity	$(-1)^{a+b}$	1%
$\text{row}_{\text{parity}}$	—	1%

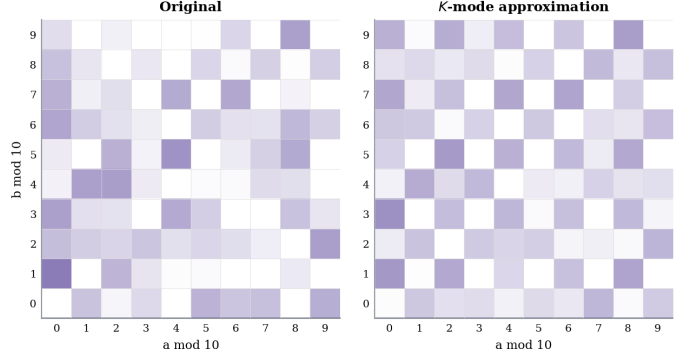


Left: original. Right: K -mode approx.

L18_F29858 $K = 8, R^2 = 0.274$ mixed : 42%; parity : 21%; diff : 14%; sum : 7%; row : 6%; col : 5%; $\text{row}_{\text{parity}}$: 4%

$$\begin{aligned}
f(a, b) \approx & 0.251 + 0.23 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.098 \cos\left(\frac{2\pi(5a)}{10}\right) \\
& + 0.094 \cos\left(\frac{2\pi(4a-4b)}{10} + 2.14\right) + 0.077 \cos\left(\frac{2\pi(2a-3b)}{10} - 0.502\right) + 0.073 \cos\left(\frac{2\pi(2a+2b)}{10} - \frac{3\pi}{4}\right) \\
& + 0.072 \cos\left(\frac{2\pi(a-4b)}{10} + 2.78\right) + 0.069 \cos\left(\frac{2\pi(a)}{10} - 0.049\right) + 0.067 \cos\left(\frac{2\pi(2a-2b)}{10} + 2.906\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	42%
parity	$(-1)^{a+b}$	21%
diff	$\cos(2\pi k(b-a)/N)$	14%
sum	$\cos(2\pi k(a+b)/N)$	7%
row	$f(a)$	6%
col	$f(b)$	5%
row _{parity}	—	4%

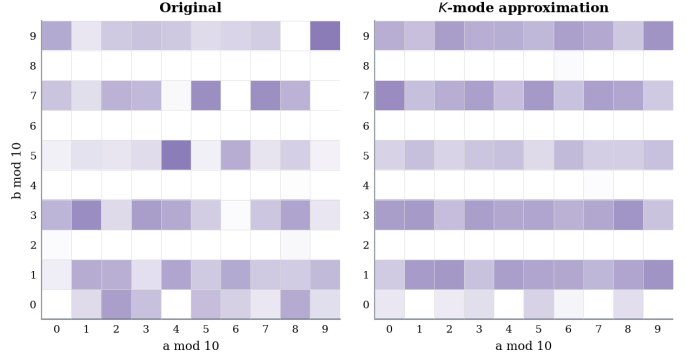


Left: original. Right: K -mode approx.

L00_F160002 $K = 8, R^2 = 0.284$ *col_{parity}: 38%; mixed: 38%; col: 14%; diff: 7%*

$$\begin{aligned}
f(a, b) \approx & 0.244 + 0.348 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.094 \cos\left(\frac{2\pi(-b)}{10} + 0.234\right) \\
& + 0.094 \cos\left(\frac{2\pi(-3b)}{10}\right) + 0.084 \cos\left(\frac{2\pi(4a-4b)}{10} - \frac{\pi}{3}\right) + 0.075 \cos\left(\frac{2\pi(4a+3b)}{10} - 0.373\right) \\
& + 0.074 \cos\left(\frac{2\pi(3a+b)}{10} + 2.422\right) + 0.069 \cos\left(\frac{2\pi(4a-3b)}{10} - 0.235\right) + 0.066 \cos\left(\frac{2\pi(3a+2b)}{10} + 1.959\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	38%
mixed	—	38%
col	$f(b)$	14%
diff	$\cos(2\pi k(b-a)/N)$	7%
sum	$\cos(2\pi k(a+b)/N)$	2%
row	$f(a)$	1%

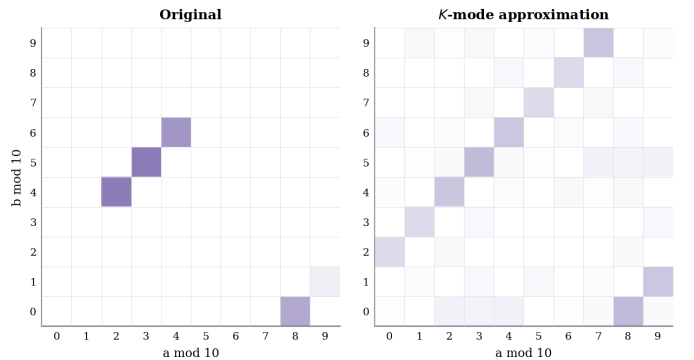


Left: original. Right: K -mode approx.

L13_F21051 $K = 8, R^2 = 0.449$ *mixed: 42%; diff: 35%; sum: 7%; row: 6%; col: 6%; parity: 4%*

$$\begin{aligned}
f(a, b) \approx & 0.036 + 0.071 \cos\left(\frac{2\pi(2a-2b)}{10} + 2.513\right) + 0.071 \cos\left(\frac{2\pi(-3a+3b)}{10} + 2.513\right) \\
& + 0.071 \cos\left(\frac{2\pi(a-b)}{10} + 1.257\right) + 0.071 \cos\left(\frac{2\pi(4a-4b)}{10} - 1.257\right) + 0.071 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.045 \cos\left(\frac{2\pi(2b)}{10}\right) + 0.045 \cos\left(\frac{2\pi(2a)}{10} + 2.544\right) + 0.045 \cos\left(\frac{2\pi(2a-4b)}{10} + 2.483\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	42%
diff	$\cos(2\pi k(b-a)/N)$	35%
sum	$\cos(2\pi k(a+b)/N)$	7%
row	$f(a)$	6%
col	$f(b)$	6%
parity	$(-1)^{a+b}$	4%

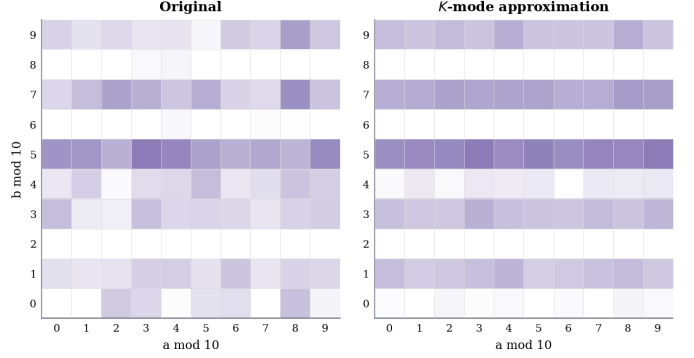


Left: original. Right: K -mode approx.

L00_F60365 $K = 8, R^2 = 0.397$ *col_parity: 43%; col: 34%; mixed: 15%; row: 4%; sum: 3%*

$$f(a, b) \approx 0.266 + 0.338 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.116 \cos\left(\frac{2\pi(b)}{10} + 3.047\right) \\ + 0.116 \cos\left(\frac{2\pi(4b)}{10} + 0.675\right) + 0.114 \cos\left(\frac{2\pi(2b)}{10} + 0.344\right) + 0.072 \cos\left(\frac{2\pi(3b)}{10} - 1.631\right) \\ + 0.052 \cos\left(\frac{2\pi(2a)}{10} + 1.955\right) + 0.05 \cos\left(\frac{2\pi(5a-b)}{10} - 0.318\right) + 0.047 \cos\left(\frac{2\pi(3a-4b)}{10} + 1.787\right)$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	43%
col	$f(b)$	34%
mixed	—	15%
row	$f(a)$	4%
sum	$\cos(2\pi k(a+b)/N)$	3%

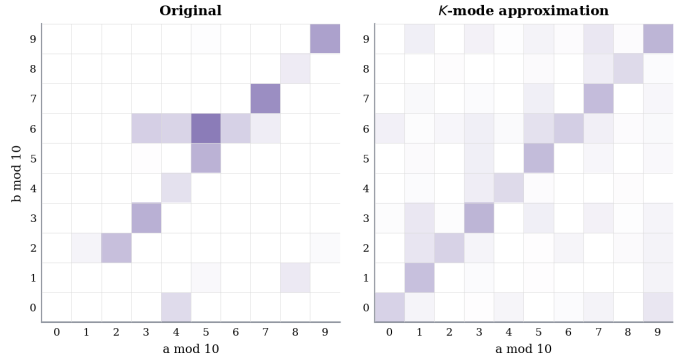


Left: original. Right: K -mode approx.

L10_F54838 $K = 8, R^2 = 0.454$ *mixed: 39%; diff: 37%; col: 9%; row: 5%; sum: 5%; parity: 2%*

$$f(a, b) \approx 0.064 + 0.095 \cos\left(\frac{2\pi(a-b)}{10} + 0.259\right) + 0.077 \cos\left(\frac{2\pi(2a-2b)}{10} + 0.279\right) \\ + 0.077 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.062 \cos\left(\frac{2\pi(4a-4b)}{10} + 0.309\right) + 0.057 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\ + 0.052 \cos\left(\frac{2\pi(4a+b)}{10} + 2.904\right) + 0.051 \cos\left(\frac{2\pi(-3b)}{10} - 1.29\right) + 0.05 \cos\left(\frac{2\pi(5a)}{10} - \pi\right)$$

Dir	Structure	Frac
mixed	—	39%
diff	$\cos(2\pi k(b-a)/N)$	37%
col	$f(b)$	9%
row	$f(a)$	5%
sum	$\cos(2\pi k(a+b)/N)$	5%
parity	$(-1)^{a+b}$	2%
$\text{row}_{\text{parity}}$	—	2%

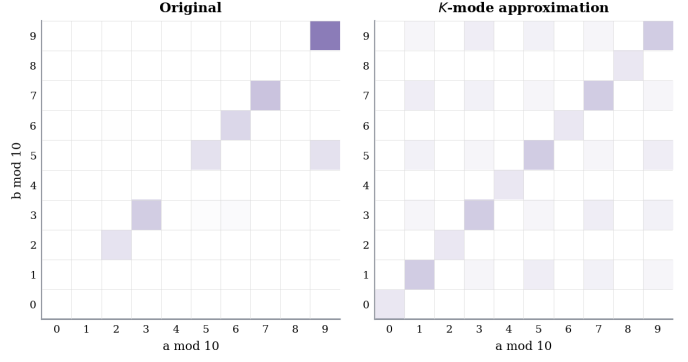


Left: original. Right: K -mode approx.

L10_F36803 $K = 8, R^2 = 0.372$ *mixed: 43%; diff: 34%; row: 6%; parity: 5%; col: 4%; sum: 4%; col_parity: 2%; row_parity: 2%*

$$f(a, b) \approx 0.029 + 0.056 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.053 \cos\left(\frac{2\pi(3a-3b)}{10} - 0.067\right) \\ + 0.052 \cos\left(\frac{2\pi(2a-2b)}{10} + 0.085\right) + 0.048 \cos\left(\frac{2\pi(4a-4b)}{10} + 0.048\right) + 0.048 \cos\left(\frac{2\pi(a-b)}{10} - 0.078\right) \\ + 0.037 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.035 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) + 0.033 \cos\left(\frac{2\pi(3a+2b)}{10} + 3.026\right)$$

Dir	Structure	Frac
mixed	—	43%
diff	$\cos(2\pi k(b-a)/N)$	34%
row	$f(a)$	6%
parity	$(-1)^{a+b}$	5%
col	$f(b)$	4%
sum	$\cos(2\pi k(a+b)/N)$	4%
col _{parity}	—	2%
row _{parity}	—	2%

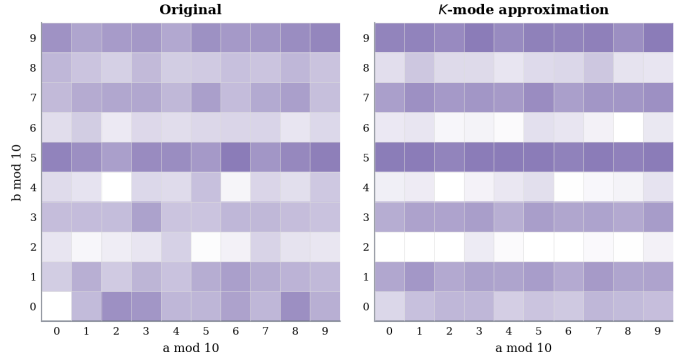


Left: original. Right: K -mode approx.

L00_F63286 $K = 8$, $R^2 = 0.406$ *col_parity: 42%; col: 36%; mixed: 15%; sum: 3%; row: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.51 + 0.313 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.139 \cos\left(\frac{2\pi(2b)}{10} + 0.474\right) \\
& + 0.112 \cos\left(\frac{2\pi(b)}{10} + 1.154\right) + 0.099 \cos\left(\frac{2\pi(4b)}{10} - 0.086\right) + 0.05 \cos\left(\frac{2\pi(2a+4b)}{10} - \pi\right) \\
& + 0.042 \cos\left(\frac{2\pi(2a-b)}{10} - 2.951\right) + 0.037 \cos\left(\frac{2\pi(3a+3b)}{10} + 2.76\right) + 0.037 \cos\left(\frac{2\pi(5a)}{10} - \pi\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	42%
col	$f(b)$	36%
mixed	—	15%
sum	$\cos(2\pi k(a+b)/N)$	3%
row	$f(a)$	2%
diff	$\cos(2\pi k(b-a)/N)$	1%

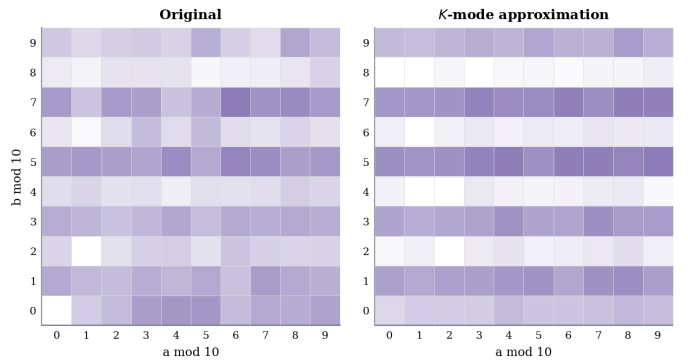


Left: original. Right: K -mode approx.

L00_F14799 $K = 8$, $R^2 = 0.378$ *col_parity: 42%; col: 30%; mixed: 18%; row: 6%; diff: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.468 + 0.304 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.128 \cos\left(\frac{2\pi(4b)}{10} + 0.206\right) \\
& + 0.102 \cos\left(\frac{2\pi(2b)}{10} - 0.708\right) + 0.073 \cos\left(\frac{2\pi(3b)}{10} - 0.413\right) + 0.054 \cos\left(\frac{2\pi(a)}{10} + 2.208\right) \\
& + 0.049 \cos\left(\frac{2\pi(2a)}{10} + \frac{2\pi}{3}\right) + 0.044 \cos\left(\frac{2\pi(4a-4b)}{10} + 2.89\right) + 0.043 \cos\left(\frac{2\pi(3a+b)}{10} - 2.629\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	42%
col	$f(b)$	30%
mixed	—	18%
row	$f(a)$	6%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%

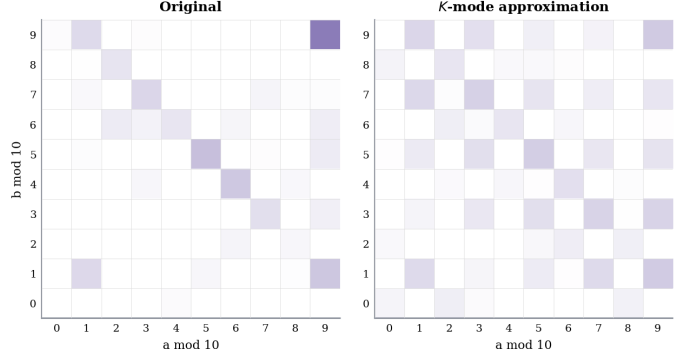


Left: original. Right: K -mode approx.

L17_F142043 $K = 8$, $R^2 = 0.305$ *mixed: 40%; sum: 23%; parity: 11%; row: 9%; diff: 6%; col: 5%; row_parity: 4%; col_parity: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.053 + 0.09 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.06 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.146\right) \\
& + 0.059 \cos\left(\frac{2\pi(-a-b)}{10} - 0.255\right) + 0.051 \cos\left(\frac{2\pi(5a)}{10} + \pi\right) + 0.041 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) \\
& + 0.039 \cos\left(\frac{2\pi(2a+3b)}{10} + \pi\right) + 0.037 \cos\left(\frac{2\pi(3a)}{10} + 2.012\right) + 0.036 \cos\left(\frac{2\pi(-a+5b)}{10} + 2.692\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	40%
sum	$\cos(2\pi k(a+b)/N)$	23%
parity	$(-1)^{a+b}$	11%
row	$f(a)$	9%
diff	$\cos(2\pi k(b-a)/N)$	6%
col	$f(b)$	5%
row _{parity}	—	4%
col _{parity}	—	2%

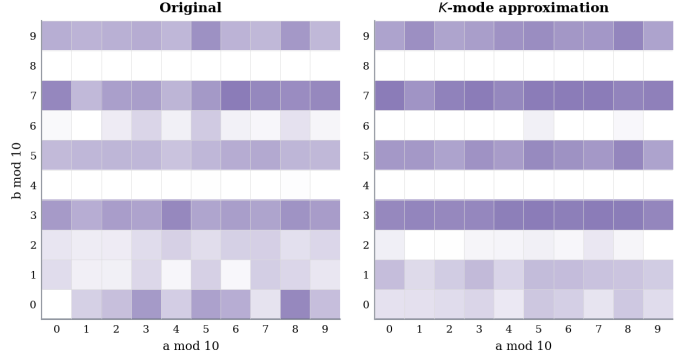


Left: original. Right: K -mode approx.

L00_F13444 $K = 8$, $R^2 = 0.448$ *col_parity: 44%; col: 41%; mixed: 10%; row: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.383 + 0.403 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.199 \cos\left(\frac{2\pi(3b)}{10} + 0.714\right) \\
& + 0.184 \cos\left(\frac{2\pi(4b)}{10} + 0.515\right) + 0.049 \cos\left(\frac{2\pi(a)}{10} + 2.405\right) + 0.042 \cos\left(\frac{2\pi(3a-2b)}{10} + 2.921\right) \\
& + 0.042 \cos\left(\frac{2\pi(4a)}{10} - \frac{\pi}{4}\right) + 0.041 \cos\left(\frac{2\pi(4a+2b)}{10} - 1.377\right) + 0.038 \cos\left(\frac{2\pi(3a-3b)}{10} + 2.245\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	44%
col	$f(b)$	41%
mixed	—	10%
row	$f(a)$	3%
diff	$\cos(2\pi k(b-a)/N)$	1%

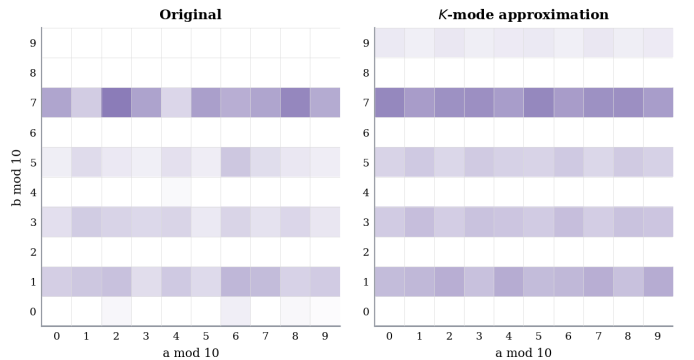


Left: original. Right: K -mode approx.

L00_F107710 $K = 8$, $R^2 = 0.480$ *col: 45%; col_parity: 43%; mixed: 7%; diff: 2%*

$$\begin{aligned}
f(a, b) \approx & 0.159 + 0.304 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.153 \cos\left(\frac{2\pi(-3b)}{10} + 0.922\right) \\
& + 0.145 \cos\left(\frac{2\pi(2b)}{10} - 2.161\right) + 0.048 \cos\left(\frac{2\pi(4b)}{10} + 0.589\right) + 0.04 \cos\left(\frac{2\pi(b)}{10} + 2.437\right) \\
& + 0.032 \cos\left(\frac{2\pi(4a-4b)}{10} - 1.529\right) + 0.03 \cos\left(\frac{2\pi(-4a-b)}{10} - 1.67\right) + 0.03 \cos\left(\frac{2\pi(4a-2b)}{10} + 2.837\right)
\end{aligned}$$

Dir	Structure	Frac
col	$f(b)$	45%
col _{parity}	—	43%
mixed	—	7%
diff	$\cos(2\pi k(b-a)/N)$	2%

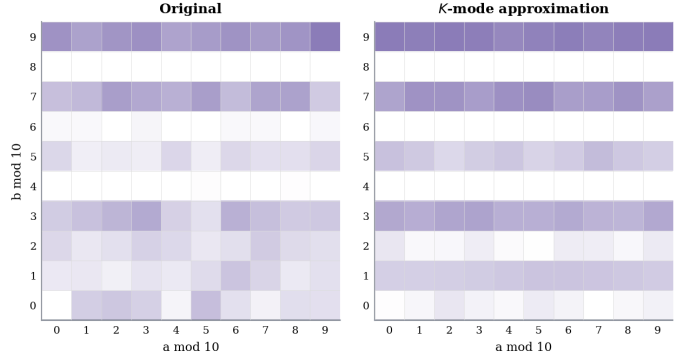


Left: original. Right: K -mode approx.

L00_F623 $K = 8, R^2 = 0.491$ *col: 46%; col_parity: 44%; mixed: 7%*

$$\begin{aligned}
f(a, b) \approx & 0.288 + 0.351 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.148 \cos\left(\frac{2\pi(3b)}{10} + \frac{\pi}{3}\right) \\
& + 0.144 \cos\left(\frac{2\pi(4b)}{10} + 1.53\right) + 0.127 \cos\left(\frac{2\pi(-b)}{10} - 0.386\right) + 0.076 \cos\left(\frac{2\pi(2b)}{10} + \frac{3\pi}{4}\right) \\
& + 0.041 \cos\left(\frac{2\pi(a-3b)}{10} - 1.845\right) + 0.039 \cos\left(\frac{2\pi(3a+3b)}{10} + 2.761\right) + 0.038 \cos\left(\frac{2\pi(3a-b)}{10} + 2.223\right)
\end{aligned}$$

Dir	Structure	Frac
col	$f(b)$	46%
col_parity	—	44%
mixed	—	7%
sum	$\cos(2\pi k(a+b)/N)$	1%

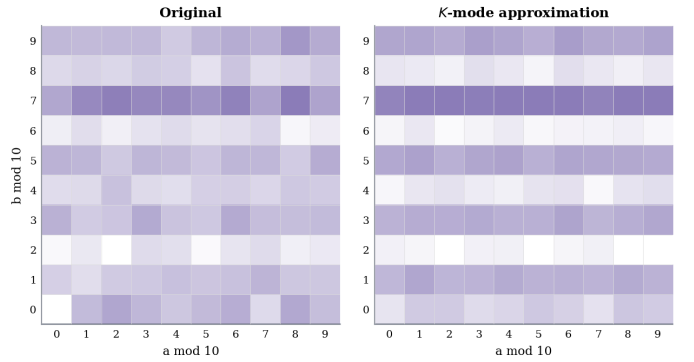


Left: original. Right: K -mode approx.

L00_F79735 $K = 8, R^2 = 0.419$ *col_parity: 41%; col: 37%; mixed: 13%; row: 4%; diff: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.438 + 0.274 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.119 \cos\left(\frac{2\pi(3b)}{10} - 0.522\right) \\
& + 0.1 \cos\left(\frac{2\pi(b)}{10} + 1.458\right) + 0.091 \cos\left(\frac{2\pi(4b)}{10} + \frac{\pi}{4}\right) + 0.039 \cos\left(\frac{2\pi(3a+2b)}{10} - 2.928\right) \\
& + 0.039 \cos\left(\frac{2\pi(3a-3b)}{10} - 2.898\right) + 0.038 \cos\left(\frac{2\pi(4a)}{10} - 2.569\right) + 0.037 \cos\left(\frac{2\pi(-3a-3b)}{10} - 2.432\right)
\end{aligned}$$

Dir	Structure	Frac
col_parity	—	41%
col	$f(b)$	37%
mixed	—	13%
row	$f(a)$	4%
diff	$\cos(2\pi k(b-a)/N)$	3%
sum	$\cos(2\pi k(a+b)/N)$	2%

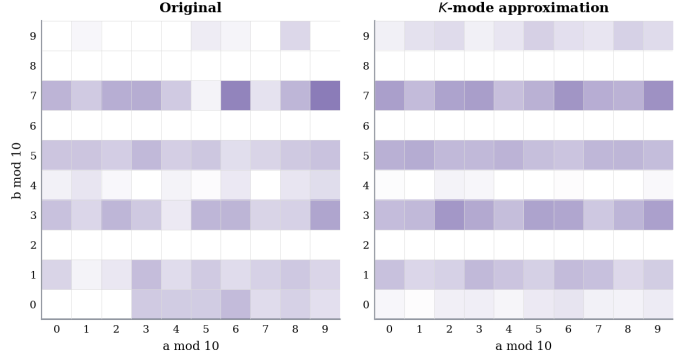


Left: original. Right: K -mode approx.

L00_F146197 $K = 8, R^2 = 0.425$ *col_parity: 36%; col: 34%; mixed: 15%; diff: 6%; row: 5%; sum: 4%*

$$\begin{aligned}
f(a, b) \approx & 0.212 + 0.278 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.139 \cos\left(\frac{2\pi(3b)}{10} - 0.622\right) \\
& + 0.115 \cos\left(\frac{2\pi(4b)}{10}\right) + 0.056 \cos\left(\frac{2\pi(-3a+3b)}{10} - 1.792\right) + 0.054 \cos\left(\frac{2\pi(b)}{10} - 2.689\right) \\
& + 0.052 \cos\left(\frac{2\pi(3a+2b)}{10} - 1.363\right) + 0.052 \cos\left(\frac{2\pi(-3a)}{10} - 1.737\right) + 0.045 \cos\left(\frac{2\pi(a+b)}{10} + 2.632\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	36%
col	$f(b)$	34%
mixed	—	15%
diff	$\cos(2\pi k(b-a)/N)$	6%
row	$f(a)$	5%
sum	$\cos(2\pi k(a+b)/N)$	4%

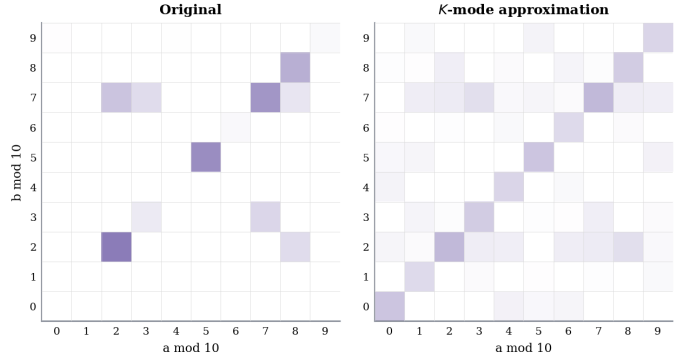


Left: original. Right: K -mode approx.

L15_F142885 $K = 8, R^2 = 0.463$ *mixed: 34%; diff: 30%; col: 11%; sum: 10%; row: 10%; parity: 5%*

$$\begin{aligned}
f(a, b) \approx & 0.05 + 0.086 \cos\left(\frac{2\pi(2a-2b)}{10} - 0.095\right) + 0.074 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.065 \cos\left(\frac{2\pi(-3a+3b)}{10}\right) + 0.063 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.081\right) + 0.062 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.474\right) \\
& + 0.054 \cos\left(\frac{2\pi(4b)}{10} + 0.728\right) + 0.052 \cos\left(\frac{2\pi(2b)}{10} - 2.706\right) + 0.051 \cos\left(\frac{2\pi(-a+b)}{10}\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	34%
diff	$\cos(2\pi k(b-a)/N)$	30%
col	$f(b)$	11%
sum	$\cos(2\pi k(a+b)/N)$	10%
row	$f(a)$	10%
parity	$(-1)^{a+b}$	5%

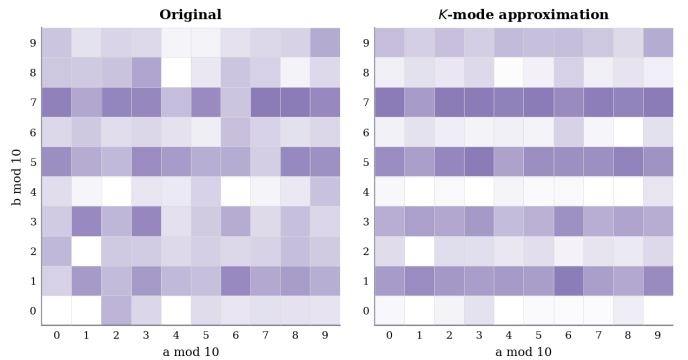


Left: original. Right: K -mode approx.

L00_F78585 $K = 8, R^2 = 0.379$ *col_parity: 35%; col: 26%; mixed: 22%; row: 7%; diff: 6%; sum: 3%*

$$\begin{aligned}
f(a, b) \approx & 0.43 + 0.317 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.139 \cos\left(\frac{2\pi(2b)}{10} - 2.002\right) \\
& + 0.089 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.583\right) + 0.078 \cos\left(\frac{2\pi(3b)}{10} - 1.898\right) + 0.078 \cos\left(\frac{2\pi(4b)}{10} + 0.109\right) \\
& + 0.073 \cos\left(\frac{2\pi(-b)}{10} - 2.713\right) + 0.072 \cos\left(\frac{2\pi(2a+4b)}{10} + 2.892\right) + 0.063 \cos\left(\frac{2\pi(-3a)}{10} - 1.426\right)
\end{aligned}$$

Dir	Structure	Frac
$\text{col}_{\text{parity}}$	—	35%
col	$f(b)$	26%
mixed	—	22%
row	$f(a)$	7%
diff	$\cos(2\pi k(b-a)/N)$	6%
sum	$\cos(2\pi k(a+b)/N)$	3%

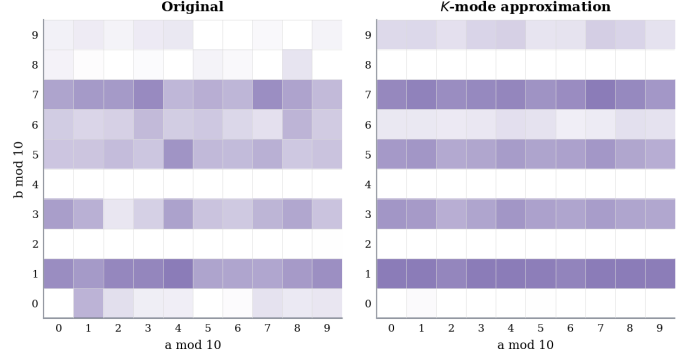


Left: original. Right: K -mode approx.

L00_F36348 $K = 8, R^2 = 0.472$ *col: 45%; col_parity: 43%; mixed: 8%*

$$\begin{aligned}
f(a, b) \approx & 0.321 + 0.401 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.228 \cos\left(\frac{2\pi(2b)}{10} - 1.516\right) \\
& + 0.126 \cos\left(\frac{2\pi(3b)}{10} - \frac{\pi}{4}\right) + 0.107 \cos\left(\frac{2\pi(4b)}{10} - 1.512\right) + 0.059 \cos\left(\frac{2\pi(b)}{10} + 3.101\right) \\
& + 0.041 \cos\left(\frac{2\pi(2a+4b)}{10} - 1.107\right) + 0.039 \cos\left(\frac{2\pi(3a)}{10} - 1.311\right) + 0.037 \cos\left(\frac{2\pi(3a+5b)}{10} + 2.606\right)
\end{aligned}$$

Dir	Structure	Frac
col	$f(b)$	45%
col _{parity}	—	43%
mixed	—	8%
row	$f(a)$	1%
sum	$\cos(2\pi k(a+b)/N)$	1%

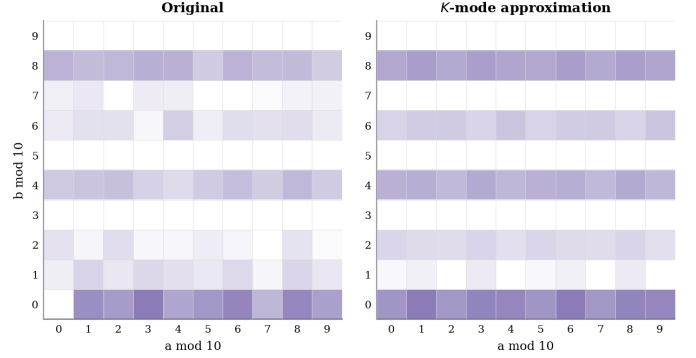


Left: original. Right: K -mode approx.

L00_F134361 $K = 8, R^2 = 0.446$ col_{parity}: 43%; col: 41%; mixed: 12%; row: 2%

$$\begin{aligned}
f(a, b) \approx & 0.229 + 0.337 \cos\left(\frac{2\pi(5b)}{10}\right) + 0.16 \cos\left(\frac{2\pi(3b)}{10} - \frac{\pi}{3}\right) \\
& + 0.12 \cos\left(\frac{2\pi(b)}{10} + 0.366\right) + 0.087 \cos\left(\frac{2\pi(2b)}{10} + 0.431\right) + 0.081 \cos\left(\frac{2\pi(4b)}{10} - 0.901\right) \\
& + 0.042 \cos\left(\frac{2\pi(4a+3b)}{10} - 2.937\right) + 0.04 \cos\left(\frac{2\pi(4a)}{10} - 2.166\right) + 0.038 \cos\left(\frac{2\pi(4a+b)}{10} - \frac{2\pi}{3}\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	43%
col	$f(b)$	41%
mixed	—	12%
row	$f(a)$	2%
diff	$\cos(2\pi k(b-a)/N)$	2%

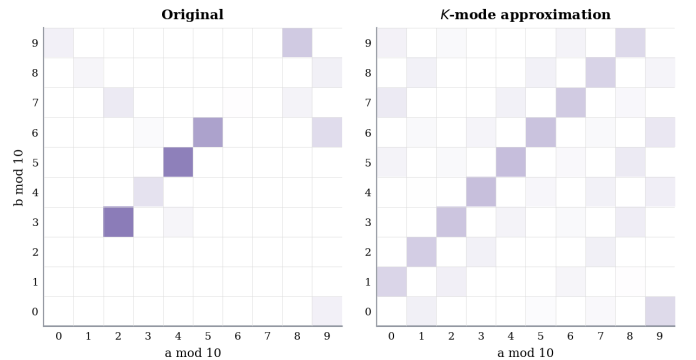


Left: original. Right: K -mode approx.

L19_F103698 $K = 8, R^2 = 0.389$ mixed: 44%; diff: 31%; parity: 7%; col: 6%; row: 6%; sum: 5%

$$\begin{aligned}
f(a, b) \approx & 0.043 + 0.087 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.065 \cos\left(\frac{2\pi(2a-2b)}{10} + 1.244\right) \\
& + 0.065 \cos\left(\frac{2\pi(3a-3b)}{10} + 1.897\right) + 0.064 \cos\left(\frac{2\pi(4a-4b)}{10} + 2.654\right) + 0.064 \cos\left(\frac{2\pi(a-b)}{10} + 0.488\right) \\
& + 0.036 \cos\left(\frac{2\pi(2a-b)}{10} - 1.508\right) + 0.036 \cos\left(\frac{2\pi(3a-4b)}{10} - 1.634\right) + 0.036 \cos\left(\frac{2\pi(-b)}{10} - 3.047\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	44%
diff	$\cos(2\pi k(b-a)/N)$	31%
parity	$(-1)^{a+b}$	7%
col	$f(b)$	6%
row	$f(a)$	6%
sum	$\cos(2\pi k(a+b)/N)$	5%

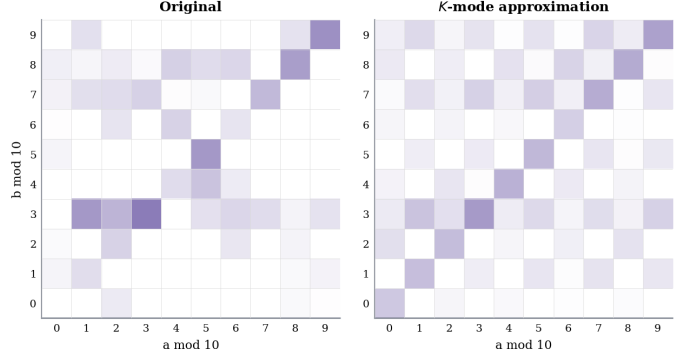


Left: original. Right: K -mode approx.

L14_F58071 $K = 8, R^2 = 0.438$ *mixed: 35%; diff: 28%; col: 17%; parity: 11%; sum: 3%; row: 3%; col_parity: 2%*

$$f(a, b) \approx 0.122 + 0.139 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.1 \cos\left(\frac{2\pi(2b)}{10} + 2.542\right) \\ + 0.091 \cos\left(\frac{2\pi(a-b)}{10} + 0.229\right) + 0.076 \cos\left(\frac{2\pi(2a-2b)}{10} + 0.16\right) + 0.076 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.218\right) \\ + 0.066 \cos\left(\frac{2\pi(3a-3b)}{10} + 0.12\right) + 0.062 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.061 \cos\left(\frac{2\pi(-a+3b)}{10} + 1.617\right)$$

Dir	Structure	Frac
mixed	—	35%
diff	$\cos(2\pi k(b-a)/N)$	28%
col	$f(b)$	17%
parity	$(-1)^{a+b}$	11%
sum	$\cos(2\pi k(a+b)/N)$	3%
row	$f(a)$	3%
col_parity	—	2%

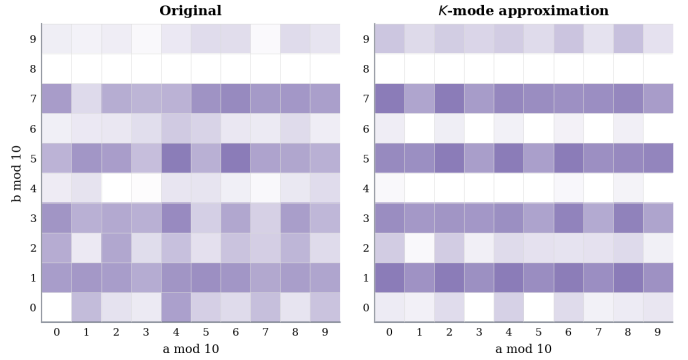


Left: original. Right: K -mode approx.

L00_F161769 $K = 8, R^2 = 0.436$ *col: 40%; col_parity: 39%; mixed: 13%; row: 2%; sum: 2%*

$$f(a, b) \approx 0.402 + 0.363 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.181 \cos\left(\frac{2\pi(2b)}{10} - 1.5\right) \\ + 0.132 \cos\left(\frac{2\pi(b)}{10} - 1.896\right) + 0.112 \cos\left(\frac{2\pi(4b)}{10} + 0.095\right) + 0.08 \cos\left(\frac{2\pi(5a)}{10}\right) \\ + 0.065 \cos\left(\frac{2\pi(3b)}{10} - 0.996\right) + 0.057 \cos\left(\frac{2\pi(4a-2b)}{10} + 2.679\right) + 0.051 \cos\left(\frac{2\pi(4a+4b)}{10} + \frac{3\pi}{4}\right)$$

Dir	Structure	Frac
col	$f(b)$	40%
col_parity	—	39%
mixed	—	13%
row	$f(a)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%
row_parity	—	2%
diff	$\cos(2\pi k(b-a)/N)$	1%

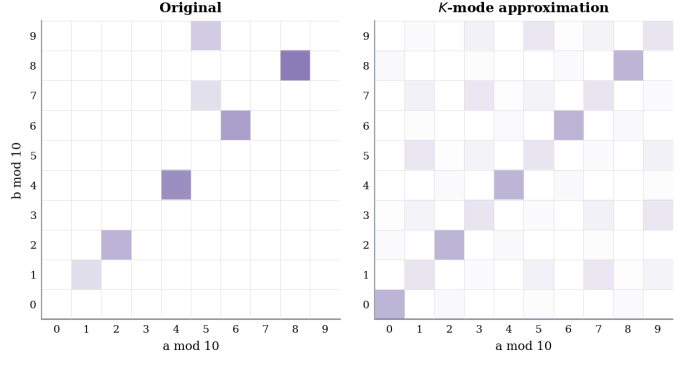


Left: original. Right: K -mode approx.

L13_F65498 $K = 8, R^2 = 0.533$ *mixed: 49%; diff: 32%; parity: 6%; row: 3%; sum: 3%; col: 3%*

$$f(a, b) \approx 0.04 + 0.08 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.067 \cos\left(\frac{2\pi(2a-2b)}{10} - 0.07\right) \\ + 0.067 \cos\left(\frac{2\pi(-3a+3b)}{10} - 0.07\right) + 0.064 \cos\left(\frac{2\pi(4a-4b)}{10} - 0.144\right) + 0.064 \cos\left(\frac{2\pi(a-b)}{10} + 0.144\right) \\ + 0.064 \cos\left(\frac{2\pi(a+4b)}{10} - 0.144\right) + 0.064 \cos\left(\frac{2\pi(4a+b)}{10} + 0.144\right) + 0.06 \cos\left(\frac{2\pi(2a+3b)}{10} + 0.078\right)$$

Dir	Structure	Frac
mixed	—	49%
diff	$\cos(2\pi k(b-a)/N)$	32%
parity	$(-1)^{a+b}$	6%
row	$f(a)$	3%
sum	$\cos(2\pi k(a+b)/N)$	3%
col	$f(b)$	3%
row _{parity}	—	2%
col _{parity}	—	2%

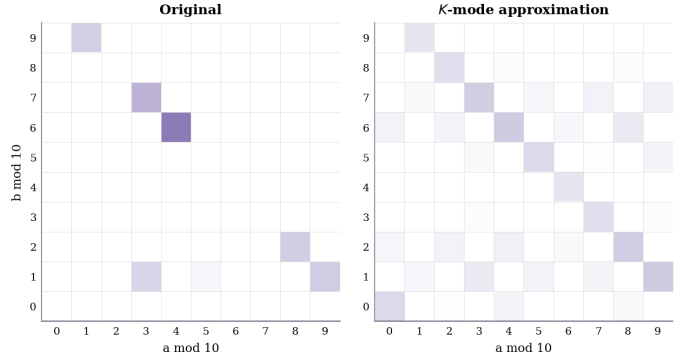


Left: original. Right: K -mode approx.

L15_F121770 $K = 8, R^2 = 0.430$ mixed: 41%; sum: 31%; diff: 8%; row: 7%; col: 7%; parity: 5%

$$\begin{aligned}
f(a, b) \approx & 0.031 + 0.062 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.057 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.086\right) \\
& + 0.057 \cos\left(\frac{2\pi(3a+3b)}{10} - 0.086\right) + 0.048 \cos\left(\frac{2\pi(4a+4b)}{10} + 0.063\right) + 0.048 \cos\left(\frac{2\pi(a+b)}{10} - 0.063\right) \\
& + 0.038 \cos\left(\frac{2\pi(2b)}{10} - 1.628\right) + 0.038 \cos\left(\frac{2\pi(5a+3b)}{10} + 1.628\right) + 0.036 \cos\left(\frac{2\pi(2a)}{10} + 1.812\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	41%
sum	$\cos(2\pi k(a+b)/N)$	31%
diff	$\cos(2\pi k(b-a)/N)$	8%
row	$f(a)$	7%
col	$f(b)$	7%
parity	$(-1)^{a+b}$	5%

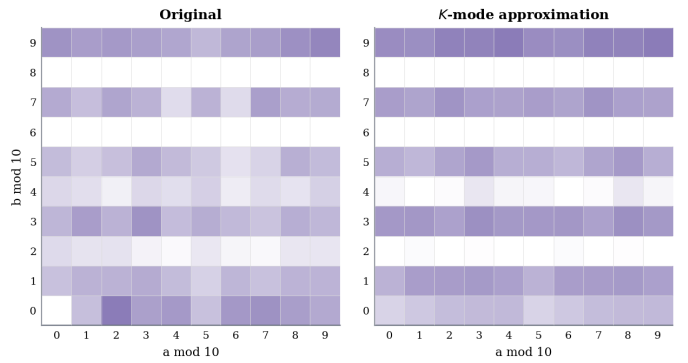


Left: original. Right: K -mode approx.

L00_F109146 $K = 8, R^2 = 0.448$ col_{parity}: 42%; col: 40%; mixed: 13%; row: 2%; sum: 2%

$$\begin{aligned}
f(a, b) \approx & 0.394 + 0.366 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.146 \cos\left(\frac{2\pi(-b)}{10} + 0.299\right) \\
& + 0.135 \cos\left(\frac{2\pi(3b)}{10} + 0.122\right) + 0.13 \cos\left(\frac{2\pi(2b)}{10} + 0.633\right) + 0.093 \cos\left(\frac{2\pi(4b)}{10} + 0.946\right) \\
& + 0.053 \cos\left(\frac{2\pi(-4a-b)}{10} - 2.772\right) + 0.049 \cos\left(\frac{2\pi(2a)}{10} + 2.659\right) + 0.044 \cos\left(\frac{2\pi(2a+2b)}{10} + 2.569\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	42%
col	$f(b)$	40%
mixed	—	13%
row	$f(a)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%

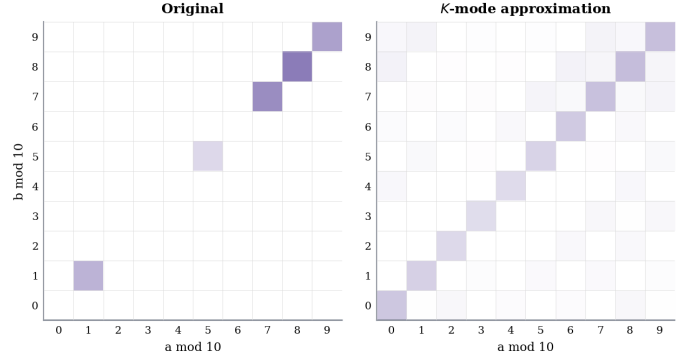


Left: original. Right: K -mode approx.

L15_F97834 $K = 8, R^2 = 0.467$ mixed: 45%; diff: 37%; row: 5%; col: 5%; parity: 5%

$$\begin{aligned}
f(a, b) \approx & 0.034 + 0.069 \cos\left(\frac{2\pi(a-b)}{10}\right) + 0.069 \cos\left(\frac{2\pi(2a-2b)}{10}\right) \\
& + 0.069 \cos\left(\frac{2\pi(3a-3b)}{10}\right) + 0.069 \cos\left(\frac{2\pi(4a-4b)}{10}\right) + 0.069 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.04 \cos\left(\frac{2\pi(b)}{10} + 1.166\right) + 0.04 \cos\left(\frac{2\pi(a)}{10} + 1.166\right) + 0.04 \cos\left(\frac{2\pi(a-2b)}{10} - 1.166\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	45%
diff	$\cos(2\pi k(b-a)/N)$	37%
row	$f(a)$	5%
col	$f(b)$	5%
parity	$(-1)^{a+b}$	5%
sum	$\cos(2\pi k(a+b)/N)$	1%

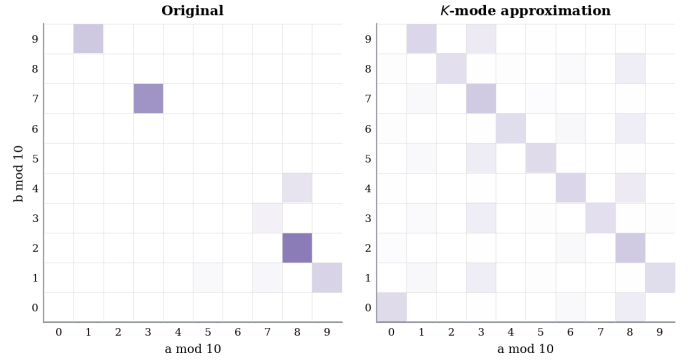


Left: original. Right: K -mode approx.

L21_F42017 $K = 8$, $R^2 = 0.384$ *mixed: 42%; sum: 28%; diff: 11%; row: 8%; col: 6%; parity: 5%*

$$\begin{aligned}
f(a, b) \approx & 0.03 + 0.06 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.054 \cos\left(\frac{2\pi(-a-b)}{10}\right) \\
& + 0.054 \cos\left(\frac{2\pi(4a+4b)}{10}\right) + 0.049 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.049\right) + 0.049 \cos\left(\frac{2\pi(2a+2b)}{10} - 0.049\right) \\
& + 0.036 \cos\left(\frac{2\pi(a+5b)}{10} + 1.512\right) + 0.036 \cos\left(\frac{2\pi(4a)}{10} - 1.512\right) + 0.036 \cos\left(\frac{2\pi(2a)}{10} + 2.548\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	42%
sum	$\cos(2\pi k(a+b)/N)$	28%
diff	$\cos(2\pi k(b-a)/N)$	11%
row	$f(a)$	8%
col	$f(b)$	6%
parity	$(-1)^{a+b}$	5%

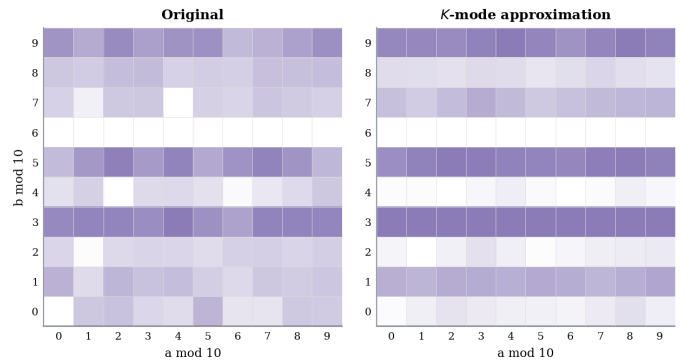


Left: original. Right: K -mode approx.

L00_F103058 $K = 8$, $R^2 = 0.456$ *col-parity: 42%; col: 42%; mixed: 11%*

$$\begin{aligned}
f(a, b) \approx & 0.445 + 0.373 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.188 \cos\left(\frac{2\pi(2b)}{10} + 1.618\right) \\
& + 0.136 \cos\left(\frac{2\pi(4b)}{10} - 0.573\right) + 0.097 \cos\left(\frac{2\pi(-3b)}{10} - \frac{3\pi}{4}\right) + 0.08 \cos\left(\frac{2\pi(-b)}{10} + 1.161\right) \\
& + 0.043 \cos\left(\frac{2\pi(2a)}{10} + \frac{3\pi}{4}\right) + 0.041 \cos\left(\frac{2\pi(-3a+2b)}{10} + 2.993\right) + 0.04 \cos\left(\frac{2\pi(2a+4b)}{10} - 2.642\right)
\end{aligned}$$

Dir	Structure	Frac
col _{parity}	—	42%
col	$f(b)$	42%
mixed	—	11%
row	$f(a)$	2%

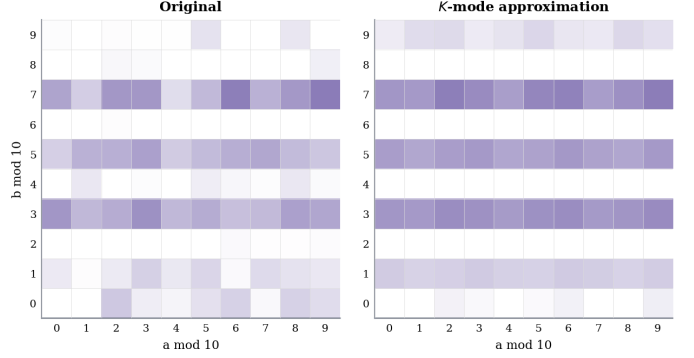


Left: original. Right: K -mode approx.

L00_F64475 $K = 8, R^2 = 0.473$ *col: 43%; col_parity: 41%; mixed: 10%; row: 3%*

$$f(a, b) \approx 0.238 + 0.362 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.18 \cos\left(\frac{2\pi(4b)}{10}\right) \\ + 0.135 \cos\left(\frac{2\pi(3b)}{10} - 0.349\right) + 0.123 \cos\left(\frac{2\pi(b)}{10} - 3.028\right) + 0.056 \cos\left(\frac{2\pi(2b)}{10} - 2.799\right) \\ + 0.052 \cos\left(\frac{2\pi(-3a)}{10} - 2.032\right) + 0.042 \cos\left(\frac{2\pi(-3a+4b)}{10} - 1.219\right) + 0.041 \cos\left(\frac{2\pi(-3a+3b)}{10} - 2.28\right)$$

Dir	Structure	Frac
col	$f(b)$	43%
col_parity	—	41%
mixed	—	10%
row	$f(a)$	3%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	1%

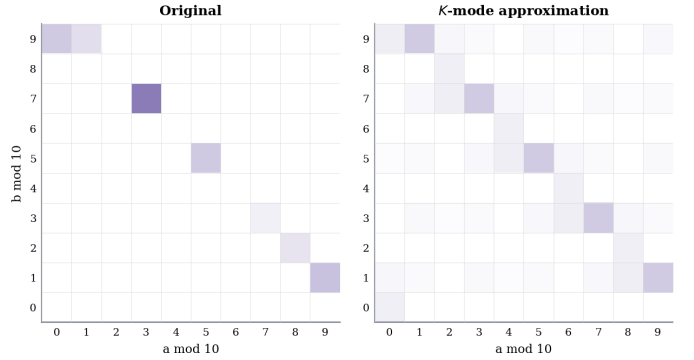


Left: original. Right: K -mode approx.

L17_F10029 $K = 8, R^2 = 0.439$ *mixed: 49%; sum: 31%; diff: 5%; row: 4%; col_parity: 4%; col: 3%; parity: 3%*

$$f(a, b) \approx 0.028 + 0.055 \cos\left(\frac{2\pi(a+b)}{10} + 0.087\right) + 0.052 \cos\left(\frac{2\pi(2a+2b)}{10} + 0.152\right) \\ + 0.049 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.047 \cos\left(\frac{2\pi(a-4b)}{10} - 3.039\right) + 0.047 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.168\right) \\ + 0.044 \cos\left(\frac{2\pi(2a-3b)}{10} - 2.962\right) + 0.042 \cos\left(\frac{2\pi(4a+4b)}{10} + 0.115\right) + 0.04 \cos\left(\frac{2\pi(5a+5b)}{10}\right)$$

Dir	Structure	Frac
mixed	—	49%
sum	$\cos(2\pi k(a+b)/N)$	31%
diff	$\cos(2\pi k(b-a)/N)$	5%
row	$f(a)$	4%
col_parity	—	4%
col	$f(b)$	3%
parity	$(-1)^{a+b}$	3%
row_parity	—	2%

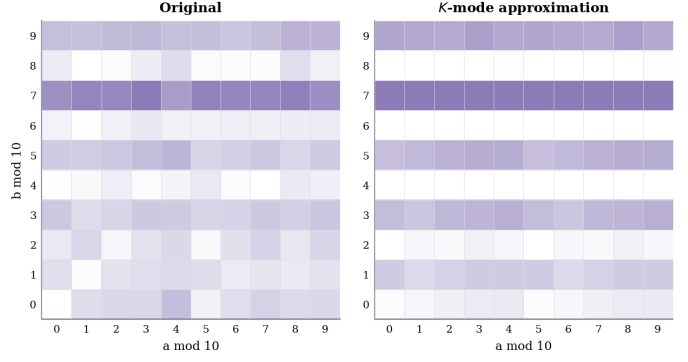


Left: original. Right: K -mode approx.

L00_F10087 $K = 8, R^2 = 0.512$ *col: 48%; col_parity: 42%; mixed: 6%; row: 2%*

$$f(a, b) \approx 0.319 + 0.322 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.177 \cos\left(\frac{2\pi(4b)}{10} + \frac{\pi}{3}\right) \\ + 0.118 \cos\left(\frac{2\pi(b)}{10} + \frac{\pi}{2}\right) + 0.094 \cos\left(\frac{2\pi(3b)}{10} + 0.181\right) + 0.071 \cos\left(\frac{2\pi(2b)}{10} - 2.777\right) \\ + 0.046 \cos\left(\frac{2\pi(2a)}{10} + \frac{2\pi}{3}\right) + 0.031 \cos\left(\frac{2\pi(4a-4b)}{10} + 2.744\right) + 0.029 \cos\left(\frac{2\pi(2a-4b)}{10} + 2.972\right)$$

Dir	Structure	Frac
col	$f(b)$	48%
col _{parity}	—	42%
mixed	—	6%
row	$f(a)$	2%

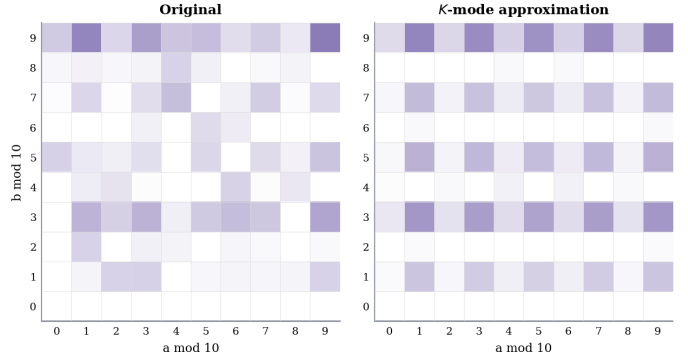


Left: original. Right: K -mode approx.

L22_F127983 $K = 8, R^2 = 0.289$ *mixed: 30%; col_{parity}: 27%; col: 22%; parity: 7%; row_{parity}: 6%; row: 4%*

$$\begin{aligned}
f(a, b) \approx & 0.172 + 0.219 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.113 \cos\left(\frac{2\pi(5a+5b)}{10}\right) \\
& + 0.108 \cos\left(\frac{2\pi(5a)}{10} - \pi\right) + 0.104 \cos\left(\frac{2\pi(2b)}{10} + 1.833\right) + 0.082 \cos\left(\frac{2\pi(3b)}{10} + 1.677\right) \\
& + 0.065 \cos\left(\frac{2\pi(5a+3b)}{10} - \frac{\pi}{2}\right) + 0.048 \cos\left(\frac{2\pi(4a)}{10} - \pi\right) + 0.046 \cos\left(\frac{2\pi(4b)}{10} + 2.916\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	30%
col _{parity}	—	27%
col	$f(b)$	22%
parity	$(-1)^{a+b}$	7%
row _{parity}	—	6%
row	$f(a)$	4%
sum	$\cos(2\pi k(a+b)/N)$	2%
diff	$\cos(2\pi k(b-a)/N)$	2%

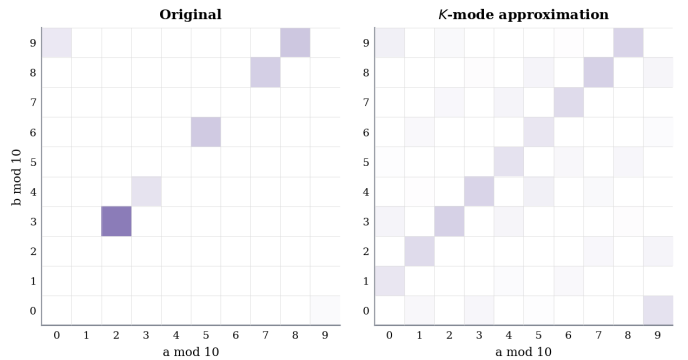


Left: original. Right: K -mode approx.

L13_F81514 $K = 8, R^2 = 0.393$ *mixed: 44%; diff: 31%; sum: 7%; col: 7%; row: 6%; parity: 5%*

$$\begin{aligned}
f(a, b) \approx & 0.026 + 0.052 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.05 \cos\left(\frac{2\pi(a-b)}{10} + 0.563\right) \\
& + 0.05 \cos\left(\frac{2\pi(4a-4b)}{10} + 2.578\right) + 0.046 \cos\left(\frac{2\pi(2a-2b)}{10} + 1.213\right) + 0.046 \cos\left(\frac{2\pi(3a-3b)}{10} + 1.928\right) \\
& + 0.029 \cos\left(\frac{2\pi(-a+3b)}{10} + \frac{\pi}{2}\right) + 0.029 \cos\left(\frac{2\pi(4a-2b)}{10} - \frac{\pi}{2}\right) + 0.028 \cos\left(\frac{2\pi(5a+3b)}{10} + \frac{\pi}{3}\right)
\end{aligned}$$

Dir	Structure	Frac
mixed	—	44%
diff	$\cos(2\pi k(b-a)/N)$	31%
sum	$\cos(2\pi k(a+b)/N)$	7%
col	$f(b)$	7%
row	$f(a)$	6%
parity	$(-1)^{a+b}$	5%

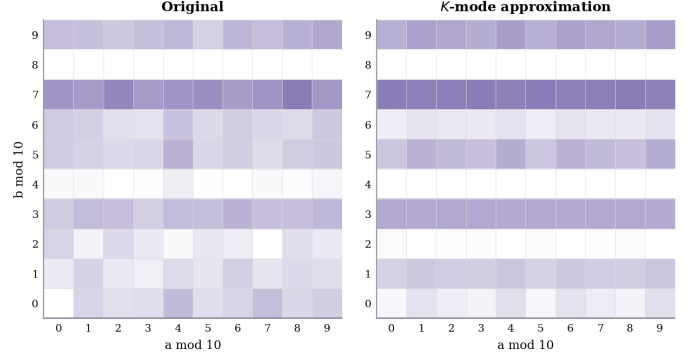


Left: original. Right: K -mode approx.

L00_F48963 $K = 8, R^2 = 0.501$ *col: 47%; col_{parity}: 42%; mixed: 7%*

$$f(a, b) \approx 0.329 + 0.317 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.175 \cos\left(\frac{2\pi(3b)}{10} + 0.54\right) \\ + 0.103 \cos\left(\frac{2\pi(4b)}{10} + 1.107\right) + 0.103 \cos\left(\frac{2\pi(b)}{10} + 1.804\right) + 0.073 \cos\left(\frac{2\pi(2b)}{10} - 2.048\right) \\ + 0.036 \cos\left(\frac{2\pi(4a)}{10} + 2.951\right) + 0.034 \cos\left(\frac{2\pi(4a+2b)}{10} + 3.09\right) + 0.03 \cos\left(\frac{2\pi(4a-2b)}{10} + 2.679\right)$$

Dir	Structure	Frac
col	$f(b)$	47%
col _{parity}	—	42%
mixed	—	7%
row	$f(a)$	2%

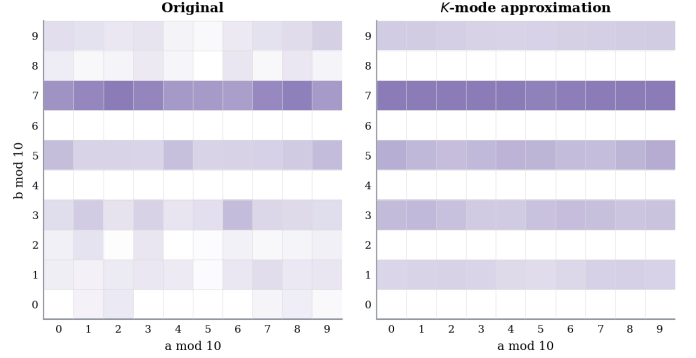


Left: original. Right: K -mode approx.

L00_F126663 $K = 8, R^2 = 0.526$ col: 51%; col_{parity}: 43%; mixed: 4%

$$f(a, b) \approx 0.214 + 0.33 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.157 \cos\left(\frac{2\pi(4b)}{10} + \frac{\pi}{4}\right) \\ + 0.133 \cos\left(\frac{2\pi(-b)}{10} - \frac{2\pi}{3}\right) + 0.121 \cos\left(\frac{2\pi(2b)}{10} - 2.666\right) + 0.088 \cos\left(\frac{2\pi(3b)}{10} - 0.723\right) \\ + 0.029 \cos\left(\frac{2\pi(-2a+3b)}{10} + \frac{2\pi}{3}\right) + 0.028 \cos\left(\frac{2\pi(a)}{10} + 0.356\right) + 0.026 \cos\left(\frac{2\pi(-2a-b)}{10} + 2.557\right)$$

Dir	Structure	Frac
col	$f(b)$	51%
col _{parity}	—	43%
mixed	—	4%
row	$f(a)$	1%

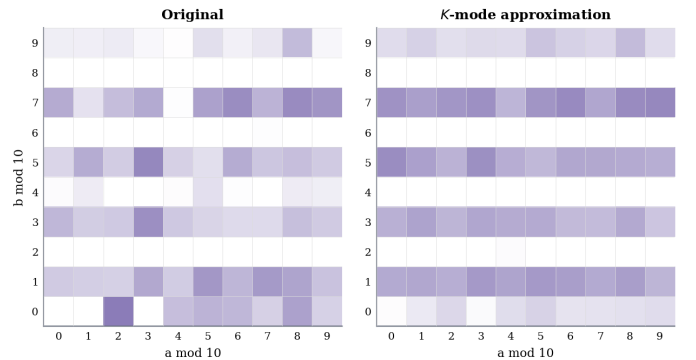


Left: original. Right: K -mode approx.

L00_F62283 $K = 8, R^2 = 0.373$ col_{parity}: 37%; col: 29%; mixed: 23%; row: 5%; sum: 4%; diff: 2%

$$f(a, b) \approx 0.268 + 0.346 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.152 \cos\left(\frac{2\pi(3b)}{10} - \frac{\pi}{4}\right) \\ + 0.114 \cos\left(\frac{2\pi(4b)}{10} - 0.078\right) + 0.097 \cos\left(\frac{2\pi(2b)}{10} - 0.919\right) + 0.069 \cos\left(\frac{2\pi(4a+5b)}{10} + 1.953\right) \\ + 0.062 \cos\left(\frac{2\pi(3a+b)}{10} - 2.785\right) + 0.061 \cos\left(\frac{2\pi(-a-b)}{10} - 2.65\right) + 0.052 \cos\left(\frac{2\pi(3a-3b)}{10} + 2.801\right)$$

Dir	Structure	Frac
col _{parity}	—	37%
col	$f(b)$	29%
mixed	—	23%
row	$f(a)$	5%
sum	$\cos(2\pi k(a+b)/N)$	4%
diff	$\cos(2\pi k(b-a)/N)$	2%

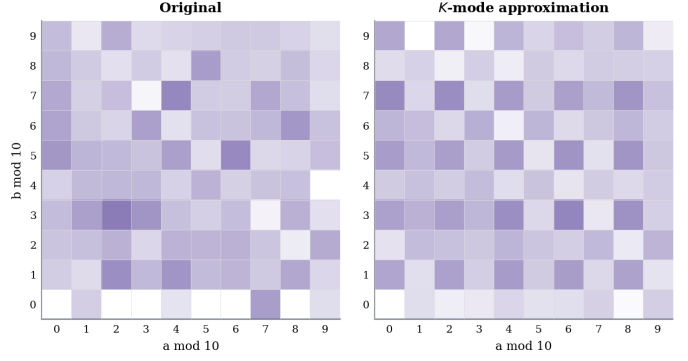


Left: original. Right: K -mode approx.

L18_F142773 $K = 8, R^2 = 0.349$ *mixed: 33%; col: 20%; sum: 17%; parity: 10%; diff: 9%; col_parity: 4%; row: 3%; row_parity: 3%*

$$f(a, b) \approx 0.44 + 0.137 \cos\left(\frac{2\pi(5a+5b)}{10} + \pi\right) + 0.1 \cos\left(\frac{2\pi(4a+4b)}{10} + 3.081\right) \\ + 0.092 \cos\left(\frac{2\pi(5b)}{10} + \pi\right) + 0.091 \cos\left(\frac{2\pi(b)}{10} - 2.768\right) + 0.073 \cos\left(\frac{2\pi(2b)}{10} - 2.398\right) \\ + 0.07 \cos\left(\frac{2\pi(5a)}{10}\right) + 0.069 \cos\left(\frac{2\pi(4a+b)}{10} + 1.734\right) + 0.068 \cos\left(\frac{2\pi(-a-b)}{10} - 2.819\right)$$

Dir	Structure	Frac
mixed	—	33%
col	$f(b)$	20%
sum	$\cos(2\pi k(a+b)/N)$	17%
parity	$(-1)^{a+b}$	10%
diff	$\cos(2\pi k(b-a)/N)$	9%
col_parity	—	4%
row	$f(a)$	3%
row_parity	—	3%

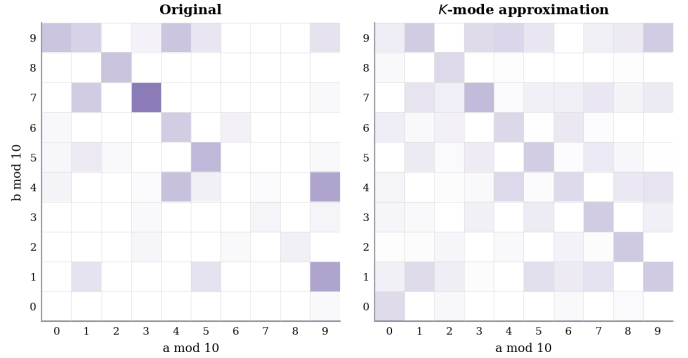


Left: original. Right: K -mode approx.

L20_F12435 $K = 8, R^2 = 0.390$ *mixed: 41%; sum: 27%; col: 8%; row: 8%; diff: 7%; parity: 5%; col_parity: 2%; row_parity: 2%*

$$f(a, b) \approx 0.08 + 0.082 \cos\left(\frac{2\pi(5a+5b)}{10}\right) + 0.074 \cos\left(\frac{2\pi(4a+4b)}{10} - 0.385\right) \\ + 0.068 \cos\left(\frac{2\pi(a+b)}{10}\right) + 0.067 \cos\left(\frac{2\pi(3a+3b)}{10} + 0.296\right) + 0.059 \cos\left(\frac{2\pi(2a+4b)}{10} - 2.498\right) \\ + 0.058 \cos\left(\frac{2\pi(4b)}{10} + 2.455\right) + 0.054 \cos\left(\frac{2\pi(2a-2b)}{10} - 0.139\right) + 0.053 \cos\left(\frac{2\pi(5b)}{10} + \pi\right)$$

Dir	Structure	Frac
mixed	—	41%
sum	$\cos(2\pi k(a+b)/N)$	27%
col	$f(b)$	8%
row	$f(a)$	8%
diff	$\cos(2\pi k(b-a)/N)$	7%
parity	$(-1)^{a+b}$	5%
col_parity	—	2%
row_parity	—	2%

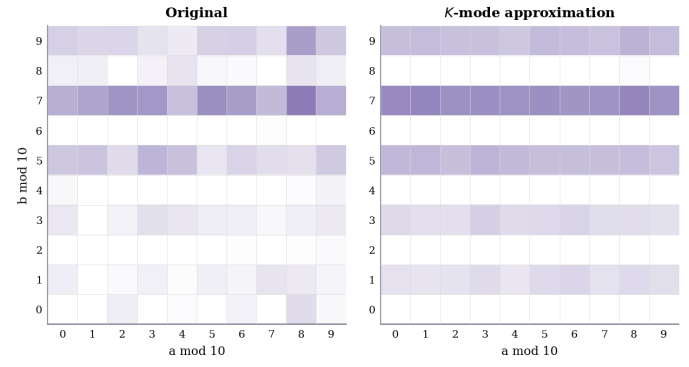


Left: original. Right: K -mode approx.

L00_F127339 $K = 8, R^2 = 0.483$ *col: 45%; col_parity: 40%; mixed: 9%; row: 2%*

$$f(a, b) \approx 0.182 + 0.293 \cos\left(\frac{2\pi(5b)}{10} - \pi\right) + 0.158 \cos\left(\frac{2\pi(b)}{10} + 1.809\right) \\ + 0.141 \cos\left(\frac{2\pi(4b)}{10} + \frac{\pi}{3}\right) + 0.048 \cos\left(\frac{2\pi(3b)}{10} - \frac{\pi}{4}\right) + 0.041 \cos\left(\frac{2\pi(2b)}{10} - 2.965\right) \\ + 0.033 \cos\left(\frac{2\pi(4a)}{10} - 1.31\right) + 0.032 \cos\left(\frac{2\pi(a+b)}{10} + 1.661\right) + 0.032 \cos\left(\frac{2\pi(3a-b)}{10} + 2.45\right)$$

Dir	Structure	Frac
col	$f(b)$	45%
col _{parity}	—	40%
mixed	—	9%
row	$f(a)$	2%
diff	$\cos(2\pi k(b-a)/N)$	2%
sum	$\cos(2\pi k(a+b)/N)$	2%



Left: original. Right: K -mode approx.